CONTAMINATION-RESILIENT CRT DESIGN · PUBLIC-DATA EDITION · STATISTICS IN MEDICINE

Educational companion

A guided reading of the public-data manuscript, one passage at a time

How to read this

Each entry below takes one passage of the manuscript and does six things with it: says where it sits, what it is doing in the argument, what it means in ordinary language, what kind of evidence stands behind it, what it would be a mistake to conclude from it, and one thing you can do to check it. The evidence label is deliberate. A number produced by a simulation is marked as such and is never presented as a measurement, and an open question stays open.

Entries follow the manuscript in reading order. An entry's number is a fixed label, not its position, so the numbers skip. File names that begin supporting/ refer to the OSF deposit (DOI 10.17605/OSF.IO/XPA4S): CSV files are in its outputs/ folder, R scripts in its code/ folder, and CCR_MAP_directed_3D_elevation.html at its top level.

These six statements hold across the whole companion and the whole paper.

- **NC1.** The design reduces cross-arm exposure and the measured residual bias. It does not deliver freedom from SUTVA violation.
- **NC2.** The trial is simulated. The geography, the survey counts and the connectivity are real; the arms, the intervention and its effect are not.
- **NC3.** The design does not restore nominal coverage for the naive or the boundary-censored estimator.
- **NC4.** The composite ranking is a selection statistic, not a test. No p-value is reported for it.
- **NC5.** The spillover-adjusted estimator needs the mobility kernel. Its good behavior does not survive a kernel the analysis does not assume.
- **NC6.** Randomization inference is recommended for the primary test and is not evaluated in this paper.

E1 The headline coverage numbers: a range, not a single win

Abstract, the sentence reporting boundary-censored coverage of 0.26 to 0.29 under simple randomization against 0.28 to 0.48 under the two-criterion contamination-resilient design, allocating individual sites, across eight candidate pools (nominal coverage is 0.95), together with the later abstract sentence saying these figures assume no between-site variance.

WHAT THE PASSAGE IS DOING

Reports the key inference-level result as a same-estimator comparison and as a range over independent candidate pools, so the reader sees design variability next to the gain, and concedes that, at moderate contamination, coverage is not restored by any naive or boundary-censored analysis.

IN ORDINARY LANGUAGE

Coverage is the share of trials in which a 95 percent confidence interval holds the true effect. A confidence interval is a range of plausible values. The target here is 0.95. The abstract compares like with like. It uses one estimator, the boundary-censored one, under two designs. Boundary-censored means we drop the 30 percent of sites most exposed to spillover before we do the math. Spillover is infection risk that crosses a site boundary. Contamination is infection risk mixing across the border between the two arms, that is, the treated group and the comparison group. All of these numbers are for moderate contamination. Under simple randomization, coverage is 0.26 to 0.29. Under the contamination-resilient design, the two-criterion design built to limit spillover, it is 0.28 to 0.48. These numbers come from allocating single sites, not whole clusters. The range comes from drawing the pool of candidate allocations eight times. So the design helps. Some draws show a big gain and some show almost none. Even the best case, 0.48, is far below 0.95. The design does not bring coverage back to the target.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> #### DO NOT ACCIDENTALLY CONCLUDE
>
> Do not read 0.48 as the design's typical performance; it is the top of the range and the single realization shown in Table 1. Do not conclude the design restores valid inference; when individual sites are allocated, no naive or boundary-censored analysis reaches nominal coverage at moderate contamination. Do not read these figures as covering real between-site variation either: they assume none. The appendix adds it as an intracluster correlation, a measure of how much sites differ from one another beyond chance. Once it is present, the binomial analyses fail even without contamination. The analysis of each site, the unit assigned to an arm when sites are allocated, stays calibrated at 0.02 to 0.10. When whole clusters are allocated, at 0.02 to 0.05, the analysis of whole clusters comes close, with a false-alarm rate of up to 0.084. Under contamination, when single sites are allocated, the boundary-censored coverage gain persists. The gain in RMSE, a measure of typical error size, is smaller once the intracluster correlation reaches 0.05. These are fixed-seed simulation outputs laid over real geography, not field results.

TRY IT

Compute the width of each range, 0.29 minus 0.26 and 0.48 minus 0.28, and say which design's performance depends more on the luck of the candidate draw.

E2 What the public-data edition is and is not

Abstract, the closing sentences declaring that assignments and effects are simulated while the geography and survey counts are real, and naming the Yemen cholera and Louisiana COVID-19 corroborating geographies.

WHAT THE PASSAGE IS DOING

Declares the evidential architecture of this edition: every analysis runs on openly available, real measured geography and baseline disease rates, while the trial, the intervention, and its effect are simulated on top, and no conducted trial is being reported.

IN ORDINARY LANGUAGE

Every data set in this edition is open and free to download. The Uganda malaria survey points are real, dated public records. So are the Yemen cholera counts and the Louisiana COVID counts. The map and the baseline disease rates are measured facts. Population is handled differently in each testbed. In Uganda the number of people examined at each survey site stands in for population. In Yemen the population is back-calculated from case counts and attack rates, a relative stand-in rather than a census. Only the Louisiana parishes use real populations, U.S. Census Bureau 2019 estimates. The trial is not real. The paper splits the sites into two arms, that is, two treatment groups. It then simulates the intervention and its effect on top of this real geography. So any trial performance number, such as bias or coverage, comes from simulation. Bias is how far off an estimate sits on average. Coverage is how often a confidence interval holds the true value. The abstract states that the assignments and effects are simulated. It states that the geography and survey counts are real.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

> #### DO NOT ACCIDENTALLY CONCLUDE
> Do not conclude the operating characteristics are field results from real trials; equally, do not dismiss the geography as toy data. The correct split for the malaria testbed: coordinates, survey denominators, and parasite rates are measured and dated, while everything downstream of the simulated intervention is modeled. In the Yemen and Louisiana testbeds the outcome's denominator is simulated as well, a hypothetical survey of 800 per governorate or parish. Also, reported case counts reflect testing intensity as well as transmission, a caveat the manuscript carries through its later descriptive readings.

TRY IT

Make a two-column list for the malaria testbed: real (coordinates, parasite rates, survey denominators) versus simulated (arm assignment, intervention, treatment effect). Check your list against the abstract sentence that draws exactly this line.

E4 Contamination and bias toward the null

Introduction, the opening paragraph, where bidirectional spillover across cluster boundaries is described as biasing treatment effect estimates toward the null.

WHAT THE PASSAGE IS DOING

Sets up the core problem the paper addresses: in spatially connected settings, treatment effects leak across cluster boundaries in both directions, and this leakage systematically shrinks the estimated effect toward zero rather than merely adding noise.

IN ORDINARY LANGUAGE

A cluster randomized trial assigns whole communities, such as villages, to treatment or control. It does not assign single people. Contamination means infection risk mixes across community borders between the trial's arms. Mosquitoes move and people travel. So control villages near treated villages get some protection. Treated villages also keep taking in parasites from untreated neighbors. Both leaks make the two arms, treated and control, look more alike. The effect we measure then shrinks toward zero. That shift is called bias toward the null. The intervention itself is not weaker. Only the comparison between the two arms is blurred.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> #### DO NOT ACCIDENTALLY CONCLUDE
>
> Do not conclude that contamination is merely a power problem that a larger sample would fix, or that a null result from a contaminated trial proves the intervention failed. The passage argues the damage is systematic bias, not random noise, so more data alone converges to the wrong answer. Also do not conclude bias toward the null is safely conservative. It can bury a genuinely effective intervention, and the paper's simulations show the opposite error too: on the malaria geography at moderate contamination under simple randomization, the naive test's Type-I error, its rate of declaring an effect when the treatment effect is zero, averages 0.730 over the eight candidate pools (Limitation 4), because at zero treatment effect the simulated contamination still mixes baseline infection risk across the two arms according to the assignment, and the naive analysis leaves that mixing out. That figure measures the omitted mixing of baseline risk, not spillover of the treatment effect, and it assumes no variation between sites beyond binomial sampling.

TRY IT

List the two leak directions named in the Introduction's opening paragraph, positive spillover into control villages and efficacy leakage into intervention villages. For each, write one line showing why it moves the between-arm difference toward zero.

E5 SUTVA, partial interference, and the estimand

Introduction, the opening paragraph stating SUTVA together with the later paragraph that adopts the exposure-mapping machinery of the interference literature (not partial interference, which the paper sets aside in the estimand subsection) to define the boundary-insulated effect as the target estimand.

WHAT THE PASSAGE IS DOING

Defines the assumption that contamination violates (SUTVA), the formulation the paper actually works under (an exposure-mapping summary of spatial interference, explicitly not partial interference), and the causal quantity the design is built to target (the boundary-insulated effect).

IN ORDINARY LANGUAGE

SUTVA is the Stable Unit Treatment Value Assumption. Its no-interference part says each community's outcome depends only on its own treatment. It must never depend on what its neighbors get. Infectious disease breaks this rule, because infections cross borders. So the paper does not assume that spillover stops at some group boundary. Instead, it scores each site's exposure to spillover with an assumed mobility model. Each site gets a cross-arm exposure score. That score adds up the pull from every site in the other arm. Each pull is weighted by how many people were surveyed at each site and by how close the two sites are. The paper states its design does not rely on partial interference, the assumption that spillover happens within defined groups of clusters but never between them. In any allocation the trial can use, both arms hold sites, so no site reaches zero exposure. Some sites simply have far less of it than others. An estimand is the exact quantity a study sets out to estimate. The paper's estimand is the boundary-insulated effect. That is the treatment effect a site would show if it had zero cross-arm exposure. Zero exposure happens only when every other site shares its arm. So the target compares every site treated with every site untreated. It is a total effect. It is not a direct effect with the neighbors held fixed. No site actually has zero exposure. So this is a limit the paper approaches among its least-exposed sites, the core zone. It is not the effect in a group of fully shielded sites.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that the framework restores SUTVA or makes interference disappear. It instead defines an estimand that remains meaningful under interference and then designs the trial to reduce cross-arm exposure. Later sections state the estimand is approximately recovered among core-zone units: bias is reduced, not eliminated.

TRY IT

Write the potential outcome of one village as a function of its own assignment and its neighbors' assignments. Mark which dependence terms SUTVA forbids. Then say which of those the paper's exposure score keeps, since no allocation the trial can use gives any site zero cross-arm exposure.

E6 Balance is not contamination control

Introduction, the paragraph criticizing geographic buffering and Euclidean distance-based partitioning together with the following literature paragraph, where covariate balance is said not to control contamination bias.

WHAT THE PASSAGE IS DOING

States the paper's central negative claim: the field-standard balance remedy, covariate-constrained randomization, controls baseline balance, but balance and contamination are different failure modes, and controlling one does not control the other. The claim is asserted here and demonstrated later by simulation.

IN ORDINARY LANGUAGE

A cluster randomized trial assigns whole groups, such as villages, to two arms. Each such group is a cluster. A covariate is a trait measured before the trial starts, such as height above sea level or the disease rate at the start. Balance means the two arms start out alike on their covariates. The standard tool for this is covariate-constrained randomization. It draws a random split, then throws that split out if the arms differ too much at the start. Balance guards against a bad draw that leaves the starting groups lopsided. Contamination is a different failure. It is infection risk mixing between the arms across their shared borders after the groups are assigned. A design can be balanced and still put every cluster next to a cluster in the other arm. The paper's central negative claim is that balance alone does not control bias from contamination. The claim is stated here. The support is the study of operating characteristics reported later, which shows how each design performs over many simulated trials.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> #### DO NOT ACCIDENTALLY CONCLUDE
>
> Do not conclude that balance is useless or can be discarded. Balance still guards against baseline confounding, and it is one of the two criteria of the two-criterion design (CRR) that the operating-characteristics study uses. The manuscript later shows the trade-off cuts both ways across geographies, so that two-criterion design is a robustness compromise, not a claim that contamination control always dominates balance. And do not conclude that the framework's own contamination scores follow the environmental corridors, such as waterways and elevation, that the passage faults distance-based designs for missing: they are built from distances between sites, baseline prevalence and survey size alone, and the environmental covariates enter its balance, topology and free-energy terms, not its contamination scores.

TRY IT

Sketch a checkerboard assignment of villages to two arms. Balance on almost any covariate can be excellent, yet every village borders the opposite arm. State which failure mode this design addresses and which it maximizes.

E7 Positioning among prior work and the anchoring limit

Introduction, the literature-synthesis paragraph that runs through the Leung boundary-estimator lineage and the Watson et al. identification result and closes with the spatial-analysis contamination models, which it presents as analysis-side alternatives to the gravity-flux model, neither nesting the other.

WHAT THE PASSAGE IS DOING

Locates the framework at the junction of three literatures and concedes a proved identification limit: the cited Watson et al. result is a theorem in that work, and the paper positions its own core-zone estimator as an anchored estimator subject to that limit.

IN ORDINARY LANGUAGE

This passage places the framework in three lines of past work. Then it names a hard limit. Causal inference under interference gives the estimand, which is the exact thing the design aims to measure. Interference means that a village's result can depend on how other villages were treated. Design-based work gives buffering, which leaves unused space between the arms. It also gives randomization that is constrained on covariates. Mobility research gives the gravity model, which says flow between two places grows with their sizes and shrinks with the distance between them. The design keeps that predicted flow between the arms low. Watson and colleagues prove a limit. It holds for cluster randomization with a fixed exposure set size. A spillover kernel is the curve for how an effect leaks with distance. The data identify the shape of that curve, but not its level. So every direct-effect estimator rests on an anchoring assumption. That is a chosen reference point that fixes the level. The paper's boundary-censored core-zone estimator drops the 30 percent of sites with the highest cross-arm exposure and treats the rest as approximately unexposed. The paper's target is a total effect, not a direct effect. Still, this estimator fixes the spillover level by assumption, so it is anchored too. The claim here is that the design cuts exposure. It does not pin down the effect free of assumptions.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> #### DO NOT ACCIDENTALLY CONCLUDE
>
> Do not conclude that the boundary-censored estimator identifies the target effect free of assumptions, or that this paper escapes the shape-versus-level limit. It does not claim to. What stays anchored is the level: the paper fixes it by treating the lowest-exposure sites as approximately unexposed, and the residual bias its contamination sweep measures is the footprint of the level term that censoring cannot remove. The shape term, error from misspecification of the kernel itself, is probed directly, in the kernel-misspecification sensitivity analysis, where the contamination is generated by kernels the design and the analysis do not assume, and where the spillover-adjusted estimator loses its nominal coverage. Also do not read the paragraph as claiming novelty over Leung's boundary exclusion; the paper presents its estimator as the analysis-side analogue of that device. And do not read the finding that fewer, larger clusters reduce perimeter contamination as free: it is the bias side of a trade-off whose variance side the main simulations do not charge, and Appendix D, which does charge it, finds the coarser pools giving up power.

TRY IT

Pose this question to any effect estimate in the paper: what assumption pins the spillover level here? For the core-zone estimator, find the paper's answer and check it against the shape-versus-level distinction stated in this paragraph.

E71 The framework's objective and its three components

Introduction, the short paragraph that closes the framing material and states in one sentence what the optimization framework is and what it optimizes, just before the Interaction-Aware Feature Space subsection.

WHAT THE PASSAGE IS DOING

This is the thesis sentence of the Introduction. It turns the preceding catalog of risks into a single stated objective, baseline covariate exchangeability set against proxy measures of topological fragmentation and cross-arm contamination exposure, and it names the machinery that supplies that objective: graph-theoretic spatial clustering, Persistent Homology from Topological Data Analysis, and an ensemble decision engine, through which the framework jointly considers operational geometry, spatial topology, and statistical identity. It is the Introduction's statement of what the whole apparatus is for, ahead of the Methods section that groups the ten metrics into three pillars, and it attaches a limited claim to it: robustness to measured spatial and mobility-related contamination routes rather than their removal.

IN ORDINARY LANGUAGE

This sentence says what the tool is. The tool searches many ways to split the study sites into two arms. A site is one village or survey point that can be assigned. An arm is one of the two groups in a trial. The tool gives each split a score, and the score has three parts. The first part is balance on covariates. A covariate is a trait measured before the trial starts, such as height above sea level. Balanced arms start out alike on those traits. The older method here is constrained randomization. It draws a split at random, then throws it out when the arms differ too much at the start. This framework keeps that goal and adds two more parts. The second part is shape. The tool uses clustering, which gathers nearby sites into groups. It also uses persistent homology, a way to count the holes and gaps in a pattern of points. That method comes from topological data analysis, the study of shape in data. The third part is exposure to contamination. Contamination is infection risk that mixes across the border between the two arms. The tool cannot watch leakage happen. It uses stand-ins, or proxies, built from distance and from travel between places. An ensemble engine then blends the three parts into one ranking. Ensemble here means the parts are pooled by set rules rather than by taste. The promise made is careful. The chosen split should hold up better against the leak routes the data can see.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read the words optimization and maximize as a guarantee about the design that comes out. The framework ranks candidate configurations on a composite score and keeps a high-ranking one; that composite is a selection statistic, not a test, so a high score is not evidence that the arms are exchangeable or that spillover has been designed away. Take seriously what the sentence itself concedes: the topology and contamination terms are proxy measures of fragmentation and exposure, not observed leakage, and the robustness claimed reaches only measured spatial and mobility-related sources. The design reduces cross-arm exposure and residual bias without delivering freedom from SUTVA violation; it does not restore nominal coverage for the naive or boundary-censored estimator; and the spillover-adjusted estimator paired with it still depends on the mobility kernel, which the paper tests at site-level allocation

only. Randomization-based inference is recommended in this paper but not evaluated in it.

TRY IT

Open supporting/PROPOSED_design_selection_MAP.csv and read only its header row. Try sorting the ten metric columns into the three pillars of the manuscript's 10-Metric Diagnostic Suite: d_balance, d_wasser and d_density_symmetry belong to Pillar 1, Statistical Interchangeability, the statistical identity axis this sentence names; d_euler, d_persistence_gap, d_vfe and d_spatial_mixing belong to Pillar 2, Topological and Spatial Cohesion, which carries both the spatial topology and the operational geometry of the sentence, since the paper reports the Euler Characteristic, Persistence Gap and free-energy terms as operational-feasibility diagnostics; and d_contam_auc, d_frontier_auc_exposure and d_mobility_auc_leakage belong to Pillar 3, Contamination Resilience. The composite_score column is the ensemble output the sentence promises. Check your sorting against the metrics section of the manuscript, which also notes that d_spatial_mixing, though placed in Pillar 2, is a second functional of the same cross-arm risk-asymmetry profile as the Contamination AUC, so the pillars group the metrics without partitioning what each one measures.

E52 What the expanded feature space does not guarantee

Introduction, the Interaction-Aware Feature Space subsection, the paragraph stating what the expanded covariate space fixes and what it leaves unresolved.

WHAT THE PASSAGE IS DOING

Fixes the comparison scale for the balance metrics, so candidate treatment and control arms are scored inside compound ecological strata rather than on marginal averages alone. It then concedes that this does not equalize the arms, notes that the selected design retains a baseline prevalence difference reported in Results, and points to the mitigation appendix for the countermeasures, framing the benefit as bounding one route to spatial confounding.

IN ORDINARY LANGUAGE

Balance means the two arms start out alike on background traits measured before the trial. The added product terms change what the balance scores compare. Arms are now scored on combinations of conditions, such as low ground near water, not only on each trait's separate average. In this public build, though, nearness to water is not mapped. It is a stand-in built from each site's elevation rank. The paper is careful here. Scoring on combinations does not make the arms equal, and the chosen design still starts with a gap in baseline malaria rate. That gap is reported later, and an appendix lists countermeasures that a trial adopting the design could use to address it; the paper itself does not apply them.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

DO NOT ACCIDENTALLY CONCLUDE

Do not conclude that the arms are comparable at baseline because the feature space was expanded. The passage says only that the expansion sets what balance is scored on, and it states plainly that a baseline prevalence difference survives in the selected design. Treat the protection as bounded to the measured traits and combinations actually included, not as protection against unmeasured drivers of transmission.

TRY IT

Open supporting/PROPOSED_selected_design_arms_MAP.csv and compare the mean_baseline_prevalence values in the treatment and control rows. The difference you see is the residual baseline gap this passage admits, so read Results and the mitigation appendix for how the paper handles it.

E53 Building the analytic site set, and what “water proximity” really is

Study Data and Context, the opening paragraph that builds the analytic site set stage by stage and discloses the elevation-rank stand-in for water proximity.

WHAT THE PASSAGE IS DOING

Specifies the construction of the analytic sample step by step: a survey-year restriction, aggregation of co-located points to a single site, then a binomial usability filter, with the resulting counts stated for each stage and the median-latitude split into two strata. It also discloses that in the public build water proximity is not a hydrological layer but a normalized rank of elevation, and states the reason the two exclusions are treated as baseline rather than outcome-related.

IN ORDINARY LANGUAGE

The author builds the site list in stages. They keep surveys from 2009 onward and merge points at the same spot, which gives 110 merged points. They then keep only points where at least 30 people were tested and the rate is neither 0 nor 1, leaving 57. One map is drawn on all 110; every analysis uses the 57. Elevation is real, pulled from a public terrain service. But water proximity here is not water data at all: it is a rank of elevation, so low-lying sites score high.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read “water proximity” as measured hydrology anywhere in this paper. In the public build it is a rank of elevation, so the elevation-by-water interaction is a function of elevation alone, a substitution the passage discloses. Do not treat the 110-point map and the 57-site analysis as describing the same sample. And do not conclude the filters are harmless because the paper calls them baseline properties rather than outcome-related selection: they still narrow the geography on which the design was built and scored.

TRY IT

Open supporting/PROPOSED_selected_design_arms_MAP.csv and add the two arms’ site counts, then add their sites_stratum_south counts. Check that the totals match the analytic site count and the southern-stratum count stated in this passage, with the remainder falling in the northern stratum.

E8 The public malaria geography: real sites, simulated trial

Study Data and Context, the opening paragraph describing the Malaria Atlas Project survey points, the filters that yield 57 analytic sites, and the simulated allocation imposed on that real geography.

WHAT THE PASSAGE IS DOING

Establishes the public testbed: real Malaria Atlas Project survey sites with real baseline prevalence, filtered to 57 aggregated sites, onto which a fully synthetic two-arm cluster-randomized allocation is imposed. It also states the honesty register: this is an illustrative design exercise, not a record of any conducted trial.

IN ORDINARY LANGUAGE

The testbed is a set of real malaria survey points in Uganda. Each point has real coordinates, a real count of people tested, and a real count testing positive. The parasite rate is the share of tested people who were positive. The author keeps surveys from 2009 onward, merges surveys at the same spot, and requires at least 30 people tested. That filtering leaves 57 sites. The tested count is the survey's sample size, not a census of the community. The trial itself is invented. A cluster-randomized trial assigns whole communities, not individuals, to treatment or control. Here the arms and the treatment effect are simulated on top of the real map. These sites sit in dense, connected settlements, so infection risk could mix between nearby arms. That mixing is called contamination, and it is the design problem this paper attacks.

EVIDENCE STATUS **MEASURED DATED** a real measurement from dated real data

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that any trial was run, or that any later operating characteristic is a field measurement. The coordinates, examined counts, and prevalence are real and dated (PR point dataset version 202406, surveys 2009 onward), but the arms and effects are simulated, so every downstream performance number is a model output. Also do not assume the 57 sites are the paper's only representation of this data: a later descriptive map uses a larger, differently filtered set of survey points, and the two counts serve different purposes.

TRY IT

Install the malariaAtlas R package, pull the Uganda PR points, and apply the stated filters: year at least 2009, at least 30 examined, prevalence strictly between 0 and 1, co-located surveys merged. Check whether you reproduce 57 sites.

E9 Engineered eco-epidemiological interaction covariates

Study Data and Context, the paragraph defining the baseline covariate matrix, the numbered list of the two eco-epidemiological interaction terms, and the sentence after it saying where those terms enter.

WHAT THE PASSAGE IS DOING

Defines the baseline covariate matrix and justifies adding two spatial interaction terms (elevation times water proximity, latitude times water proximity), so that candidate arms are compared on compound environmental risk profiles, through the Wasserstein Distance and the Density Symmetry metric, rather than only on marginal averages. The Population Balance metric uses the four

marginal covariates alone.

IN ORDINARY LANGUAGE

A covariate is a background variable measured before the trial, such as elevation. The design tries to balance the two arms on these variables. The author adds two product terms. One multiplies elevation by a water-proximity term. The other multiplies standardized latitude by that same term. The motivating idea is that mosquito breeding risk spikes where low ground sits next to water. In the public build, though, there is no hydrology layer at all. Water proximity is a rank of elevation that scores how low-lying a site is, from 0 to 1. So the first product is a nonlinear function of elevation alone, and the second is standardized latitude times that elevation rank. Both stand in for what a real trial would compute from actual water data. Both products enter the Wasserstein Distance, a score that compares the whole spread of the arms' covariates, as two extra features. The first product also sets the weight on each site's area in the Density Symmetry score. The Population Balance score leaves both out. So candidate arms get compared on combinations, not just on each variable's separate average.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that the paper estimated or verified these ecological synergy effects on the Uganda data. The interaction terms are design inputs motivated by vector-ecology reasoning. Balancing on them lowers the chance that sites high on these computed combinations end up concentrated in one arm, which bounds one route to confounding if the combinations matter. It does not guarantee equality between the arms, and it does not demonstrate that the combinations matter.

TRY IT

Ask what balancing on an interaction costs if the interaction is actually pure noise. Reason it through: some balancing effort is spent on an irrelevant column, but no bias is introduced, which is why such terms function as cheap insurance.

E10 Two distance rulers: standardized coordinates for search, great-circle distance for scoring

Study Data and Context, the closing paragraph on the dual-resolution distance architecture and the standardized clustering space.

WHAT THE PASSAGE IS DOING

Describes the two distance spaces: great-circle (haversine) distances on a sphere form the matrix D_geo, which anchors the evaluation suite, with every metric scored on D_geo, on UTM-projected areas or on a kilometer point cloud, while candidate partitions are generated on longitude and latitude standardized to unit variance, which is disclosed as a deliberate anisotropic rescaling rather than an approximation to a metric.

IN ORDINARY LANGUAGE

The framework uses two distance rulers. To score a plan, it uses great-circle distance in kilometers, the shortest path over a sphere. On this geography, that distance stays within 0.6 percent of the true ellipsoid distance. The fast search step uses a different ruler. It centers longitude and latitude, then divides each by its spread across the 57 sites; statisticians call that standardizing. Latitude varies less than longitude here. So one degree of latitude counts 1.33 times one degree of longitude in the search space. That favors clusters stretched east to west. The rescaling only changes which groupings get proposed. Every metric is scored in kilometers, so no score inherits it.

EVIDENCE STATUS **ANALYTIC** follows by mathematics

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude the search space is a flat, undistorted map; the paper now says plainly that it is stretched by a factor of 1.33 along latitude. Do not conclude that stretch reaches any result: scoring runs on the kilometer matrix, projected areas or the kilometer point cloud. And do not read the distances as WGS 84 geodesics; they are spherical haversine distances, which the paper states.

TRY IT

Take two sites one degree apart in longitude and two sites one degree apart in latitude. Compute their distances in the standardized search space using the two standard deviations quoted in the passage, 1.035 and 0.780 degrees, and confirm the 1.33 ratio.

E54 The workflow figure: the pipeline before its parts

Methods, the opening sentence of the section together with the workflow figure it sends the reader to before any single stage is defined.

WHAT THE PASSAGE IS DOING

The sentence hands the reader the architecture of the whole procedure in one figure before the subsections define each stage. The figure fixes the order of the steps, from the public survey sites through candidate cluster generation, metric evaluation, score standardization, ensemble weighting and final design selection, and shows the random-allocation null baseline entering as a side branch rather than as the last step.

IN ORDINARY LANGUAGE

A workflow figure is a map of the steps a method takes, in order. This one starts with the real survey sites. It ends with a single featured design, the top-ranked member of an elite pool of the 40 highest-scoring candidates. A trial would not simply adopt that top design: the allocation it implements is drawn at random, with equal chances, from the whole elite pool. In between, sites are grouped into clusters. Each candidate grouping is scored on a fixed set of design metrics. Those scores are then combined into one ranking. A side branch builds a set of plain random allocations. That lets each candidate be judged against what chance alone would give. Read the figure first and the rest of the Methods is much easier to follow. Each later part of that section explains one box.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

DO NOT ACCIDENTALLY CONCLUDE

Do not read the diagram as evidence that the pipeline works, or that this ordering of steps was compared against alternatives and won. A flow chart records the procedure the author ran; whether that procedure improves inference is settled later, by the operating-characteristics simulation, not by the figure.

TRY IT

Trace the figure from top to bottom and, for each box, find the Methods subsection that defines it. Note where the null baseline branch joins the main path, and check that it enters before the ensemble weighting rather than after the final ranking.

E72 Within-stratum partitioning across convex and non-convex families

Methods, the Algorithmic Selection and Clustering Strategy subsection: its opening sentence, which says why more than one clustering family is evaluated, together with the paragraph that routes the standardized distance matrix into those families.

WHAT THE PASSAGE IS DOING

States why the candidate search runs several clustering families rather than one: so the selected design is not an artifact of a single distance metric or geometric assumption. It then names what the candidate-generation stage actually consumes, the standardized planar distance matrix over the analytic sites, and not the true-surface great-circle matrix that the evaluation stage uses. It also fixes, for every algorithm family at once, that partitioning runs inside each stratum, so no candidate cluster spans a stratum boundary. Hierarchical and centroid-based schemes impose compact, convex-leaning shapes, while the graph-theoretic scheme applies a Gaussian kernel to those same distances and imposes no convexity, so the candidate set spans both geometries.

IN ORDINARY LANGUAGE

The search step needs a table of distances between sites. It uses the flat, rescaled distances from the equation just above it. It does not use the curved-earth distances in kilometers. Those are saved for the scoring step. The table covers the 57 sites kept for analysis. Grouping runs inside each stratum. A stratum is a pre-set subgroup of sites, here a band of latitude. So no group of sites ever crosses a band boundary. Several kinds of recipe then run on that same table. The paper runs more than one on purpose. That way the chosen design is not an artifact of one distance rule or one idea of shape. Hierarchical recipes merge nearby groups step by step. Centroid-based recipes grow groups around center points. Both tend to make round, compact blobs. The third recipe is graph-based. It turns each distance into a link strength with a Gaussian kernel. That is a rule where the link fades smoothly to zero as distance grows. It then cuts the weakest links. Its groups may bend or curve. They need not be convex. Convex means the shape has no dents in its outline. Running all of them gives the search blob shapes and bent shapes alike.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

DO NOT ACCIDENTALLY CONCLUDE

Do not read “a comprehensive range” as an exhaustive one. These families, run inside each stratum on one rescaled flat ruler, propose only some of the groupings this geography allows, and a grouping that spans a stratum boundary is never proposed at all; nothing reported here

> can speak for a design the search never generated. Do not read the parenthetical about the true-surface matrix as making the search ruler harmless either. It cannot enter a metric score, but it does decide which candidates the scores ever get to see.

TRY IT

Open supporting/PROPOSED_design_selection_MAP.R and read the design-space comment block near line 178, which states that clusters are formed within each latitude-band stratum and that the per-stratum cluster count is drawn from {4, 6, 8}. Then sort the lat column of data/map_uganda_sites_frozen_20260905.csv, find its median, and confirm that every cluster the engine can build must lie wholly on one side of that median.

E43 Equation 1: candidate partitions are generated on standardized coordinates

Methods, the Algorithmic Selection and Clustering Strategy subsection, the paragraph defining the standardized-coordinate clustering distance and the equation that follows it.

WHAT THE PASSAGE IS DOING

States the distance used to generate candidate partitions, the Euclidean distance on longitude and latitude each standardized to unit variance, discloses the anisotropy that standardization introduces, and confines it to candidate generation.

IN ORDINARY LANGUAGE

A cluster here is just a group of nearby survey sites treated as one unit. Before forming clusters, the code centers longitude and latitude and divides each by its spread. The clustering distance is then plain straight-line distance in that rescaled plane. The sites spread out more in longitude than in latitude. So one degree of latitude counts 1.33 times one degree of longitude. That makes east-west stretched clusters a little more likely to be proposed. This affects only which candidate layouts get proposed. A great-circle distance is a distance measured along the curved surface of the Earth. Every score is computed on great-circle distances, projected areas, or kilometer coordinates. So no metric inherits the stretch.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that the paper clusters on raw map coordinates; an earlier version said so and was wrong. Do not conclude that the 1.33 stretch biases any reported metric; it can only change the candidate set, and the null baseline is built from the same candidate configurations.

TRY IT

Sketch a square of four sites in longitude and latitude, then redraw it after stretching the north-south axis by 1.33. Say which pair of sites a distance-based clustering would now join first.

E11 Pre-set design gates: cluster counts and singleton rejection

Methods, the Algorithmic Selection and Clustering Strategy subsection, the bulleted list of pre-scoring operational constraints on cluster count and singleton rejection.

WHAT THE PASSAGE IS DOING

States the operational constraints every candidate partition must satisfy before scoring: total cluster counts restricted to k in {8, 12, 16} so that clusters split evenly across the two arms within each of the two strata, an allocation balanced in cluster count but not in site count or in examined individuals, k=4 excluded, and automatic rejection of any partition containing a singleton cluster.

IN ORDINARY LANGUAGE

Before scoring, every candidate design must pass two gates. First, the number of clusters k must be 8, 12, or 16. Each of these is divisible by 4, so the clusters split evenly across the two arms within each of the two latitude bands. That even split is in cluster count only. Because clusters differ in size, the arms still end up with unequal numbers of sites and unequal numbers of people examined. A stratum is a pre-defined subgroup, here a latitude band, within which the randomization is balanced. Four clusters was ruled out because so few units cannot support covariate balancing. Second, no cluster may contain only one village. A singleton cluster can break mixed-effects models, which are regression models with cluster-level random terms. Partitions with singletons are dropped automatically, with no hand fixing.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that the k values were tuned to the outcomes, or that these three values are universal limits of the framework. They are stated operational constraints for this two-arm, two-stratum public geography. Also do not read the eventual selection of a spectral k=8 design as evidence that k=8 is generically optimal: the later composite score is a design-selection ranking on this geography, not a hypothesis test.

TRY IT

Work out why an even multiple of 4 yields perfectly balanced allocation across 2 arms times 2 strata. Then determine which k values would remain admissible if a third stratum were added.

E55 Hard filters, and the guardrail the composite runs without here

Methods, the Algorithmic Selection and Clustering Strategy subsection, the operational-constraint list and the sentence closing it that calls the two constraints hard filters and describes the further thresholds the same layer would accept.

WHAT THE PASSAGE IS DOING

States that the cluster-count and singleton constraints act as admissibility filters applied before any candidate is ranked, and that the same filter layer can carry additional non-inferiority thresholds, for example a ceiling on cross-arm mobility leakage at the null median or a cap on the component count. Those thresholds are offered as the protection a compensatory composite lacks against a candidate that scores well on most metrics and unacceptably on one, and the passage reports that none is applied here: the featured design is positive on all ten metrics, but a trial draws its

allocation from the elite pool, and 37 of that pool's 40 allocations score worse than the null on at least one metric, by up to 1.36 null standard deviations.

IN ORDINARY LANGUAGE

The two gates are pass or fail rules, and a design that fails either one is dropped before any scoring. The same slot can hold more rules, such as a ceiling on cross-arm mobility leakage, the modeled travel of people between the arms. Rules like that guard against a weakness in the overall score. That score adds the ten metrics together as a weighted sum. Each metric is first rescaled against a random-allocation benchmark. Each then gets a weight derived from the data rather than an equal share. In such a sum, a large gain on one metric can hide an unacceptable result on another. The author applies no extra rule here. Their featured design came out positive on all ten metrics. But a real trial draws its allocation at random from the 40 best designs, and 37 of those 40 score worse than the random benchmark on at least one metric. So the author recommends such a rule whenever a single metric matters for safety.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

> #### DO NOT ACCIDENTALLY CONCLUDE
>
> Do not conclude that the reported designs were screened by any leakage ceiling or component cap. The passage says none is applied here. Although the featured design is positive on all ten metrics, 37 of the 40 elite-pool allocations a trial could draw are not. The selection-score section states that no additional penalty is applied to a configuration that performs well on most metrics and poorly on one. So the compensatory composite ran without that guardrail. Do not read a design that is positive on all ten metrics as a design shown to improve inference either. That question is settled by the operating-characteristics simulations, not by the filters.

TRY IT

Open supporting/PROPOSED_selected_design_profile_MAP.csv and read the Z column across its ten metric rows to confirm none is negative. Then find the metric with the smallest margin above zero, and say whether a ceiling set at the null mean, where $Z = 0$, would have kept or removed this design on that metric. The paper's example ceiling sits at the null median, which the deposited files do not report.

E12 Five candidate generators, one graph-based outlier

Methods, the Algorithmic Selection and Clustering Strategy subsection, the catalog of the five candidate-generating algorithms and the retry-and-jitter paragraph that closes it.

WHAT THE PASSAGE IS DOING

Catalogs the five clustering algorithms used to generate candidate partitions (Ward's, average linkage, CLARA, K-means, spectral), so the candidate pool does not rest on any one algorithm's grouping rule or convexity assumption. All five work from the same standardized distance, so the distance metric itself is not varied, and on this geography only four yield admissible partitions. Spectral clustering is singled out as the only graph-theoretic member able to follow non-convex boundaries in the settlement pattern.

IN ORDINARY LANGUAGE

Clustering means grouping nearby sites into units. Each unit is then sent to the treatment arm or the control arm as a whole, at random. The paper runs five different grouping recipes. Ward's method and average linkage build groups by merging them step by step. They favor groups that are compact or evenly spread. K-means places group centers and grows round, tight groups around them. CLARA works like K-means, but each center must be a real site. Such a center is called a medoid. With only 57 sites, split into two latitude strata of 28 and 29, the code runs its exact core, PAM, on each full stratum. It keeps the CLARA name. Spectral clustering is different. It turns the map into a network, in which nearby sites get strong links. It then cuts the network where the links are weakest. So the borders can follow the gaps between groups of sites instead of forcing circles. The network is built from the distances between sites alone, with no terrain layer. So a valley or ridge can shape the borders only through where the sites sit. If a method's first try leaves a singleton cluster, the code tries again. It tries up to 80 times in all. After the first try, it adds small random shifts to the site coordinates. It keeps the first partition, that is, one full split of the sites, with no singleton cluster. A singleton is a cluster that holds just one site. In the deposited run, average linkage never found such a partition. Ward and CLARA found one only at K=8, the setting with eight clusters in all. Which combinations pass depends partly on chance in the search. A CLARA K=12 partition is admissible within 80 attempts on 3 of 20 independent restart streams. But the deposited run found none within its 80 attempts.

EVIDENCE STATUS **DEFINITION** defines a term

> #### DO NOT ACCIDENTALLY CONCLUDE
> Do not conclude that the five algorithms are competing statistical analyses, or that the eventual winner validates its whole family. They only generate candidates; the downstream composite score ranks them, and that score is a design-selection index, not a p-value. On this geography the single top-ranked design (Spectral, K=8, composite 0.86) and the strongest typical composite scores (K=8 median composite 0.27 versus about 0.00 at K=12 and -0.35 at K=16) coincide at the same resolution, and 37 of the 40 elite configurations are K=8 partitions drawn from four algorithm families (Spectral 13, CLARA 10, K-Means 9, Ward 5); the value of the multi-paradigm pool here is that the winning resolution is not an artifact of any single algorithm family, not that it reveals a split between the typical and the optimal K.

TRY IT

Take the affinity kernel $W_{ij} = \exp(-d^2/(2\,\sigma^2))$ and reason about the two extremes of sigma. Predict what the graph, and hence the spectral partition, would look like as sigma shrinks toward zero and as it grows very large.

E13 HDBSCAN tried, then excluded on operational grounds

Methods, the HDBSCAN subsubsection, the paragraphs defining the method and reporting the screening run that disqualified it.

WHAT THE PASSAGE IS DOING

Documents that a sixth method, HDBSCAN, was evaluated for completeness on the 57 public survey sites and then excluded: with a minimum cluster size of 2 it returned 16 clusters but left 4 sites unassigned as noise (12 clusters and 6 unassigned at size 3; 5 and 15 at size 4), and integrating unassigned sites would require subjective post-hoc assignments that break the automated framework's reproducibility.

IN ORDINARY LANGUAGE

HDBSCAN is a grouping method built on density, that is, on how tightly packed the sites are. It forms clusters only where sites sit close together. It labels sparse, far-off sites as noise. A noise point joins no cluster at all. That is fine for map making. But a trial must put every community into a study group, called an arm. We ran it on the 57 public sites with a smallest cluster size of 2. It gave 16 clusters and left 4 sites unassigned. Asking for larger clusters only pushed more sites into the noise bin, first 6 and then 15. Placing those sites by hand would add the very judgment calls the framework exists to remove. So the method was tested, judged unusable for this design task, and dropped.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that HDBSCAN failed as an algorithm or that density-based methods are unsound. It did what it is designed to do: refuse to force sparse points into clusters. The exclusion is an operational judgment about trial-design requirements. The counts are for this geography and these settings; a denser site network could return no noise at all.

TRY IT

List two ways HDBSCAN could still contribute to a trial workflow, for example as a pre-screen flagging outlier sites, and name the new subjective decision each use would reintroduce.

E56 Core distance, mutual reachability, and which ruler HDBSCAN uses

Methods, the HDBSCAN subsection, the paragraph that introduces the mutual reachability distance and, immediately after the equation, defines its core-distance and Euclidean terms and names the kilometer coordinate space the algorithm is run in.

WHAT THE PASSAGE IS DOING

Unpacks the two ingredients of the mutual reachability distance given in the preceding equation, namely the core distance to a point's k-th nearest neighbor and the ordinary Euclidean distance between two points. It then disambiguates the coordinate space, stating that HDBSCAN is run in the kilometer coordinates named in the next paragraph rather than in the standardized clustering plane of Eq. (1) or the great-circle distances of D_geo.

IN ORDINARY LANGUAGE

Each site first gets a core distance: how far you must travel to reach its k-th nearest neighbor, for a fixed k. A site in a crowded area has a small core distance, and a lone site has a large one. The mutual reachability distance between two sites is the largest of three numbers: the two core distances and the ordinary straight-line distance between them. This inflates distances around isolated sites, so they are not quietly folded into a dense group. The line also names the ruler being used, which is kilometer coordinates on the ground. That is not the stretched coordinates used to propose clusters, and not the great-circle distance used for scoring.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read mutual reachability as a real distance between two villages that anyone could measure. It is a working quantity inside one algorithm, and it changes when k changes. Do not conclude that HDBSCAN contributed to the reported design; this line only sets up a method the paper goes on to reject. And do not carry the kilometer ruler over to the other clustering methods, which the paper searches on the standardized coordinates instead.

TRY IT

Open supporting/PROPOSED_hdbscan_screen_MAP.R and find the two lines that project the sites and then divide the coordinates by 1000. Confirm the algorithm is handed kilometer coordinates, as this line of the manuscript states, and not the standardized coordinates of Eq. (1).

E14 CRITIC and entropy weights fused in log space; the third leg sets only the exponent

Methods, the Ensemble Weighting Framework subsection through its Geometric Mean Fusion subsubsection, from the CRITIC and entropy definitions to the fused weights and their reported range.

WHAT THE PASSAGE IS DOING

Defines how the weights on the evaluation criteria are set without hand-tuning: CRITIC and Shannon entropy weights are computed and fused by a geometric mean in log space. A third leg, bootstrap rank stability, is computed by the deposited code but is constant by construction, so the paper describes a two-paradigm fusion and says why the third contributes no column-specific information, although its presence fixes the fusion exponent at 1/3.

IN ORDINARY LANGUAGE

A composite score is one number built from several measures. It must decide how much each measure counts. The paper does not set these weights by hand. It computes two sets of weights from the data and blends them. CRITIC gives more weight to a measure that varies a lot but repeats little of what the others say. That overlap is judged by signed correlation. Shannon entropy asks how evenly spread a measure's values are. Values that are nearly even cannot tell candidate designs apart, so that measure gets less weight. The two sets are blended by a geometric mean in log space. That means the code adds the logs of the two weights and multiplies the sum by one third, so neither set can take over. The one third comes from a third weight that the code also computes from a bootstrap, which resamples the rows at random. That third weight is the same for every column, so it cancels as a factor once the weights are rescaled, but not as an exponent. So the reported weights are the normalized cube roots of the CRITIC and entropy product. They are not the square roots. The paper says so rather than describe the third weight as carrying information about the columns. The final metric weights run from 0.049 for mobility leakage, a score of predicted travel between the arms, to 0.220 for the free-energy metric, a score that rewards a simpler split. The same recipe is also run on the site-by-covariate table, to set how much each of the four covariates counts inside the Population Balance score. The two weight sets are never swapped.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude the resulting weights are objective in a unique, assumption-free sense; choosing these two paradigms and fusing in log space are design decisions. Do not conclude that the bootstrap leg adds robustness; it carries no column-specific information and survives only as the 1/3 exponent, and an earlier version of the paper described it as if it were informative. The metric weights rank candidate trial designs, and the covariate weights only scale the four covariates inside the Population Balance score; neither says anything about the causal importance of any covariate.

TRY IT

Write down the ranks of any five numbers you like and take their mean. Change the numbers and take the mean again. Explain, from what you see, why the variance of that mean across resamples is zero and why the stability weight therefore cancels.

E15 Three pillars of the ten-metric suite

Methods, the subsection "The 10-Metric Diagnostic Suite", the framing paragraph that names the three pillars and says which of its metrics carry the inferential results, together with the itemized pillar lists that follow it.

WHAT THE PASSAGE IS DOING

Organizes the ten design-evaluation metrics into three pillars and declares up front which metrics carry the later inferential results (Pillars 1 and 3, together with the Spatial Mixing term of Pillar 2) and which serve operational feasibility (the three topological metrics of Pillar 2). This framing stops the reader from crediting every metric with the bias and coverage gains reported later.

IN ORDINARY LANGUAGE

The framework scores every candidate trial layout on ten measures. A candidate layout is one way to group villages into clusters. It also assigns each cluster to treatment or control. The ten measures fall into three groups, called pillars. Pillar 1 checks that the two arms start out alike. It compares baseline covariates. Those are traits measured before the trial starts, such as the local malaria rate. Pillar 2 checks shape and terrain. It uses topological data analysis. That is math describing shape, such as connected pieces, holes, and loops. Pillar 3 checks contamination risk. Contamination means infection risk mixes between arms. Mosquitoes or traveling people can carry it. The paper is clear about this split of work. Pillars 1 and 3 carry most of the bias and coverage results. So does one measure listed under Pillar 2, Spatial Mixing, which is a second summary of the same cross-arm contamination profile that the Contamination AUC of Pillar 3 scores. Coverage is how often a confidence interval contains the true effect. The other three Pillar 2 measures, the topological ones, mainly keep the design practical. They favor connected zones, matched terrain, and no scattered checkerboard layout.

EVIDENCE STATUS **DEFINITION** defines a term

> #### DO NOT ACCIDENTALLY CONCLUDE
>
> Do not conclude that all ten metrics contribute equally to the inferential gains reported later. The passage assigns the bias and coverage behavior mainly to the covariate-balance metrics and the contamination-geometry metrics, Spatial Mixing included. Removing the three topological metrics (Euler Characteristic, Persistence Gap, Variational Free Energy) leaves the boundary-censored estimator's coverage unchanged within Monte-Carlo error at weak and moderate contamination and measurably worse at strong contamination, raises its bias from moderate contamination on, and costs the naive estimator coverage at moderate and strong contamination. Those terms act through the resolution of the top-ranked designs, because they favor coarser partitions. Their contribution is real but indirect, and they are reported alongside selection rather than driving the inferential claim.

TRY IT

Before reading further, write the ten metric names under their three pillars from the itemized lists in this region. Then mark which metrics the opening paragraph says carry most of the bias and coverage behavior (two whole pillars and one metric listed under the remaining pillar), and check your marking against the cluster-level comparison of selection objectives in the Discussion.

E16 Contamination AUC across mosquito flight scales

Methods, the 10-Metric Diagnostic Suite, Pillar 3 "Contamination Resilience", the Contamination AUC item and its flight-scale sweep.

WHAT THE PASSAGE IS DOING

Defines the first contamination metric: the asymmetry in mosquito-borne exposure between arms, computed at every plausible vector flight scale and then integrated, so that no single flight-distance assumption is load-bearing for the design choice.

IN ORDINARY LANGUAGE

Mosquitoes can carry infection from one arm of the trial into the other. The problem is that nobody knows exactly how far they fly. Field studies report routine flights of hundreds of meters up to about one kilometer, with observed maxima near three kilometers. Wind can carry vectors much farther. So the metric does not pick one distance. It sweeps a decay scale, called lambda, from 0.5 to 10 kilometers. At each lambda, every pair of villages gets a weight that shrinks exponentially with the distance between them. Using these weights, the method computes each arm's average exposure to the other arm's baseline malaria prevalence. Prevalence is the fraction of people infected. It then takes the absolute difference between the two arms' exposures. That difference is the asymmetry at that flight scale. Finally, it adds up the asymmetry across all scales into an area under the curve, or AUC. A small AUC means exposure stays balanced whether mosquitoes fly short or long distances.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read the 10 kilometer upper bound as a claim that mosquitoes routinely disperse 10 kilometers. The passage states it is a conservative ceiling for the integration sweep, chosen so the exposure profile extends well beyond routine flight, and explicitly says it is not presented as a maximum dispersal distance; the cited routine range is far shorter.

TRY IT

Compute the kernel weight $e^{(-d/lambda)}$ for a village pair about three kilometers apart, once with lambda at the 0.5 km lower bound and once at the 10 km upper bound. The contrast shows how strongly the assumed scale changes who counts as a neighbor, which is why the metric integrates over lambda instead of fixing it.

E57 Integrating frontier risk over buffer radii

Methods, the ten-metric suite under Pillar 3 'Contamination Resilience', the Frontier Risk (Buffer Depth AUC) item, the closing step that sweeps the buffer radius, integrates the frontier ratio trapezoidally into one number, and defines the symbols of that integral.

WHAT THE PASSAGE IS DOING

Removes the arbitrary choice of a single buffer width by evaluating the frontier exposure ratio across the whole radius spectrum and integrating it trapezoidally into a scalar, AUC_frontier, so one summary value enters the composite score rather than one radius-specific reading.

IN ORDINARY LANGUAGE

A buffer is a circle of a chosen radius drawn around each village. The paper does not fix that radius. Instead it sweeps the radius from 0.5 to 5.0 kilometers and records the frontier risk at each step. Frontier risk is the share of the trial's disease burden that sits in villages having a neighbor in the opposite arm. Those swept values are then added up with the trapezoid rule, a standard way to get the area under a curve. The result is a single number, the frontier AUC, and a smaller value means less border-sitting burden across the radii tested.

EVIDENCE STATUS **DEFINITION** defines a term

DO NOT ACCIDENTALLY CONCLUDE

Do not read a small frontier AUC as evidence that little contamination actually crosses the boundary. The integral summarizes one geometric ratio over radii no wider than 5.0 kilometers; on a map whose villages sit farther apart than that sweep, the value can be small because of scale rather than because the layout is well insulated.

TRY IT

Open supporting/PROPOSED_design_selection_MAP.csv and sort the d_frontier_auc_exposure column. That column is not the raw AUC of the equation above; it is the standardized gain against the random-allocation baseline, oriented so that higher is better, and it runs from -7.7585 to 0.2384. Count how many of the 331 candidate designs share the largest value, 0.2384, which is the gain earned by a raw frontier AUC of exactly zero. The count is 297. A tie that large tells you the 0.5 to 5.0 kilometer sweep is separating few candidates, so this integral is doing little work in the ranking on this geography.

E17 Mobility leakage via a gravity model

Methods, the 10-Metric Diagnostic Suite, Pillar 3 “Contamination Resilience”, the Mobility Network Leakage AUC item and its gravity-model friction sweep.

WHAT THE PASSAGE IS DOING

Defines the human-movement counterpart of the contamination metric: a gravity model of travel between villages, with the fraction of simulated flow that crosses arms integrated over a range of travel-friction assumptions to cover demographic uncertainty.

IN ORDINARY LANGUAGE

People moving between villages can spread treatment effects across arms, just as mosquitoes can. The metric simulates this with a gravity model. A gravity model says flow between two places grows with their sizes and shrinks with the distance between them. Here size is epidemiological mass, meaning baseline prevalence times the surveyed population, so populous high-burden villages exchange the most flow. Distance enters raised to a friction exponent, which the paper calls beta. With beta equal to 1, people travel far easily, as with good roads. With beta equal to 3, travel is highly local, as with hard terrain. The true exponent is unknown, so the method sweeps beta from 1 to 3. At each beta, it computes the share of all simulated flow that crosses between treatment and control. It then adds that share up across the sweep into a leakage AUC, an area under the curve. A layout with a small AUC keeps cross-arm traffic low under many mobility regimes.

EVIDENCE STATUS **DEFINITION** defines a term

DO NOT ACCIDENTALLY CONCLUDE

Do not conclude that leakage is computed from observed travel data. The flows are simulated from a formula using pairwise distance and baseline epidemiological mass; a low leakage AUC means the layout is robust within that model family, not that real human movement was measured and found to be contained.

TRY IT

For two villages with fixed masses about three kilometers apart, compute the ratio of the gravity flow at beta equal to 1 to the flow at beta equal to 3. The ratio shows why distant pairs dominate leakage under low friction and nearly vanish under high friction, which is the uncertainty the sweep integrates over.

E18 Directed flux: sources, sinks, and net leakage

Methods, the paragraph headed “Directed contamination connectedness” that follows the 10-Metric Diagnostic Suite and refines the symmetric mobility metric into a source/sink accounting.

WHAT THE PASSAGE IS DOING

Refines the symmetric mobility metric into a direction-aware accounting: it separates each site’s capacity to export infection from its capacity to import, yielding partition-independent source and sink roles and the net direction of cross-arm leakage. It is reported as a diagnostic alongside metric 10, not as an eleventh metric, and it does not enter the selection score, whose mobility term remains the symmetric flux.

IN ORDINARY LANGUAGE

The mobility metric above measures how much flow crosses arms, but not which way. Its formula forces the flow from village i to village j to equal the reverse flow. The refinement breaks that tie. Each village gets a production mass, its capacity to export infection, based on prevalence and population. It also gets an attraction mass, its capacity to import infection, based on its susceptible population. Flow from i to j now multiplies i’s production by j’s attraction, so it differs from the reverse flow. Exports minus imports, summed over all other villages, gives each village a net flux. A positive net flux marks a source, a net exporter of infection. A negative net flux marks a sink, a net importer. This role is computed over all sites before any arm assignment, so it does not depend on the design. For a candidate design, cross-arm leakage then splits into two opposing currents. Their difference shows the net direction in which contamination crosses the trial perimeter. On this geography, high-prevalence sites resolve as net sources and low-prevalence sites as net sinks.

EVIDENCE STATUS

DEFINITION defines a term

> #### DO NOT ACCIDENTALLY CONCLUDE
>
> Do not conclude that the directed refinement drives the bias or coverage results, or that a source label means measured parasite export. The source and sink roles are a descriptive diagnostic computed from a gravity formula over the reported-prevalence surface; they are reported alongside the mobility metric, they do not enter the selection score, and the COVID testbed shows the directed accounting should be kept out of that score.

TRY IT

Using the net flux equation, argue that swapping every village’s arm assignment leaves each village’s net flux unchanged. The sum runs over all sites, not over arms, so the source and sink map is a property of the geography, exactly as the passage’s partition-independence claim states.

E84 Reading the directed source and sink map

Methods, Section 3.3, the figure and caption showing the three-dimensional directed connectedness map of the Uganda survey points, placed after the paragraph that defines the directed flux.

WHAT THE PASSAGE IS DOING

Supplies the reading key for the paper's only picture of the directed flux metric: which sample it is drawn on, what marker color and marker size encode, and why the markers are lifted above the terrain. It is also where the caption fences the picture's status, naming the awkward units of the net flux and denying that the map validates any transmission pathway. The caption is also where the figure itself flags, at the point of use, that its 110 drawn points are a wider set than the 57-site analytic subset the operating-characteristics analysis runs on.

IN ORDINARY LANGUAGE

This is a map of survey sites in Uganda. It is draped over terrain elevation, the height of the land at each spot. Each dot is one survey site. The dots are raised 130 meters so they clear the ground and stay easy to see. The map is drawn on 110 sites. Those are the merged sites from 2009 or later. The trial performance work uses a smaller set of 57 sites. Those sites also had at least 30 people tested and a rate that was neither 0 nor 1. Color shows the sign of net flux. Net flux is exports minus imports in the flow math of the paper. Red marks a net source, a place that sends out more than it takes in. Blue marks a net sink, a place that takes in more than it sends out. Dot size shows how big that net number is. Prevalence is the share of tested people who were positive. Sites with high prevalence come out red. Sites with low prevalence come out blue. The flow numbers carry odd units. They are products of counts of people tested, divided by squared distance in kilometers. The penalty for distance uses an exponent of 2, and any two sites closer than 0.5 km are treated as 0.5 km apart. A version you can spin ships with the paper.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
>
> Do not read the red and blue markers as observed spread. The caption states that the net flux is a descriptive property of the production accounting on baseline prevalence, not an independent validation of transmission pathways; nothing here tracks infection moving between sites. Do not treat the 110 points drawn here as the sample behind any performance number, since those come from the 57-site analytic subset. And do not read the map as a picture of the trial: the 130 meter lift is for legibility only, and the colors are arm-free site roles, so no treatment assignment is being displayed.

TRY IT

Open supporting/CCR_MAP_directed_3D_elevation.html in a browser, rotate it, and check that the reddest markers sit on the high-prevalence sites. Then run wc -l on data/map_uganda_sites_frozen_20260905.csv: it holds 58 lines, one header plus the analytic sites, which is visibly fewer points than the map draws.

E19 Metric weighting, the leave-one-out ablation, and the equal-weight check

Methods, the subsection “Objective Metric Weighting and Ablation Validation”, from its opening through the paragraph reporting the leave-one-out ablation and the equal-weight check.

WHAT THE PASSAGE IS DOING

Explains how the ten metrics are combined without hand-picked weights, then validates the suite three ways: a leave-one-out ablation shifts the ranking for every metric, though three of the ten terms (spatial mixing, mobility leakage and frontier risk) barely move it, an equal-weight composite gives the same winner, and the metrics are mostly non-redundant. It then concedes that influence on the ranking reaches inference only through resolution.

IN ORDINARY LANGUAGE

How much should each of the ten metrics count? The weights come from the data, not from a person. Two weighting schemes are fused to make them. To check that no metric is dead weight, a leave-one-out test drops each metric’s term in turn. It then re-ranks the 331 candidates, leaving the other weights as they are. Old and new rankings are compared by Spearman rank correlation, a score for how well two orderings agree. It runs from 0.82 when the free-energy metric is dropped to 0.997 for spatial mixing. The mean is 0.96. Three terms, spatial mixing, mobility leakage and frontier risk, barely move the ranking. A typical candidate moves about 5 percent of the way through the list. The featured design stays first under seven of the ten reduced objectives. It is second under the other three. Giving all ten metrics equal weight also leaves it first. The two elite pools, the top 40 designs each, share 34 members. The metrics are mostly non-redundant, so they do not simply repeat each other. Their mean absolute correlation is 0.16. One near-duplicate pair sits at 0.82. The suite behaves like about 7.0 effective dimensions out of 10. It takes 8 principal components to carry 90 percent of its variation. The metrics complement each other without being fully independent. Now the concession. Contamination here means infection risk mixing across the trial arms. The free-energy metric moves the ranking most, followed by density symmetry, a balance metric. The three topological metrics are Euler characteristic, persistence gap and free energy. Removing them changes the operating characteristics only at moderate and strong contamination. At those levels, the boundary-censored estimator’s bias rises and the naive estimator loses coverage. The boundary-censored estimator’s coverage falls only at strong contamination. The topological metrics act only by favoring coarser partitions, that is, fewer and larger clusters.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
>
> Do not conclude that because every metric shifts the selection ranking, every metric matters equally for inference. Influence on which design gets picked is not the same as influence on inference; the Discussion’s objectives comparison is where the inferential effect of the topological terms is measured. And do not read “equal weights give the same winner” as proof the weights are irrelevant; the elite pool overlaps in 34 of 40 places, not 40.

TRY IT

Reconcile two numbers in this passage: a mean absolute inter-metric correlation of 0.16, but Population Balance versus the Wasserstein distance at 0.82. Then explain why the free-energy metric, whose removal drops the rank correlation to 0.82, carries the largest weight, 0.220, in the weight column of outputs/PROPOSED_loo_ablation_MAP.csv.

E20 The composite is a ranking, not a test

Methods, the subsection 'Selection Score and Global Optimality Index Computation', the paragraph standardizing each metric against the Monte Carlo null together with the Selection Score definition that follows it.

WHAT THE PASSAGE IS DOING

Defines how ten incommensurable metric scales are made comparable: each is Z-scored against a Monte Carlo null of unoptimized allocations, then combined with the ensemble weights into a single linear Selection Score that ranks candidate designs.

IN ORDINARY LANGUAGE

The ten metrics sit on very different scales. The metric the paper calls the Euler characteristic counts the separate connected pieces of the treatment arm, without subtracting holes, so it is a small whole number. The Wasserstein distance measures how far apart distributions lie, and it varies smoothly. Averaging them raw would be meaningless. So each metric is turned into a Z-score. A Z-score is the number of standard deviations a value sits from a reference distribution. Here the reference is a Monte Carlo null. Monte Carlo means drawing many random repeats to map out a distribution. An allocation is one way of splitting the villages between the treatment arm and the control arm. This null is built from many random, unoptimized allocations of the same villages. Nine of the ten metrics are better when small, so their Z-scores are flipped in sign, and without that flip the score would reward contamination. After the flip, a positive Z-score always means better than the null. So a design's Z-score says by how many null standard deviations it beats ordinary allocation on that one metric. A negative Z-score says by how many it falls short. The weighted sum of the ten Z-scores is the Selection Score. That one number is used to rank the candidate designs. One caution matters here. This machinery borrows the language of null distributions. But it is a device for choosing a design. It is not a hypothesis test, and nothing in it is a p-value.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not treat the Z-scores against the random-allocation null as significance tests, or a high Selection Score as evidence that inference improves. Standardization against random allocations measures separation from unoptimized designs on design metrics only; whether bias, coverage, and test size actually improve is settled separately, by the operating-characteristics simulations in the next section.

TRY IT

State in one sentence what a Z-score of zero on a given metric means for a candidate under this standardization. Then state why even a very large Selection Score could not, in principle, function as a p-value about any treatment effect.

E21 The optimality index is a reporting scale, not a second criterion

Methods, the subsection 'Selection Score and Global Optimality Index Computation', the closing paragraph that defines the Global Optimality Index as a min-max rescaling of the Selection Score, immediately before the estimand section.

WHAT THE PASSAGE IS DOING

Defines the Global Optimality Index as the Selection Score rescaled onto a bounded 0 to 100 range, and states that the selected design is the highest-scoring candidate on this geography.

IN ORDINARY LANGUAGE

The Selection Score is a weighted sum of the ten standardized metrics. Standardized means each metric is first put on a common scale. The Selection Score is the criterion that decides. The Global Optimality Index does not add a second test. It takes the Selection Score and stretches it onto a 0 to 100 range. The lowest-scoring configuration the search evaluated sits at 0, and the highest sits at 100. That transformation only stretches and shifts. So it ranks every candidate in exactly the same order as the Selection Score. On this geography the selected design is a spectral clustering partition with eight clusters. Spectral clustering groups sites using the structure of a similarity graph, not straight-line distance alone. That partition scores 100 out of 100. By the construction above, that means it is the top-scoring candidate in the pool. It does not mean it reached some absolute standard. Nor is it necessarily the allocation a trial would run. As in covariate-constrained randomization, the trial's allocation is drawn at random, with equal chance, from the elite pool. That pool holds the 40 highest-scoring configurations. This partition is the one the paper reports because it ranks first in that pool.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
>
> Do not read 100 out of 100 as an absolute standard met, or as statistical significance. The scale is defined by the best and worst candidates this particular search evaluated, so the top candidate scores 100 by construction, whatever its absolute quality. Do not read the index as a safeguard against a lopsided design either. The underlying weighted sum is compensatory: a large gain on one metric offsets a deficit on another, and no extra penalty applies to a design that is strong on most metrics and weak on one. The passage itself declines to claim uniform robustness for the selected design. Elsewhere the paper does run a contamination-only baseline, but against CRR, the two-criterion contamination-plus-balance design of the operating-characteristics tables, which the paper states is deliberately not the ten-metric composite that selects the featured design here. On this geography that baseline is directionally rather than conclusively better at the boundary-censored estimator, comparable in precision at the spillover-adjusted one, and worse at the naive one on bias and, at the two-MCSE threshold, on coverage, and the paper calls that two-criterion objective a robustness compromise.

TRY IT

Check the claim that the index adds nothing the Selection Score does not. If the index is an increasing rescaling of that score, the two must order every candidate identically, so ask what would have to be true for the two rankings to disagree. Then ask what the score of 100 would become if a single better candidate were added to the search.

E58 The deduplicated candidate set: identical allocations counted once

Methods, the subsection ‘Selection Score and Global Optimality Index Computation’, the closing paragraph that defines the structurally deduplicated candidate set and rescales the Selection Score min-max onto the bounded 0-100 optimality index, then defines the elite pool from which a trial’s allocation is drawn at random.

WHAT THE PASSAGE IS DOING

Defines the reference set for the bounded index. Candidate configurations that produce the same site-level arm-assignment vector, whether they came from different clustering algorithms or from different cluster counts K, are collapsed to a single highest-scoring representative, and the min-max rescaling of the Selection Score is taken over that reduced set.

IN ORDINARY LANGUAGE

A candidate is one way of grouping villages into clusters. Whole clusters are then sent to the treatment or control arm. Two different grouping recipes, or two different cluster counts, can still send every village to the same arm. The paper treats those as one design. It keeps only the single highest-scoring copy. That trimmed list is the set the 0 to 100 index is stretched across. So the scale is set by distinct allocations. It is not set by near-copies of one allocation crowding its ends. The 40 highest-ranked designs on that list form the elite pool. The featured design is the top one. A trial would not simply adopt it, though. As in covariate-constrained randomization, the allocation a trial uses is drawn at random from the 40, with equal chances.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read this collapse as evidence that the surviving candidates are well spread or independent. Only exactly identical arm-assignment vectors are merged; two candidates differing by one village are both kept, so the set can still be full of near-copies. Do not assume the collapse is score-neutral either: the same arm assignment can carry different metric scores depending on how the clusters were drawn, and the passage keeps the best of them, so a surviving row reports a maximum over its representations, not a unique value.

TRY IT

Open supporting/PROPOSED_design_selection_MAP.csv, the deposited deduplicated candidate list, which holds 331 rows plus a header. Check that the smallest optimality_index in the file is 0 and the largest is 100 to within the small guard constant, which is what a min-max rescaling over exactly this set has to produce, then count how many rows come from each method and k_value to see what survived the collapse.

E22 The decisive question: does the design improve inference?

The opening of the section 'Estimand and Operating Characteristics', the paragraph posing the decisive question of whether the optimized design improves inference and announcing the simulation on real malaria survey geography.

WHAT THE PASSAGE IS DOING

Pivots the paper from design selection to inference. It asks whether the optimized design actually reduces bias and improves confidence-interval coverage relative to simple randomization and to covariate-constrained randomization, and what it leaves unfixed, and announces a simulation study on real public malaria survey geography to answer it.

IN ORDINARY LANGUAGE

So far the paper has picked a trial layout by scoring candidate maps. This section asks the question a trial statistician cares about. Does the optimized layout give better statistical answers, not just better-looking maps? 'Inference' here means the effect estimate, its bias, its confidence interval, and its tests. The paper answers by simulation. A simulation builds many fake trials on the real village map and checks how each layout performs. The map and the baseline disease rates are real. Travel between villages is not observed: it comes from an assumed gravity formula built from the real survey counts and distances. The trials and their effects are invented on a computer.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that the operating characteristics that follow are field measurements from a trial that was actually run. The geography and baseline prevalence are real and dated, but every trial, effect, and error rate is simulated, so those numbers are model outputs, never observed trial results.

TRY IT

Before reading further, write down which of the four properties named here (bias, coverage, test size, power) you expect design optimization to improve most, and which least. Check your guess against the tables that follow this section.

E23 Exposure mapping and the boundary-insulated effect

The subsection 'The target estimand under spatial interference', the opening paragraph defining the cross-arm exposure from the gravity kernel and the displayed estimand.

WHAT THE PASSAGE IS DOING

Defines the formal target of the trial. A cross-arm exposure built from a human-mobility gravity kernel, together with the gravity-weighted mean baseline log-odds of the opposite-arm neighbors, summarizes interference; the second drops out at the target, and the estimand, which the paper names the boundary-insulated effect, is the effect of assignment on the log-odds of infection at zero exposure; since zero exposure requires every site to share one arm, it is the contrast between the all-treated and all-control allocations, a total effect rather than a direct effect at fixed neighbor assignment.

IN ORDINARY LANGUAGE

An estimand is the exact quantity a trial is trying to measure. Here every village gets a treatment or control label. Each of those groups is called an arm. “Interference” means a village’s outcome can depend on other villages’ labels. That is because people travel between them. The paper does not assume “partial interference”, the idea that a village is affected only by others inside its own group. It adopts an “exposure mapping” for spatial interference instead. Under it, the outside influence collapses into two numbers per village. One is the gravity-weighted average baseline log-odds of the opposite-arm neighbors. It drops out at the zero-exposure target. The other is the exposure. It is a weighted count of every village assigned to the opposite arm, however far away. The weights come from a “gravity kernel”. Larger and closer village pairs exchange more people, and the weight falls off with squared distance. The target is called the boundary-insulated effect. A village has zero cross-arm exposure only when every other village shares its arm. So the target compares a village’s outcome when every village is treated with its outcome when no village is. That is what the interference literature calls a total effect. It is not a direct effect with the neighbors’ labels held fixed. The name refers to shielding from cross-arm exposure. In the simulation it equals the effect with no interference at all. In a real vector-control trial it would also carry any protection a treated village gets from treated neighbors. It is stated on the log-odds scale. That is a standard scale for changes in infection risk.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that the gravity kernel is known to be the true mobility law for these villages. It is a modeling choice, and the anchoring paragraph later in this section treats it as one. Error from misspecifying the kernel’s form is probed directly by the sensitivity analysis of Appendix C (Table 13). It generates the contamination under three kernels the design and analysis do not assume. Two weaken with distance to the first or third power instead of the second, and one decays exponentially. Limitation 1 reports what that probe found. The design’s bias reduction survives all three departures, while the spillover-adjusted estimator’s nominal coverage does not, falling to between 0.51 and 0.89.

TRY IT

Place three equal-population villages on a line at equal spacing. Using the inverse-square gravity rule, compute the middle village’s exposure when both neighbors are in the opposite arm, and again when only one of them is. Then repeat for an end village, whose far neighbor sits twice as far away.

E24 Approximate recovery: bias reduced, not eliminated

The subsection ‘The target estimand under spatial interference’, the closing sentences just after the estimand equation, on approximate recovery in the core zone, the attenuated naive contrast, and the three tools that attack the contamination term.

WHAT THE PASSAGE IS DOING

States how close the estimators get to the target. The target effect is approximately recovered among core-zone villages, the naive contrast is attenuated toward the null by spillover, and three tools each attack the contamination term.

IN ORDINARY LANGUAGE

‘Core-zone’ villages sit away from the boundary between the trial’s arms, treated and control. So their exposure to the other arm is small. Among them the target effect is approximately recovered. That word ‘approximately’ does real work. The simulations show bias is reduced, not removed. The ‘naive ITT’, or intention to treat, is the plain treated-versus-control comparison. It ignores spillover entirely. ‘Spillover’ here means infection risk mixing across the boundary between the arms. It pulls each boundary village toward the other arm’s outcome. So the naive comparison shrinks toward no effect. Three tools work against that contamination term. One is a layout chosen to minimize contamination. One is an estimator that drops boundary villages. One is an estimator that adjusts for each village’s measured exposure.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that restricting analysis to core-zone villages identifies the target effect exactly. The paper says approximately recovered, and its own sensitivity table shows residual bias at every positive contamination strength; the estimand is approached, never attained.

TRY IT

Follow the reference to the sensitivity table and check the core-zone rows at the strongest contamination setting. Confirm the pattern this passage predicts: bias smaller than the naive estimator’s, but still not zero.

E25 Answering the caution against partial interference

Estimand and Operating Characteristics, the target-estimand subsection, the paragraph headed ‘On the interference structure.’

WHAT THE PASSAGE IS DOING

Responds to a published caution that partial interference may be untenable for irregularly distributed units exchanging across boundaries. The paper argues its exposure summary and core-zone restriction target exactly that regime, and that its simulations quantify the consequences of contamination under the assumed interference structure, with a sensitivity appendix probing departures from it, while stating that no simulation can establish that structure’s adequacy for a real trial.

IN ORDINARY LANGUAGE

Spillover, or interference, means a village's outcome depends on its neighbors' treatment. Behind it is infection risk that mixes across the boundary. The paper calls that mixing contamination. A recent critique warns that "partial interference" may not hold. That is the assumption that spillover stays inside fixed, predefined groups and never crosses between them. It can fail when sites are scattered unevenly and mix across the boundary. That is exactly this survey-site setting, and the paper says so. Its reply has two parts. First, it does not assume the problem away. The results lean on core-zone villages, far from the boundary. There, spillover from the other group is smallest. Second, the simulation shows what contamination costs under the assumed interference structure. An appendix probes departures from it. But no simulation can show that the structure fits a real trial. At zero contamination the estimator has no bias and near-nominal coverage. Nominal coverage means the confidence interval covers the truth as often as it claims. As contamination grows, bias and coverage get worse in a smooth way. So the cost of a failed assumption is visible. The zero-contamination check is a weak test, though. With no variation between sites beyond binomial sampling, the analysis model at zero contamination is the model that generated the data. When such variation is added, the same binomial analysis has coverage of only 0.200 to 0.622, even with no contamination. In that setting, single sites rather than whole clusters are assigned to arms. The intracluster correlation is 0.02 to 0.10.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that the critique has been refuted. The paper accepts it and works inside it. The smooth degradation it points to quantifies the assumption's failure cost within the paper's own simulation model, generated from its own gravity kernel, not in nature.

TRY IT

Ask what result at zero contamination would have falsified this setup, for instance bias or badly off-nominal coverage with no interference present. Then verify the paper reports exactly that calibration check and passes it, and find the appendix paragraph showing that the binomial analysis passes it only when sites carry no variation beyond binomial sampling.

E26 The aliasing result: kernel shape is identified, level is not

Estimand and Operating Characteristics, the target-estimand subsection, the opening sentences of the paragraph headed 'On anchoring.'

WHAT THE PASSAGE IS DOING

Imports an identification result from Watson et al.: under cluster randomization with a fixed exposure set size, the data determine the spillover kernel's shape but not its level, so every direct-effect estimator rests on an anchoring assumption, and any anchored estimator's bias decomposes into level, shape, and leakage terms.

IN ORDINARY LANGUAGE

'Identification' asks what the data alone could pin down with unlimited samples. The cited result says cluster-randomized data pin down the shape of the spillover curve, meaning its relative pattern across units. They cannot pin down its level, the overall height of that curve. The level is 'aliased' with the intercept and the direct effect: the data cannot tell them apart. This holds at any sample size and under any outcome model. So every direct-effect estimator must fix the level by assumption, and that fixing step is the 'anchor'. The result also splits any anchored estimator's bias into three parts. They are a level error, a shape error from a wrong kernel, and a leakage term set by the realized geometry.

EVIDENCE STATUS **ANALYTIC** follows by mathematics

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that a larger trial or more simulation replicates could remove the aliasing. The result holds at any sample size and under any outcome model; no volume of data separates the kernel's level from the direct effect without an anchoring assumption.

TRY IT

State the implicit anchor of the simplest estimator you know, the plain difference in arm means. Ask yourself what exposure level it silently assigns to every unit, and compare that to the core-zone estimator's anchor described next.

E27 The core-zone estimator is an anchored estimator

Estimand and Operating Characteristics, the target-estimand subsection, the middle of the paragraph headed 'On anchoring,' where the boundary-censored estimator is read as an anchored estimator.

WHAT THE PASSAGE IS DOING

Reads the paper's own machinery through the anchoring framework. The boundary-censored estimator anchors by treating the 70 percent of sites with smallest cross-arm exposure as approximately unexposed, and the contamination-resilient design acts before data collection to reduce the level term that anchor must absorb, driving the between-arm spillover mean, and with it the level term, toward zero.

IN ORDINARY LANGUAGE

The paper now turns that framework on itself. Cross-arm exposure is spillover between sites in opposite arms. Its boundary-censored estimator keeps the 70 percent of sites with the least of it. Treating those sites as roughly unexposed is its anchoring assumption. Level error means error in the overall size of the spillover. Data alone cannot pin it down. The residual bias in the contamination sweep is that error's footprint. Censoring cannot remove it. The design contributes before any data exist. It picks a layout that lowers realized cross-arm exposure. That makes the anchor closer to true. Less real exposure means the roughly unexposed assumption bends less. In the framework's terms, the design pushes the between-arm spillover mean, and so the level error, toward zero.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

DO NOT ACCIDENTALLY CONCLUDE

Do not conclude that the design plus censoring together deliver anchor-free identification. The estimator remains anchored; the design only shrinks the anchor's error, and the sweep still records residual bias at every positive contamination strength.

TRY IT

Write the two roles in one line each: censoring fixes the level after randomization, the design shrinks realized exposure before it. Then ask which of the two contributions the sensitivity table can actually separate, and which it measures only jointly.

E28 The shape term, now probed, and the claim's exact boundary

Estimand and Operating Characteristics, the target-estimand subsection, the closing sentences of the paragraph headed 'On anchoring.'

WHAT THE PASSAGE IS DOING

Names the second of the anchoring decomposition's three bias terms, the shape error from a wrong kernel form, points to the appendix that now probes it directly, and fixes the claim's scope: design-side exposure reduction, an ex-ante reduction of the quantity the anchor must absorb, and a simulation-measured residual bias across contamination strengths and kernels, not identification free of anchoring.

IN ORDINARY LANGUAGE

A kernel is the rule for how strongly travel links two places. The shape error is the mistake you make if that rule has the wrong form, not just the wrong height. An appendix now tests it. It generates contamination with three other kernels that the design and the analysis do not assume. The design's bias reduction holds under each of those kernels, for every estimator. Coverage is how often the interval catches the true effect. The spillover-adjusted estimator's nominal coverage does not survive. It falls to between 0.51 and 0.89 when the kernel is wrong. The design does not protect that coverage either. Take a kernel with exponential decay. There the spillover-adjusted estimator covers less often under the contamination-resilient design than under simple randomization, 0.510 against 0.729. The passage then sets the limits of the claim. The design lowers the cross-arm exposure the chosen assignment really creates. That shrinks, before any data are collected, the quantity the censoring anchor must absorb. Simulations measure the bias that is left, across contamination strengths and kernels. One thing is not claimed: that we can pin down the target effect with no anchoring assumption.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

DO NOT ACCIDENTALLY CONCLUDE

Do not conclude that the residual bias in the contamination sweep already contains the shape error; that sweep uses the assumed kernel, and the shape error is measured separately in Appendix C. Do not conclude the spillover-adjusted estimator is safe to use when the kernel is unknown; the appendix shows exactly how it fails. And do not conclude the design supplies the level of the spillover curve that the data cannot identify: exposure minimization sits with buffers and spacing, which add no level information and instead validate the anchor, while the

features that do supply it, such as sentinel units and randomized saturation, are not evaluated in this paper.

TRY IT

Open Appendix C, Table 13. For the contamination-resilient design, write down the spillover-adjusted coverage under each of the four generating kernels and say which one you would trust a field trial to resemble.

E29 What the operating-characteristics simulation is

Estimand and Operating Characteristics, the operating-characteristics simulation subsection, its opening specification paragraph.

WHAT THE PASSAGE IS DOING

Specifies the simulation testbed: real Malaria Atlas Project survey sites, baseline rates and denominators, with a simulated treatment effect and contamination process, and names the three designs and three estimators being compared.

IN ORDINARY LANGUAGE

An operating-characteristics study asks how a design and its analysis behave across many repeated trials. Those trials are run on a computer, not in the field. Here the map is real. The Uganda survey sites keep their true places, baseline malaria rates, and sample sizes. The trial is invented. The computer assigns each site to an arm, meaning the treated group or the control group. It then applies a known effect of log 0.5 on the log-odds scale, the scale a logistic model uses. Then it adds contamination. Contamination pulls each site's infection risk toward that of its neighbors in the other arm. That mixing happens as people move across the arm boundary. Three designs compete: simple randomization, covariate-constrained randomization, and the contamination-resilient design. The middle one forces the arms to match on baseline traits. Three analyses compete as well. Naive intention-to-treat scores every site as assigned. Core-zone drops the 30 percent of sites most exposed at the boundary, under the allocation each simulated trial draws. So the dropped sites are chosen after randomization. They change whenever the allocation changes. Spillover-adjusted adds the exposure, its interaction with arm, and a baseline gap to the model. The arm coefficient is then the effect at zero exposure. Spillover-adjusted needs the mobility kernel, the map of how people move, known up to its overall scale, and the exact form of the contamination formula. Core-zone needs less: it uses the kernel only to rank sites by exposure. Only naive needs no kernel at all. The study uses 1000 simulated trials. They come from one random pool of six thousand candidate allocations. An appendix reports how much the results move when that pool is redrawn.

EVIDENCE STATUS **DEFINITION** defines a term

DO NOT ACCIDENTALLY CONCLUDE

Do not read any downstream number as a measured malaria-trial outcome. The geography and baselines are real, but the assignment, the effect, and the contamination are all simulated, so every operating characteristic that follows is a model output, not an observation. The simulated counts also carry no variation between sites or clusters beyond binomial sampling, and apart from Appendix D, which adds such variation, every coverage, Type-I error and power figure in the paper holds under that assumption (Limitation 9).

TRY IT

Find the true effect, delta = log 0.5, about -0.693, restated in the Table 1 caption. Every bias entry in the tables measures how far an estimator sits from that known value.

E44 How the three designs are built, and the caveat on what is randomized

Estimand and Operating Characteristics, the operating-characteristics simulation subsection, the paired paragraphs “Design implementation, and what is randomized here” and “A caveat on the unit of randomization.”

WHAT THE PASSAGE IS DOING

Specifies exactly how simple, covariate-constrained and contamination-resilient allocations are drawn from one shared candidate space, shows that the constrained design uses exactly the resilient design’s balance criterion, under a looser constraint, and discloses that these headline allocations randomize individual sites rather than whole clusters, which is the arrangement under which contamination is worst.

IN ORDINARY LANGUAGE

An allocation is one way of splitting the survey sites between the trial arms. All three designs draw from the same pool of six thousand candidate allocations. Each one assigns individual survey sites, not whole clusters, to arms within the two latitude strata. Strata are the north and south map bands used to divide the sites. Simple randomization draws from all six thousand. Covariate-constrained randomization draws from the best quarter on a balance score. That score adds up the difference in arm means over two standardized baseline covariates. A covariate is a trait measured before the trial starts. The resilient design draws from the 40 with the best combined rank on that same balance score and on cross-arm flow. So the constrained design is not a straw man. It uses exactly the resilient design’s balance score, under a looser cut. It takes the best quarter, against the resilient design’s best 40 by combined rank. One allocation is drawn per simulated trial, so results average over design variability. Here is the caveat. These allocations randomize sites, not clusters. Neighboring sites often land in opposite arms. Under simple randomization of sites, the average realized cross-arm exposure is 0.57. Allocating whole clusters, as the design engine does, cuts that to about 0.22. The headline tables are therefore a pessimistic stress test. The simulations behind both allocation units have no variation between sites or clusters beyond sampling chance. So allocating a few large clusters carries no variance penalty here. Real differences between sites or clusters would make the reported intervals too narrow.

EVIDENCE STATUS **DEFINITION** defines a term

> #### DO NOT ACCIDENTALLY CONCLUDE
> Do not compare the site-level coverage of Tables 1 to 3 with the cluster-level coverage in the Discussion; they are different randomization units, and the cluster-level pool is the counterpart of the site-level two-criterion objective rather than the same construction, so the two sets of figures are read for their direction and not as matched cells; the naive estimator is better covered at the cluster level at every positive contamination strength and the boundary-censored estimator at c = 0.35, but the spillover-adjusted estimator is at or near nominal coverage at both units. Do not conclude the constrained design failed because its balance criterion was weak; it reaches imbalance 0.159 against 0.090 for the resilient design and 0.422 for simple

> randomization.

TRY IT

Explain in two sentences why two adjacent sites in opposite arms raise realized exposure more than two adjacent clusters in opposite arms, and why that makes the site-level tables a pessimistic stress test. Then name the two things the paper says that pessimism does not extend to.

E30 Monte-Carlo precision and the three-MCSE rule

Estimand and Operating Characteristics, the operating-characteristics simulation subsection, the paragraph headed “Monte-Carlo precision.”

WHAT THE PASSAGE IS DOING

Quantifies simulation noise and shows the headline design gains are many multiples of it. Its rule: three or more MCSEs of the difference is established, two to three is at the two-MCSE threshold, and less is directional. It then lists the comparisons reported as not established: one that clears the rule but does not reproduce, and others that fall below it.

IN ORDINARY LANGUAGE

Every simulated number carries noise, because only a finite number of trials, called replicates, were run. The Monte-Carlo standard error, or MCSE, measures that noise. Coverage is the share of trials whose 95 percent confidence interval contains the true effect. The target is 0.95. Bias is the average gap between the estimate and the truth. With one thousand replicates the coverage MCSE is at most 0.016. The bias MCSE is at most 0.009 for the naive and core-zone estimators and for every malaria row. It is far larger for the spillover-adjusted estimator on the cholera geography. There it is up to 0.033, and up to 0.054 in the contamination-only cholera run. So the paper reads a difference between two designs as established only when it is at least three standard errors of that difference. That standard error combines both designs’ noise. A difference of two to three is at the two-MCSE threshold. A smaller one is directional. The paper applies that rule to itself. The boundary-censored estimator drops the sites most exposed to the other arm. Its coverage gain, 0.259 to 0.481, is 10.6 MCSEs of the difference, within the realized candidate pool. That pool is the one batch of arm assignments actually drawn. Between independent pools, the resilient design’s coverage varies far more than that simulation noise. It runs from 0.283 to 0.479 over eight pools (Appendix C). So the abstract quotes a range, 0.28 to 0.48, instead. The naive estimator ignores spillover between arms. Its coverage gain on malaria, 0.168 to 0.225, is 3.2 MCSEs, above the three-MCSE bar. Yet the paper does not call it established. It does not come back when the same pool is simulated again, 0.190 to 0.186. And it does not survive redrawing the pool. Its coverage gain on COVID, 0.627 to 0.663, is 2.7 MCSEs, below that bar. So is the boundary-censored estimator’s gain on COVID, 0.816 to 0.845, also 2.7 MCSEs. All three are reported as not established. The two COVID gains sit at the two-MCSE threshold. The spillover-adjusted estimator is judged by RMSE, or root mean squared error, rather than coverage. The reason is that once correctly specified it is near nominal, meaning close to 0.95, under every design. RMSE itself is reported without an MCSE, and RMSE differences of a few thousandths are not interpreted.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

DO NOT ACCIDENTALLY CONCLUDE

Do not read a difference of three or more MCSEs as a significance test. It is a precision floor, not a hypothesis test, and it says nothing about whether the difference matters. And do not read clearing it as enough. The naive estimator's malaria coverage gain, at 3.2 MCSEs, is reported as not established because it does not reproduce when the same pool is simulated again. The naive and boundary-censored coverage differences on COVID, at 2.7 MCSEs each, sit at the two-MCSE threshold and fall below the rule. So do the naive coverage reversals at strong contamination, the contamination-only design's naive coverage penalty on malaria and its bias penalties on cholera. And do not treat the within-pool MCSE as the only uncertainty: the between-pool spread of the resilient design, in Appendix C, is larger.

TRY IT

Compute 0.225 minus 0.168 and divide by 0.0177, the standard error of that difference at 1,000 replicates. You should get about 3.2. That clears the paper's three-MCSE rule. Then find the reason the paper still reports this gain as not established, and say whether you agree.

E31 Balance control is not contamination control

Estimand and Operating Characteristics, the operating-characteristics simulation subsection, the design-contrast paragraph on flux and imbalance together with the moderate-contamination operating-characteristics table it introduces.

WHAT THE PASSAGE IS DOING

Reports the central design contrast: covariate-constrained randomization reaches good covariate balance, comparable to but short of the contamination-resilient design's (0.159 against 0.090) yet leaves cross-arm contamination at the simple-randomization level, and Table 1 gives the resulting operating characteristics at moderate contamination.

IN ORDINARY LANGUAGE

Covariate balance means the two arms look alike on baseline traits. Contamination control means treated and control clusters cannot easily reach each other. The simulation shows that these are different targets. Constrained randomization cuts imbalance from 0.422 to 0.159. But its mobility leakage, 4.10×10^6, sits slightly above simple randomization's 3.89×10^6. Leakage here is a gravity-model score of cross-arm movement, not a field measurement. The contamination-resilient design meets both goals. Its imbalance is 0.090 and its leakage is 5.32×10^5. The payoff shows up in the analysis, at moderate contamination. Core-zone bias falls from 0.208 under simple randomization to 0.113. Coverage, the share of intervals that hold the true effect, rises from 0.259 to 0.481. The spillover-adjusted rows behave differently. That estimator is unbiased under every design, with coverage 0.94 to 0.96. So what the resilient design buys it is precision. Its RMSE, the typical size of its error, drops from 0.149 to 0.095. Table 1 is one draw of the candidate pool. Its caption points to the appendix table that reports the spread over eight.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

DO NOT ACCIDENTALLY CONCLUDE

Do not conclude covariate-constrained randomization is a poor design; it delivers exactly the balance it promises. The narrow point is that balancing covariates does nothing about spatial

> leakage, so a balance-only criterion cannot substitute for a contamination-aware one. And do not quote the core-zone CRR coverage of 0.481 as the design's typical value: it is the most favorable of the eight pools in Appendix C. And do not carry any coverage in Table 1, the adjusted estimator's near-nominal 0.94 to 0.96 included, beyond the simulation's own model, where a site's risk is fixed by its baseline, its arm and the contamination, with no extra random variation between sites or clusters; the paper lists this as its ninth limitation.

TRY IT

Divide 3.89×10^6 by 5.32×10^5. The ratio is about seven, which quantifies how much leakage the resilient design removes while holding balance at least as good.

E32 Calibration at c=0, Type-I inflation, and why randomization inference

Estimand and Operating Characteristics, the operating-characteristics simulation subsection, the contamination-strength sensitivity and calibration table through conclusion (i) of the four-conclusions paragraph.

WHAT THE PASSAGE IS DOING

Establishes that the machinery is calibrated at zero contamination, shows the model-based test's size inflating badly once contamination is on, and grounds the recommendation that the primary treatment-effect test use randomization inference, while warning that a naive permutation test is not automatically valid under the simulation's null.

IN ORDINARY LANGUAGE

A simulation needs a null check. Coverage means how often a confidence interval holds the true value. At contamination c = 0, the naive estimator is unbiased under both designs this table runs. They are simple randomization and the resilient design. Coverage for the naive estimator is 0.951 under simple randomization. It is 0.945 under the resilient design, against the 0.95 target. The cluster-level comparison of objectives reports the same calibration for every estimator under all six objectives. So, within the simulation, later failures come from contamination, not from broken machinery. That calibration holds under the simulation's own model. There, a site's risk is fixed by its baseline, its arm and the contamination. There is no extra random variation between sites or clusters. As contamination grows, bias climbs and coverage collapses for both designs. Even so, the resilient design keeps lower bias at every positive level. On coverage it leads at c = 0.2. It also leads at c = 0.35, but only in this run, since that difference is not reproduced when the same pool is re-simulated. At c = 0.5 and 0.7 it falls behind, at the two-MCSE threshold, 0.092 against 0.126 and 0.066 against 0.097. That is a narrower interval around a still-biased estimate. Type-I error is the rate of rejecting a true null. The nominal rate here is 0.05. Under moderate contamination, the naive model-based test rejects far too often. It rejects 0.702 of the time under simple randomization. Under the resilient design it still rejects 0.483 of the time. So the reported power of 1.000 does not mean guaranteed detection. The tests are not operating at a common size. Randomization inference builds the null by re-randomizing the arm assignments instead of using the model-based test. The paper recommends it for the primary test but does not run it. It also warns that the remedy is not automatic. The paper calls the contamination rule in these simulations the blend. The blend still mixes baseline risk across the arms even when the treatment effect is zero. So outcomes still depend on the assignment. A naive permutation test is then not

automatically valid under that null. The paper points instead to conditional randomization tests built for interference. It says it is not established that they can test this null.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not compare power across designs at face value: with realized sizes of 0.702 versus 0.483, the power columns are not on a common footing. And do not treat the naive model-based test as usable just because it is calibrated at c = 0; the paper still recommends randomization inference for the primary test. And do not assume randomization inference repairs the size problem: the paper does not evaluate it, and its Limitations say a sharp-null randomization test is not automatically valid under the blended null. A two-sided randomization test confined to the featured eight-cluster partition cannot reject at 0.05, because that partition admits only 36 allocations. So the paper names the two-stage cluster-level analysis as the practical primary test at that resolution.

TRY IT

Read the naive SR row of Table 1: a Type-I error of 0.702 means roughly seven of ten null trials falsely reject. Then read Table 2's c = 0 row, where coverage is 0.951, near nominal.

E33 The four conclusions and the complementarity claim

Estimand and Operating Characteristics, the operating-characteristics simulation subsection, the paragraph beginning "Four conclusions follow" that runs through the complementarity of design and analysis to the central-result claim and the three boundaries placed on it.

WHAT THE PASSAGE IS DOING

Distills the tables into four take-home conclusions and frames the design-to-analysis payoff, that optimizing the design measurably improves the planned analysis, as the central result of the operating-characteristics study, with three boundaries placed on that claim.

IN ORDINARY LANGUAGE

The paragraph draws four lessons. First, at zero contamination the tools are calibrated under the simulation's own model. There, a site's risk is fixed by its baseline, its arm and the contamination. There is no extra random variation between sites or clusters. Contamination means infection risk mixing across the boundary between the arms as people travel. Calibration holds in the site-level tables and in the cluster-level test of objectives alike. Second, balance on background traits is not the same as control of contamination. Third, the three estimators split cleanly. The naive contrast just compares the arms, and it stays badly biased. The core-zone estimator uses only sites far from the other arm. It cuts that bias by about a third, from 0.289 to 0.208 under simple randomization and from 0.164 to 0.113 under the resilient design. In the realized pool, the one draw of candidate designs the tables use, the design nearly halves it there, from 0.208 to 0.113. On average across eight independent pools the cut is from 0.210 to 0.142. And of the three estimators, this is the one whose coverage gains the most. Even so, it never comes close to nominal. Coverage is how often the interval holds the true effect. Nominal is the rate the method promises. The spillover-adjusted estimator is unbiased under every design. Its coverage there is near nominal as well. So the design buys it precision, not validity. It cuts root-mean-square error by 36 percent. That

validity depends on the kernel, the assumed pattern of how far people travel. The analysis may assume the wrong kernel. When that happens, its coverage falls to between 0.51 and 0.89. The cut in bias from the design still holds under every kernel tested. Fourth, no one layer removes the contamination bias on its own. So design and analysis work together. Neither can replace the other. The paper regards one claim as the central result of its operating-characteristics study. That claim is that optimizing the design improves how the planned analysis performs. Three limits come with that result. The naive estimator's coverage gain is not part of it. That gain is not reproduced when the same pool is re-simulated. It does not survive redrawing the pool. And it reverses at strong contamination and under the one-over-distance kernel. When whole clusters are allocated, the bias and coverage gains belong to the leakage-only pool and the full ten-metric composite. Against comparators drawn at the composite's own resolution, its boundary-censored coverage gain disappears. Its naive coverage gain is established only at strong contamination, and its naive bias stays lower. And all of it is measured with no extra variation between sites. It is also measured under an analysis that weights each site by how many people were examined there. The appendix adds that variation, at an intracluster correlation (ICC) of 0.02 to 0.10. The ICC is roughly the share of outcome variation that comes from real differences between sites. The appendix also analyzes at the unit of randomization. The core-zone coverage advantage then persists. The bias reduction is smaller, because the site-level analysis weights sites equally. A model with a random effect for each site measures the adjusted estimator's gain in root-mean-square error. That gain is about 40 percent in the same appendix run with no variation. With the variation, it falls to between 16 and 28 percent.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude the contamination problem is solved. The best core-zone coverage in the realized pool is 0.481 against a 0.95 target, and 0.28 to 0.48 across pools. Do not conclude the spillover-adjusted estimator makes the design unnecessary: it needs the mobility kernel, and where that kernel is wrong it fails while the design still helps. And skip over the power lift from 0.90 to 1.00 for the naive estimator; its size inflation makes that uninterpretable.

TRY IT

List the three estimators and, for each, one sentence on what it assumes and what the resilient design buys it. Then say which of the three you would use if you did not trust the mobility kernel.

E34 The contamination-only ablation: directionally ahead at one estimator, comparable at one, worse at the third

Estimand and Operating Characteristics, the paragraph decomposing the design objective into a contamination-only baseline, through the sentence reporting the naive-estimator penalty on the malaria geography.

WHAT THE PASSAGE IS DOING

Reports an ablation that removes the balance term from the design objective and keeps contamination alone (CC), ranked on the identical candidate pool, and shows that the result is a trade rather than a clean win: CC is directionally better at the boundary-censored estimator, comparable in precision at the spillover-adjusted one, worse at the naive one on bias and, at the two-MCSE threshold, on coverage, and costs a large increase in baseline imbalance.

IN ORDINARY LANGUAGE

An ablation drops one ingredient to learn what it added. The composite score has two ingredients, contamination and balance. Contamination is infection risk mixing between the arms. Balance means the two arms start out alike on baseline traits. The CC design keeps contamination only. It ranks the identical candidate pool by leakage alone. On the malaria geography, CC cuts leakage about fourteenfold. That is 3.67×10^4 against 5.32×10^5 gravity units, the mobility model's scale. What it buys with that is uneven. Coverage is how often the interval traps the true effect. Nominal coverage means it does so as often as promised. The boundary-censored estimator drops the 30 percent of sites most exposed to the other arm. There CC's coverage is 0.523 against the composite's 0.481, a difference of 1.9 Monte-Carlo standard errors. The paper calls a gap established only at three or more Monte-Carlo standard errors, so it calls this one directional. The spillover-adjusted estimator models the leak instead. There both designs already sit at nominal coverage, 0.945 against 0.944. The two are comparable in precision. CC's RMSE, a typical error size, is 0.091 against the composite's 0.095, a gap too small for the paper to interpret. The naive estimator just ignores the leak. There CC is worse. Its coverage is 0.179 against 0.225, a gap of 2.6 Monte-Carlo standard errors. That is short of three, so the paper places it at the two-MCSE threshold, not established. Its bias, the average error, is 0.198 against 0.164. And the price is balance. Simple randomization assigns arms by coin flip. CC's imbalance on baseline traits is 0.496, worse than simple randomization's 0.422. It is also more than five times the composite's 0.090.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that the balance term is useless, and do not conclude that CC is simply the better design. On this one geography it is ahead only directionally at the boundary-censored estimator (1.9 Monte-Carlo standard errors) and is only comparable in precision at the spillover-adjusted one; it loses at the naive one on bias and, at the two-MCSE threshold, on coverage; and it buys that edge with baseline imbalance worse than simple randomization, which is the kind of thing a trial reviewer objects to before reading any operating characteristic.

TRY IT

Compute 0.523 minus 0.481 and divide by 0.0224, the Monte-Carlo standard error of that coverage difference at 1,000 replicates. Then do the same for the naive row, 0.225 minus 0.179 over 0.0179. The first comes to 1.9 and the second to 2.6. Both fall below the three Monte-Carlo standard errors the paper requires before it calls a difference established. So it calls the first directional and places the second at the two-MCSE threshold. Now say in one sentence what a design team should do with an ablation that points both ways.

E35 COVID and cholera: the same balance price, different naive orderings

Estimand and Operating Characteristics, the later half of the contamination-only-baseline paragraph, where the COVID and cholera geographies are reported and the three implications are drawn.

WHAT THE PASSAGE IS DOING

Reports the contamination-only baseline on the two other geographies, shows that its balance cost recurs on all three while its ordering against the composite at the naive estimator does not, and draws three implications, chiefly that the composite objective is a robustness compromise across geographies and not a uniform optimum.

IN ORDINARY LANGUAGE

CC is the design picked on contamination alone, that is, on leakage between arms. On every map it pays the same price in balance. On COVID its baseline imbalance is 0.70 against 0.37 under simple randomization. On cholera it is 1.57 against 0.74. What CC gains differs by map. On COVID it beats the composite at the naive estimator, the reverse of malaria. Its bias there is 0.246 against 0.272, and its coverage is 0.732 against 0.663. But at the spillover-adjusted estimator it is markedly less precise than both the composite and simple randomization. Its RMSE is 1.712, against 1.136 and 1.487. Cholera has only twenty clusters, and their baseline risk differs sharply. There, dropping the balance term backfires. Its point estimates of bias are larger at the naive and boundary-censored estimators. Neither gap reaches three Monte-Carlo standard errors, so the paper places both at the two-MCSE threshold, not established. At the spillover-adjusted one it also looks larger, but that gap is inside one Monte-Carlo standard error, so it is not a measured difference. Its coverage does look higher, but that comes from wider intervals around a worse-centered estimate. So no single ordering of CC and the composite holds at the naive estimator. What holds on all three maps is the balance price.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not extract a validated switching rule from three geographies. The paper reports cholera's twenty clusters and steep baseline gradient as the setting of that result, not as a tested criterion for when contamination-only selection is safe. Do not read the COVID naive result as CC being the better design there; at the spillover-adjusted estimator it is markedly less precise than both the composite and simple randomization. And do not score CC's higher cholera coverage as a win; the text ties it to uniformly larger root-mean-square error, meaning wider intervals around a worse estimate.

TRY IT

Line up the imbalance figures: 0.496 for CC against 0.422 for simple randomization on malaria, 0.70 against 0.37 on COVID, and 1.57 against 0.74 on cholera. Then line up the naive bias against the composite: 0.198 against 0.164 on malaria, 0.246 against 0.272 on COVID, and 0.268 against 0.256 on cholera. Say which of the two patterns the paper says holds on every map.

E73 Bold marks and the Type-I column in the ablation table

Estimand and Operating Characteristics, the contamination-only baseline subsection, the table that closes it together with its caption.

WHAT THE PASSAGE IS DOING

Presents the ablation as a table: CC and CRR side by side, three measures for each of the three estimators, on the malaria geography at a single contamination strength. Its distinct job is the caption's two reading rules, that bold flags the better point estimate cell by cell and that a gap smaller than the stated Monte-Carlo standard error is not a gap. It is also the only place where the two designs' Type-I error rates appear together for all three estimators, and where the caption records that CC was estimated in an additional run of the same engine on the identical candidate pool.

IN ORDINARY LANGUAGE

The table sets two designs side by side. CRR is the two-criterion design. It ranks candidate splits on both leakage and baseline balance. CC is the stripped-down version. It ranks on leakage alone. Leakage means infection risk mixing across the border between the arms of the trial. Each design gets three columns. Bias is the average gap between the estimate and the truth. Coverage is how often the interval holds the truth. Type-I error is how often the test flags an effect that is not there. Three estimators fill the rows. The naive one ignores leakage. The core-zone one drops sites near an arm border. The adjusted one models the leak and corrects for it. Bold marks the better number in each pair. The caption then warns you off some of those bold marks. Each cell carries noise from a finite run. Both designs ran for one thousand replicates. A replicate is one simulated trial. That noise is called the Monte-Carlo standard error. It is at most 0.016 on coverage and 0.009 on bias. So the bias pair in the adjusted row, 0.003 against 0.000, is not a real gap. Nor is its coverage pair, 0.944 against 0.945. The naive row is different. There CC is worse on bias, 0.198 against 0.164. It is worse on coverage too, 0.179 against 0.225, a gap of 2.6 Monte-Carlo standard errors of the difference. The paper places that gap at the two-MCSE threshold, since it reads only gaps of three or more as established. The one naive cell where CC wins is Type-I error, 0.414 against 0.483. Both of those sit far above the rate a test should hold. The whole table is read at one contamination strength, c equal to 0.35.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
>
> Do not tally the bold marks and declare CC the better design. Bold flags the better point estimate cell by cell, and several of those bolds sit inside the noise the caption itself states, including both the bias and the coverage pair in the adjusted row. The table also omits the measure on which CC does worst, its baseline covariate imbalance, so it is neither a scoreboard nor a test between the two designs; the composite ranking is a selection statistic, not a test. Do not read the adjusted row's near-nominal coverage as validity that travels, since that estimator leans on the mobility kernel being right, and every coverage figure outside the last appendix, that one included, assumes no variation between sites or clusters beyond binomial sampling chance. And do not read either column of this table as a restored naive or core-zone estimator; under both designs those two remain badly biased with intervals and tests that miss their

advertised rates.

TRY IT

In Table 3, subtract the CC coverage from the CRR coverage in the adjusted row (0.945 minus 0.944) and compare that gap with the caption's coverage Monte-Carlo standard error of 0.016. Do the same for the naive row (0.225 minus 0.179). Then say which of the table's bolded coverage cells you are entitled to call a difference. Keep in mind that the paper measures a gap against the Monte-Carlo error of the difference, not of one cell, and puts the naive coverage gap at 2.6 such errors, below its rule of three for an established difference.

E36 Why two more geographies

Main text, the opening of the section on generalization to other connected systems: the subsection heading and its introductory paragraph.

WHAT THE PASSAGE IS DOING

Opens the generalization section by announcing that the malaria-geography findings will be re-tested on two further real connected systems of different spatial scale, the cholera one under the same symmetric gravity form as the malaria sites and the COVID one under a directed flux, holding the estimand, simulation engine, and estimators fixed.

IN ORDINARY LANGUAGE

A result proved on one map might be a quirk of that map. So the paper reruns the simulation on two more real geographies. Only the part that tests the estimates and their intervals is rerun. The ten-metric scoring suite and the design-picking engine are not applied on these two maps. So what they can back up is the leakage-control claim, not the whole framework. One is the 2017-2018 cholera outbreak across Yemen's governorates. The other is early-2020 COVID-19 across Louisiana's parishes. Contamination here means infection risk mixing across cluster borders, so each area's risk is pulled toward that of its neighbors in the other arm. The target quantity, called the estimand, stays the same. So do the simulation engine and the three estimators. An estimator is a rule that turns trial data into an effect estimate. What changes is the map, the simulated survey size and the masses that weight the leakage. On the COVID map the leakage kernel also changes: it is directed, so the flow from one parish to another need not match the flow back. If the gains survive, they are not an artifact of the malaria geography.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

DO NOT ACCIDENTALLY CONCLUDE

Do not conclude that three geographies prove the design works on every connected system. Varying the geography probes robustness, but on all three maps the design scores allocations on leakage computed from the kernel that generates that map's simulated contamination; other geographies can still rank designs differently, and the paper itself reports settings where a simpler comparator wins on one geography. The composite objective is a robustness compromise, not a universal optimum.

TRY IT

List how each testbed differs from the malaria setting: both differ in spatial scale, and only the COVID one also changes the contamination form, to a directed flux. Then confirm from the section's opening sentence that the estimand, simulation engine, and estimators are identical, so any change in results must trace to geography and kernel.

E74 The cholera testbed's main-text verdict

Main text, the section on generalization to other connected systems, the first testbed paragraph, headed as a real waterborne epidemic at Yemeni governorate scale.

WHAT THE PASSAGE IS DOING

Compresses the whole Yemen cholera testbed into a single body paragraph, so a reader who never opens the appendix still learns which parts of the setting are real and dated, which parts are simulated, and how the designs compared. It is the only place in the main text that reports the leakage reduction the contamination-resilient design buys on this map, and the only one that marks a contamination strength of 0.5 as the level where inference is poor under every design. It also reports that the naive test's mildly inflated size at moderate contamination is absent at zero contamination on the same twenty governorates, so the inflation comes from the simulation's blend rather than from the small number of clusters, and it carries that finding into the paper's standing recommendation that the primary treatment-effect test use randomization inference.

IN ORDINARY LANGUAGE

Yemen had a large cholera outbreak in 2017 and 2018. Health workers counted suspected cases in each governorate, which is a province. The paper takes twenty of those provinces and their real case burdens. The last update in that record is dated 18 February 2018. Cholera moves with people and shared water. The model covers only the movement of people; shared water systems are not modeled. Leakage, meaning infection risk mixing across a border, follows a gravity model. Flow grows with population and falls with distance. Here it falls with distance raised to the power 2. Then the paper invents a trial on top of that real map. The map, the links, and the starting illness rates are real. The program and its effect are made up. The question is whether the earlier results show up again here. They mostly do. An estimator is a rule that turns trial data into an effect. With no leakage, the boundary-censored estimator is centered on the truth under every design. It works by dropping the sites most exposed to leakage first. Its coverage is then 0.947 to 0.958. Coverage is how often the interval catches the true effect. With moderate leakage, balancing traits between arms does not help. The leakage-aware design does help. Bias is the average gap from the truth. For the boundary-censored estimator it falls from 0.28 under simple randomization to 0.22. Coverage rises from 0.51 to 0.65. That lead grows up to a leakage level of 0.35. At 0.5 every design does badly. At moderate leakage the simple test fires false alarms too often, near 0.08 to 0.11, and the boundary-censored and spillover-adjusted estimators bring that rate close to where it should be. With no leakage, the simple test on the same twenty provinces fires false alarms at a rate of 0.044 to 0.052. Because the provinces are the same, the paper attributes the excess to the way the simulation mixes baseline infection risk across the arm border, not to having only twenty clusters. The paper asks for randomization inference instead, a test built from the way the arms were randomly assigned.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

DO NOT ACCIDENTALLY CONCLUDE

Do not read the roughly sevenfold fall in leakage as a sevenfold gain in inference. Leakage is the quantity the design was selected to minimize, and it is computed under one assumed gravity kernel at friction exponent 2, so it is a selection statistic and not a trial result. What actually moved is modest: boundary-censored bias from 0.28 to 0.22 and coverage from 0.51 to 0.65, the coverage still far short of the nominal 0.95 the intervals claim, and the design does not restore nominal coverage for the naive or the boundary-censored estimator. Nor does the phrase about the central findings reproducing amount to external validation; the outbreak surveillance is real and dated, the intervention and its effect are simulated, so this is reduced cross-arm exposure on a real map, not freedom from interference. Nor is this a test of the ten-metric framework: the testbed uses the two-criterion balance-plus-leakage design, so it corroborates the contamination-control thesis, not the full multi-objective framework. The randomization test named in the last sentence is recommended here, not evaluated here.

TRY IT

Open Table 10 in the generalization-testbeds appendix and put the coverage row for simple randomization beside the row for the contamination-resilient design. Subtract the two in the c = 0.35 column and again in the c = 0.50 column. Confirm that the gap narrows between those two columns while both numbers stay well below the nominal 0.95 the intervals advertise.

E59 The Louisiana verdict: bias transfers, precision improves, direction does not pay

Main text, the section on generalization to other connected systems, the paragraph on the early-2020 COVID-19 Louisiana testbed, whose second half gives the source/sink reading of the directed flux and what each design achieves on that geography.

WHAT THE PASSAGE IS DOING

Closes the directional testbed by separating what generalizes from what does not. Bias reduction under the contamination-resilient design transfers to this geography at the naive and boundary-censored estimators, covariate-constrained randomization again fails to reduce it, the spillover-adjusted estimator is nominal under every design so the design's gain there is RMSE rather than coverage, and the coverage gains are weaker, at the two-MCSE threshold at moderate contamination. The directed objective is then shown to lower net flux imbalance while raising total cross-arm flux, which the paper says governs attenuation bias, so it is demoted to a descriptive source/sink diagnostic rather than a design driver.

IN ORDINARY LANGUAGE

Contamination means infection risk mixing across borders. So each area's risk is pulled toward that of its neighbors in the other arm. The flow math marks New Orleans the top exporter and a suburb the top importer. That is a feature of the counting, not proven spread. Bias is the average gap between an estimate and the truth. Here the leak-aware design cut bias at two of three estimators. A design that only balanced background traits did not. It matched plain chance draws. The third one was near zero bias already. So the design bought precision there, cutting typical error by about a quarter. Coverage, how often the interval holds the truth, gained less. And the version tuned to flow direction did worse on bias than the leak-aware design. Still, it did no worse than plain chance draws.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> #### DO NOT ACCIDENTALLY CONCLUDE
>
> Do not read these figures as findings about COVID-19 in Louisiana. The geography and reported case counts are real, but the intervention, the survey and the effect are simulated. So only the relative behavior of designs is at issue. Do not treat the coverage results as established where the paper does not. The naive estimator's moderate-contamination gain, 0.627 to 0.663, and the boundary-censored gain, 0.816 to 0.845, are each about 2.7 Monte-Carlo standard errors. The paper places both at the two-MCSE threshold and reports them as not established. The naive estimator's clear gain arrives only at strong contamination, where the Louisiana appendix adds that inference is in any case poor for all designs. And do not conclude that the directed source/sink accounting is discredited. The passage keeps it as a description of the incidence surface while denying it any role in driving bias or coverage.

TRY IT

Open supporting/PROPOSED_covid_directional_oc.csv and read the bias column for the SR and CR rows (CR is the files' code for CCR) at the naive and corezone estimators (0.296 against 0.298, and 0.268 against 0.272). Balancing covariates moves bias essentially not at all, which is the span's claim that balance is not contamination control. Then read the rmse column on the adjusted rows and see the ordering the span reports, 1.487 under simple randomization against 1.136 under the contamination-resilient design and 1.311 under the covariate-constrained one, which is where the design pays on a geography whose adjusted estimator has no bias left to remove.

E91 AI tools, stated where the methods are

The close of the Estimand and Operating Characteristics section, the short subsection Use of AI Tools, just before Results.

WHAT THE PASSAGE IS DOING

Places the account of AI use inside the methods, where the journal's policy asks for it, and says which assistant wrote and ran which code, so a reader can trace every reported analysis to deposited scripts that can be rerun. It gives the same account as the closing declaration on AI-assisted technology.

IN ORDINARY LANGUAGE

This short subsection says how AI tools were used. The work began in 2019 as a manuscript the author wrote without AI, on trial data. This public-data version was rebuilt from it starting in 2025. Three kinds of assistant took part: ChatGPT in 2025, Gemini and local open-weight models in early 2026, and Claude in 2026. They helped search the literature, drafted parts of the text and reviewed it. Claude also wrote and ran the R code. That code covers the design-selection engine on the public map and the operating-characteristics simulations. It also covers the two extra testbeds and the heterogeneity study in the last appendix. All of that code is in the public archive. It runs on public data, and every run starts its random draws from a fixed point, so it gives the same numbers each time. Supplement S2 shows how to rerun the main table with one command. The tools did not make up or change any observed data. The author directed the work, checked the results and made every scientific and editorial decision.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read this as saying someone outside the author reran the code; it describes properties of the code. Do not read the list of AI tasks as moving responsibility for the results away from the author. And do not read 'no observed data were made up or changed' as saying the trials are real: the arms, the intervention and its effect are simulated by that code on real geography.

TRY IT

Open the deposit's README and find each script behind the four analyses this subsection lists. Then read this subsection beside the closing declaration on AI-assisted technology near the end of the paper and check that the two name the same tools and the same code.

E60 How the candidate pool was built, and which method-by-K pairs survived

Results, the first subsection 'Combinatorial Yield and Multi-Objective Selection', its opening paragraph describing how candidate allocations were enumerated and which clustering-method by resolution combinations remained feasible.

WHAT THE PASSAGE IS DOING

Describes the search that produced the scored candidate set: balanced within-stratum splits enumerated for every candidate spatial configuration across the five clustering paradigms at K in {8, 12, 16}, with enumeration capped at 50 splits per configuration and uniform sampling beyond the cap, the survivors then scored by the ten-metric suite against a stratified random-allocation baseline. It then reports that the singleton-rejection gate, kept to preserve within-group degrees of freedom for the mixed-effects model a real trial would fit, makes several method by resolution combinations infeasible on this sparse geography.

IN ORDINARY LANGUAGE

Before scoring anything, the engine must build a list of candidate designs. For each grouping of sites it splits the groups between the two trial arms, keeping the two latitude bands, the study's strata, even. Many groupings allow more than 50 such splits, so the count was capped at 50 and the rest sampled at random. This public map has only two bands, of 28 and 29 sites, so the clusters are small. The rule that drops any cluster holding a single site then leaves some method and K pairs with no legal design. Average linkage has none at any K, and Ward and CLARA none above K=8, so K-means and spectral supply most of the comparison.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read the method-by-K comparison that follows as an even contest between five algorithms: they enter with very different numbers of feasible candidates, and average linkage contributes none at all, so a method's score distribution reflects what survived the gate as much as how well it groups sites. Do not treat the candidate set as a complete enumeration of possible allocations; past 50 splits per configuration it is a random sample of them.

TRY IT

Open supporting/PROPOSED_design_selection_MAP.csv and tally the rows by the method and k_value columns. Check that no row is average linkage, that Ward and CLARA appear only at k_value 8, and that no method-by-K cell exceeds the stated cap of 50 splits.

E75 How to read the composite-score boxplot

Results, the Combinatorial Yield and Multi-Objective Selection subsection, the composite-score boxplot figure and its caption, placed just after the paragraph describing how the candidate pool was enumerated.

WHAT THE PASSAGE IS DOING

This figure supplies the picture behind the selection claim: the whole distribution of composite scores, broken out by clustering method and by K, rather than the single winning number the text goes on to report. Its caption tells the reader how to read height against the dashed naive baseline, and it names the two facts the surrounding argument leans on, that K=8 carries both the strongest median composite and the highest-scoring elite designs while K=16 sits poorly against the random-allocation baseline. It is the only figure that plots the full spread of candidate scores; the text on either side of it reports how the pool was built and which single candidate won, and the median comparison the figure makes visible is restated in prose later, in the structural trade-offs subsection.

IN ORDINARY LANGUAGE

This figure is a box plot. A box plot shows the spread of many values at once. Each box covers the middle half of the scores. The line inside a box is the median, that is, the middle value. Each score is a composite score. That is one number that blends the design's diagnostic measures. Scores are set against a naive baseline. The baseline is the average score of random splits, in which whole clusters are assigned to arms at random within each of the two latitude bands. The dashed line marks that baseline. A box above the line beats random allocation on the blended score. A box below the line does worse. The boxes are grouped by clustering method and by K. K is the count of clusters the sites are split into. At K=8 the median score is the highest. The very best single designs sit there too. At K=16 the boxes sit below the dashed line; only a few single designs reach above it.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> #### DO NOT ACCIDENTALLY CONCLUDE
> Do not read the gap between the K=8 and K=16 boxes as evidence that coarser clusters yield less biased estimates, better coverage, or more power. The composite is assembled from the design's own diagnostic metrics, and the figure carries no error bars, no p-value, and no multiplicity adjustment; it is a selection statistic, not a test. Do not read a box sitting below the dashed line as a design that would fail, since that line marks only a random-allocation comparison on the blended score. Do not treat the side-by-side boxes as an even contest among the clustering methods either, because the methods do not contribute the same number of feasible candidates at every K.

TRY IT

Open supporting/PROPOSED_design_selection_MAP.csv, group the composite_score column by k_value, and take each group's median. Check that the K=8 median is the highest and above zero, that the K=16 median is below zero, and that the single largest composite_score in the file sits on a K=8 row. That reproduces both caption claims and shows the dashed baseline in the figure is drawn at zero, which you can confirm at the geom_hline call in supporting/PROPOSED_design_selection_MAP.R.

E3 The selected design: Spectral K=8, and why K=8 dominates the elite pool

Results, the Combinatorial Yield and Multi-Objective Selection subsection, the paragraph reporting the selected Spectral Clustering K=8 configuration, its Optimality Index, and the K=8 dominance of the elite pool.

WHAT THE PASSAGE IS DOING

States the output of the design-selection engine on the public geography and immediately fences it: the Optimality Index is a ranking within the search, not a significance test. It also reports that the elite pool is almost entirely one resolution, K=8, so the typical and the best candidate now agree.

IN ORDINARY LANGUAGE

The framework scored many ways to split the survey sites into trial clusters. It picked a design that uses spectral clustering with K=8, that is, 8 clusters. Spectral clustering groups sites by a graph of links between them, not by straight-line closeness. Of the 40 best-scoring candidates, 37 use K=8. So for these survey sites the coarser split, meaning fewer and larger clusters, is both the typical winner and the outright winner. In the paper's main simulations, though, those larger clusters carry no extra variance. When variation between clusters is added (Appendix D), the coarser pools give up power, the chance of detecting a real effect. An earlier build of this work had picked a K=12 design. That result came from a topology metric that used the wrong coordinate units. Topology here means the shape of each arm's pattern of sites, such as its loops and holes. The pick changed once the units were fixed. The paper describes that correction where it defines the metric.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> #### DO NOT ACCIDENTALLY CONCLUDE
> Do not read 100/100 as statistical significance or as proof this design is uniformly best; it is a ranking produced inside the paper's own scoring pipeline, and the top candidate scores 100 by construction. Do not conclude that K=8 is a universal optimum: it is the resolution this two-stratum public geography favors once the metrics are computed consistently. Do not conclude that a trial simply adopts this top-ranked design either: it is picked by maximizing a score, so it has no randomization distribution of its own, and the allocation a trial implements is drawn at random, with equal chance, from the 40 best-scoring candidates, the elite pool, which spans four clustering methods and two resolutions. The design's inferential validity rests on that draw and on the cluster-level comparison of selection objectives, not on the ranking.

TRY IT

State in one sentence which K carries the strongest typical composite score, which K holds the single top design, and whether the two agree. Then explain in one more sentence why a rank of 100/100 among candidates is not a multiplicity-adjusted significance test.

E61 The winning design, the elite pool it came from, and why the win is not a test

Results, the subsection on combinatorial yield and multi-objective selection, the paragraph naming the selected Spectral K=8 configuration, the elite pool it came from, and the ranking-not-a-test caveat.

WHAT THE PASSAGE IS DOING

Reports the output of the selection engine on the public survey geography: the top-ranked candidate among 331 structurally deduplicated partitions, its composite score against the best partition from two rival paradigms, and the composition of the 40-design elite pool by method and resolution. It then bounds the claim, stating that an argmax over a large search cannot be read as a significance test, and locating the design's inferential validity in the uniform random draw of the trial's allocation from the elite pool, which spans four clustering methods and two resolutions, and in the cluster-level objectives comparison, whose full ten-metric cell evaluates exactly that draw, rather than in the composite ranking.

IN ORDINARY LANGUAGE

The engine scored 331 distinct candidate maps. It kept the 40 best of them as an elite pool. The winner is a spectral clustering partition into 8 clusters. Spectral clustering groups sites by a graph of links, not by straight-line distance. It scores 0.86 on the combined score. The best maps from two rival methods score 0.81 and 0.80, so the margin is narrow. Most of the pool, 37 of the 40, uses 8 clusters. None of the pool uses 16. The paper then draws a firm line. This is a ranking among candidates, not a test of significance. Pick the best out of many tries and the top score looks extreme no matter what. So the paper looks elsewhere for the design's statistical validity. A trial would not simply use the top map. It would draw its map at random from the 40 in the elite pool, each equally likely. The design's validity rests on this random draw and on a separate cluster-level comparison, whose full ten-metric entry evaluates exactly that draw, one map per simulated trial.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read the gap between 0.86 and 0.81 as a demonstrated advantage; these are scores from one scoring pipeline on one geography, with no error bars attached and no test behind them. Do not conclude that a top rank means better bias, coverage, or power; the passage itself says the composite ranking is not what carries inferential validity. Do not carry this ranking over to the site-level operating-characteristics tables: those evaluate a reduced two-criterion objective at a different allocation unit, so the two sets of results are not the same comparison.

TRY IT

Open supporting/PROPOSED_design_selection_MAP.csv, which holds one row per candidate. Sort by composite_score, take the top 40 rows, and tabulate them by method and k_value. Check that the top row is a Spectral design at K=8 and that 37 of the 40 rows carry K=8. Then look at how close the second and third distinct methods sit to the winner, and decide for yourself whether that spacing would survive a rerun with a different random pool.

E62 The standardized-gain radar: favorable on all ten metrics

Results, the subsection on the configuration and performance profile, the paragraph reading the standardized-gain radar together with the radar figure it interprets.

WHAT THE PASSAGE IS DOING

Interprets the radar plot of the selected design's ten standardized gains, noting that the largest favorable Z-scores fall on covariate balance, the multivariate (Wasserstein) distance, the Euler characteristic and the variational free energy, and that the standing is positive on every one of the ten dimensions including mobility network leakage. It then frames the profile as broad rather than carried by a single metric, and labels these per-metric contrasts diagnostics rather than inferential tests.

IN ORDINARY LANGUAGE

The radar chart shows all ten design metrics for the chosen split in one picture. Each spoke is a Z-score, meaning how far the design sits from a random split in units of spread. Longer spokes are better. Here every one of the ten spokes points outward, so no metric is worse than random allocation. The longest spokes are the balance of background traits and how closely the two arms' trait distributions match (the Wasserstein distance). They also include how few separate pieces the treated area forms (the Euler count) and the free-energy score. For all four, as for nine of the ten metrics, a smaller raw value is better, so a long spoke means a small raw value. The paper reads this as breadth, not as one strong metric carrying the whole design.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read a favorable spoke as a statistically significant advantage; the passage itself calls these diagnostics, not tests, and there is no p-value anywhere on the radar. Do not conclude that scoring well on all ten design metrics means the trial will give less biased estimates or better coverage; that question is answered by the operating-characteristics simulations elsewhere in the paper, not by this figure. Do not read the long balance spoke as balance on the baseline infection rate either. The balance metric works in raw units, so elevation, measured in meters, supplies 99.1% of the squared score for this design, and the paper says its gain does not certify balance on the baseline infection rate.

TRY IT

Open supporting/PROPOSED_selected_design_profile_MAP.csv and look only at the Z column. Check that every one of the ten rows is on the favorable side, then rank the column and confirm the four largest are the balance, Wasserstein, Euler and free-energy rows named in this paragraph.

E45 The selected design in raw units, including an unequal split

Results, the configuration and performance profile subsection: the table giving the selected design in raw units together with the stabilization-of-statistical-interchangeability paragraph that follows it.

WHAT THE PASSAGE IS DOING

Gives the selected design's raw metric values, the null mean and standard deviation behind every standardized gain, the metric weights, and the arm composition, and states what the balance gain does and does not certify.

IN ORDINARY LANGUAGE

A standardized gain Z says how many null standard deviations better than a random cluster split the design is. Table 4 shows the raw numbers behind each Z. For example the raw covariate distance is 40.5 against a null mean of 57.2 and spread of 12.1, so Z is 1.38. The table also shows the arms themselves. The design gives 21 sites with 2,564 examined people to treatment and 36 sites with 11,233 to control. That happens because arms are balanced in the number of clusters, and spectral clusters differ in size. Mean baseline prevalence is 0.60 in treatment and 0.52 in control. Balance in this framework means closer than a typical random split, not equal.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read Z = 1.38 on balance as equality of the arms; the arms differ more than fourfold in examined individuals and by eight prevalence points. Nor read it as balance on baseline parasite rate: the score is in raw units, so elevation, measured in meters, supplies 99.1 percent of its squared value for this design, and the arm composition reports parasite rate directly. Do not read the small Z on spatial mixing as a weakness of the design; that metric's null distribution is heavy-tailed and its Z is small for every candidate. Frontier Risk is zero for this design and for 297 of the 331 candidates, so it separates almost nothing here.

TRY IT

Recompute one Z from the table: subtract the null mean from the design's raw value, divide by the null standard deviation, and flip the sign for a metric that is preferred low. Check it against the printed Z.

E76 The topological and contamination pillars, and the correction that changed the winner

Results, the configuration and performance profile subsection, the second and third pillar paragraphs, on topological and spatial cohesion and on contamination resilience and cross-arm exposure control.

WHAT THE PASSAGE IS DOING

These two paragraphs carry the selected design's standing on the second and third pillars of the ten-metric suite, and they are the one place in the paper where a metric is audited rather than reported. The topological paragraph discloses a coordinate-space defect in an earlier build of the Persistence Gap, states that the corrected run changed which design is selected, and points to the coverage evidence that the three topological terms are not decorative. The contamination paragraph refuses the victory lap the four positive scores would allow: it reads the Spatial Mixing ratio of 6.38 as insulation that weakens as the assumed dispersal scale widens, and it reports that Frontier Risk ties for 297 of the 331 candidates.

IN ORDINARY LANGUAGE

The paper now grades the chosen split on two more groups of metrics. These scores rank designs. They are not tests.

First comes shape. Three metrics score the map shape of the two arms. An arm is one side of the trial. The persistence gap asks how well an arm's shape tracks the land it covers. The Euler characteristic counts how many separate pieces an arm breaks into. The free energy term rewards a simpler split. Their scores are 0.7, 1.0 and 1.0. A score of this kind counts standard deviations of gain over a random split.

The paper then reports a bug it found and fixed. The old code mixed raw map degrees with height in meters. Height then swamped the distance math. Under that bug, 311 of the 662 arm shape summaries came out empty. That is 47 percent. At least one arm was empty in 302 of the 331 candidate splits. That is 91 percent. Both arms were empty in 9 of them. The old winner, a K=12 split, sat on a thin false peak at a score of 2.9.

The fix puts every point in one shared space. The false peak is gone. The metric's share of the total score falls from 0.24 to 0.08. The winner changes to the K=8 split shown here. The old favorite now ranks 63rd of 331.

A later check backs the three shape terms. Coverage is how often a confidence interval holds the true effect. The naive estimator ignores leaks. Dropping the shape terms costs it 0.105 of coverage under a moderate leak and 0.072 under a strong one. The core-zone estimator drops the most exposed sites first. It loses 0.057 under a strong leak, from 0.911 to 0.854. Under a moderate leak it loses only 0.012. That gap is inside the noise of the simulation.

Second comes leakage. Four metrics score how much the arms bleed into each other. All four favor the chosen split, at 0.3, 0.3, 0.2 and 0.2. Those are small gains. Two of the four barely tell the candidates apart.

Spatial mixing is a ratio. It weighs cross-arm exposure at a wide range against exposure at a short one. Here it is 6.38. So the leak at 10 km is about six times the leak at the 2 km mark. The shield gets weaker as the assumed travel range grows. The positive score only says the design sits far below a wildly spread random yardstick. Almost every candidate does.

Frontier risk is no sharper. The tested buffer widths stop at 5 km. The typical gap to the nearest site is 5.9 km. So 297 of the 331 candidates tie at the best value. The metric only sorts the 34 designs in its bad tail.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read four positive contamination scores as evidence that the arms are sealed off from each other; the passage blocks that reading twice. The Spatial Mixing ratio of 6.38 says the cross-arm asymmetry at 10 km is about six times its value at the 2 km reference, so the insulation weakens as the assumed travel range widens, and its positive score records only that the design sits far below a heavy-tailed random null on which nearly every candidate ties. Frontier Risk ties for 297 of the 331 candidates, so it separates almost nothing here. Do not read the corrected Persistence Gap as a cosmetic reshuffle either; it changed which design is selected. Do not read the coverage figures as showing the topological terms help at every contamination strength, since the boundary-censored gap of 0.012 at moderate contamination sits inside Monte-Carlo error. Nothing in either paragraph frees the trial from SUTVA violation.

TRY IT

Open supporting/PROPOSED_selected_design_profile_MAP.csv and find the spatial_mixing row: raw value 6.3812, null mean 530420, null standard deviation 2897700, Z 0.183. Compute (530420 - 6.3812) / 2897700 and confirm you land on that Z. The design beats the null mean by a factor of tens of thousands and still earns a Z below 0.2, because the null's spread is more than five times its own mean. Then check the persistence_gap row's weight of 0.0793 against the 0.24 the metric carried in the uncorrected build described in the same paragraph.

E77 The optimization profile and its no-trade-off reading

Results, the opening of the standardized effect sizes and structural trade-offs subsection, the paragraph that reads the optimization profile figure together with the figure itself.

WHAT THE PASSAGE IS DOING

Opens the structural trade-off subsection with the premise that multi-objective spatial design is managed compromise, then reads the optimization profile figure against that premise. It fixes the reference for the whole standardized-gain space as the stratified random-allocation baseline drawn one thousand times, and names the four largest gains at 1.4, 1.2, 1.0 and 1.0. Its distinctive move, made in no other passage, is the claim that the coarser spectral cut delivers baseline balance and contiguity together instead of trading one against the other, which is the premise the later cross-configuration comparison builds its resolution argument on.

IN ORDINARY LANGUAGE

A Z score says how far the chosen map sits from a random cluster split. The unit is the spread of the random splits themselves. Zero means no better than a random split. Positive is better. The figure is a bar chart with one bar per design metric. Each bar is labeled with its value. The baseline is one thousand random splits drawn inside the same strata. A stratum is a group of like sites that the draw is made within. The two tallest bars are balance of background traits at 1.4 and the Wasserstein distance at 1.2. The Wasserstein distance measures how far apart two whole distributions sit. Next come the count of connected pieces at 1.0 and the free energy term at 1.0. Every bar points the good way. The paper reads that as a joint win. The coarse cut into 8 clusters gives balance and contiguity at once. Contiguity here means the treated arm, all its clusters taken together, forms few separate pieces on the map. For this design it forms a single connected piece. So this design did not trade one of those away to get the other.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read every bar positive as proof that this design escaped the compromise the subsection announces. Each bar compares the design to a random split, not to the other candidates, so the trade-offs live in the comparison across configurations and not inside one design's own profile. Do not read the 1.4 balance bar as balance on the baseline infection rate: the balance metric works in raw units, so elevation, measured in meters, supplies 99.1% of the squared score for this design, and the paper says its gain does not certify balance on the baseline infection rate. Do not read a bar at 1.4 or 1.2 as a significant finding; the composite that picked this design is a selection statistic, not a test, and no p-value sits behind any bar. And do not read the positive bars on the contamination metrics as insulation between the arms; the design lowers cross-arm exposure and residual bias without delivering freedom from SUTVA violation.

TRY IT

Open supporting/PROPOSED_selected_design_profile_MAP.csv and read the Z column. Check that the four values quoted in this paragraph are the rounded table entries 1.376, 1.249, 1.019 and 0.998, then set figures/optimization_profile_MAP.png beside the standardized-gain radar earlier in the Results and confirm both carry the same ten numbers. If they do, the two figures are one dataset drawn twice, not two independent pieces of evidence.

E78 The trade-off heatmap and the case for ensemble weighting

Results, the standardized effect sizes and structural trade-offs subsection, the paragraph headed Cross-Configuration Trade-offs together with the heatmap figure it introduces.

WHAT THE PASSAGE IS DOING

Turns the reader from the one selected design to the whole admissible candidate space, showing every method-by-resolution combination that survived singleton rejection scored across all ten metrics at once. This is the passage that earns the ensemble weighting: the cells form a checkerboard with no configuration warmest everywhere, so optimizing a single metric would yield an unstable, overfitted selection. Its caption also fences the panel as candidate space rather than result, since spectral partitions are admissible at all three resolutions while only the K=8 ones reach the elite pool.

IN ORDINARY LANGUAGE

The heatmap is a grid. Each row is one way of cutting the map into clusters. A cluster is a group of sites treated as one unit. Each row pairs a clustering method with a resolution, meaning how many clusters it makes. Eight such pairs survive in the deposited run. A pair is dropped if none of its 80 tries gives a split in which every cluster holds more than one site. Which pairs survive partly depends on the random restarts: a split by the CLARA method into 12 clusters passes on 3 of 20 independent sets of restarts, though not in the deposited run. Each column is one of the ten design metrics. Each cell holds a Z score. Z says how far that pair sits from a random split, in units of spread. Warm cells beat a random split. Cool cells do not. The colors form a checkerboard. No row is warm on every column. Rows that gain on baseline balance and cluster shape give some ground on boundary leakage. Leakage is infection risk mixing across the border between the arms, meaning the treated and untreated groups. So the paper blends all ten metrics into one score. Chasing a single metric would pick a design that shines on that metric and wobbles on the rest.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> #### DO NOT ACCIDENTALLY CONCLUDE
>
> Do not read the heatmap as a scoreboard of results. It is the candidate space, and a warm spectral row at a finer resolution is not a recommendation, because only the spectral K=8 partitions reach the elite pool. Do not read the checkerboard as evidence that the ten metrics measure independent things; the paper's own ablation calls them correlated, and a checkerboard shows only that no configuration leads everywhere. Do not read controlled boundary spillover as spillover removed. The design lowers cross-arm exposure and residual bias; it does not deliver freedom from SUTVA violation.

TRY IT

Open supporting/PROPOSED_design_selection_MAP.csv, group the rows by the method and k_ value columns, and confirm there are exactly eight groups. Then take the mean of each of the ten metric columns, d_balance through d_mobility_auc_leakage, within each group and mark which group is highest on each metric. Check that those wins are spread over several groups and that no single group is highest on all ten.

E79 The perimeter trade-off that favors coarser partitions

Results, the Standardized Effect Sizes and Structural Trade-offs subsection, the closing paragraph after the metric trade-offs heatmap, which reports resolution dependency across the surviving method-by-resolution combinations.

WHAT THE PASSAGE IS DOING

This paragraph proposes a mechanism for a pattern the rest of the Results only tallies, namely why resolution rather than clustering method is what sorts good candidates from bad. It states the trade-off in physical terms, that cutting the same map into more clusters buys tighter local baseline parity while adding cross-arm boundary, and that the added perimeter costs more in spillover exposure than the parity gain returns, a reading it offers as consistent with the pattern rather than measured; the evidence offered is the median composite by resolution, with K-Means at K=16 well below the stratified random-allocation baseline, and the resolution mix of the elite pool, 37 of 40 at K=8 and none at K=16. It then blocks the reading it most invites, holding that contiguity limits perimeter exposure without making the arms independent and without removing the residual contamination bias.

IN ORDINARY LANGUAGE

Think of the study map as a sheet you cut into pieces. Each piece is a cluster, a group of survey sites that gets one arm of the trial as a block. K is the number of pieces. More pieces mean smaller pieces. Smaller pieces let you match the arms more closely on local traits. But more pieces also mean more cut edge. Wherever a cut separates a treated piece from a control piece, a treated site sits next to a control site. People move across it. Treatment can then leak into the control arm. That leak is spillover. It breaks SUTVA, the rule that one site's outcome must not depend on another site's assignment. The paper scored every candidate map with one combined score. Within each grouping recipe, maps cut into 8 pieces beat maps cut finer. The K-Means maps at 16 pieces show it plainly. Their middle score sits well below the naive baseline, which is a random split drawn inside the same strata. The paper reads this as the extra edge costing more than the tighter local match is worth. This paragraph offers that reading as fitting the pattern, not as a test of it. The picked map is also contiguous. Here that means the treated arm, all its clusters taken together, forms one connected patch, not scattered bits. The paper counts those patches: 1 for the picked map, against 1.86 on average for random splits. The Euler Characteristic gain of 1.0 is that difference measured in units of the random splits' spread. Less edge is easier to run and easier to watch. That is not the same as clean arms. Leak across the edge that remains is still there. Of the 40 top-scoring maps, 37 are 8-piece maps. None is a 16-piece map.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not turn this into a general design rule that coarser partitions are better in cluster-randomized trials. It is a pattern on one public geography under one composite, and the boundary surface-area story is offered as consistent with that pattern, not tested against it in this paragraph. Do not read the K-Means result at K=16 as proof that finer partitions damage any trial. And do not read the Euler Characteristic value of 1.0 as insulation, because contiguous arms still touch along a shared boundary, so the residual contamination bias of the estimand section stands, and the composite that produced this ranking remains a selection statistic rather than a test.

TRY IT

Open supporting/PROPOSED_design_selection_MAP.csv, which holds one row per candidate. Group composite_score by the method and k_value columns and take each group's median: every K=8 group should sit above zero, the random-allocation baseline, while K-Means at K=16 sits well below it. Then sort the whole file by composite_score, take the top 40 rows, and tally k_value; you should count 37 rows at 8 and no rows at 16. For the picture, find the K-Means row at K=16 in the metric trade-offs heatmap figure and see how many of its cells sit on the cool side of the baseline.

E80 Resolution against insulation as the standing trade-off

Discussion and Conclusions, the section heading together with its opening paragraph, the one naming the resolution-versus-insulation tension and the composite margin over the best Ward partition.

WHAT THE PASSAGE IS DOING

Opens the Discussion by converting the selection result into the paper's organizing tension, covariate resolution set against geometric insulation. It restates the head-to-head comparison as a single margin, 0.05 in composite over the best Ward partition at the same resolution, which the Results already report as composite 0.86 against 0.81 for the best Ward K=8, and states the mechanism that keeps the finest cuts disfavored, namely that 16-cluster partitions sacrifice operational contiguity, lengthen the vulnerable boundary perimeter, and raise human-vector spillover risk. It then sets the register for everything that follows by qualifying the robustness claim as robustness on the measured diagnostics.

IN ORDINARY LANGUAGE

A cluster trial treats whole groups of sites, not single people. Each group is a cluster. Cutting the map into more clusters helps match the arms on background traits. Background traits here are the baseline infection rate, elevation, a proxy for closeness to water, and how many people were tested at each site. Cutting it into fewer clusters tends to keep the treated arm in fewer pieces. A solid block shares a shorter border with the other arm. Border length matters because effects leak across it. Mosquitoes fly. People travel. So finer matching and tighter sealing pull against each other. This paragraph names that pull as the paper's core tension. The chosen design uses spectral clustering with 8 clusters. Spectral clustering groups sites by a graph of links, not by plain distance. On the combined score it sits 0.05 above the best Ward split with the same number of clusters. Ward is a rival grouping method. That combined score blends 10 design measures into one number. The paragraph also says the finest cuts lose out. A cut into 16 clusters more often leaves the treated arm in scattered pieces. Scattered pieces add border. More border means more room to leak. The

closing claim is kept narrow on purpose. The layout looks sturdier against map-driven confounders. A confounder is a trait that differs by arm and could explain the result. Sturdier here means only on the measures the paper took. The whole trial is simulated on a real malaria map.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read the 0.05 composite margin as a demonstrated win for the spectral paradigm over Ward. The composite is a selection statistic used to rank candidates, not a test; the margin carries no error bar and no p-value, and a rerun with a different candidate pool could reorder the two. Do not read "more robust to the spatial confounders" as insulation from spillover. The design lowers cross-arm exposure and residual bias, and it does not deliver freedom from SUTVA violation or restore nominal coverage for the naive or boundary-censored estimator. Do not treat the 10-dimensional diagnostic space as 10 independent dimensions either; the metrics share geometric and epidemiological primitives and are correlated. And do not read the case against the finest cuts as a measured spillover result here; in this paragraph it is stated as a perimeter argument, with the coverage evidence supplied elsewhere.

TRY IT

Open supporting/PROPOSED_design_selection_MAP.csv, keep only rows whose k_value is 8, and take the largest composite_score among the Spectral rows and among the Ward rows. Subtract them and check the gap against the margin this paragraph claims. Then sort the whole file by composite_score and count how many of the top 40 rows carry k_value 16.

E81 Marginal influence, not statistical independence

Discussion and Conclusions, the paragraph that restates the leave-one-out ablation of the ten-metric composite and then qualifies what that ablation can and cannot show, immediately after the paragraph on the selected spectral configuration.

WHAT THE PASSAGE IS DOING

Restates the leave-one-out ablation in the Discussion, ordering the metrics by how far the composite ranking moves when each one is dropped, and then blocks the reading that ordering invites. Its distinctive job is the concession: every metric shifting the ranking is marginal influence only, the metrics are built from shared geometric and epidemiological-mass ingredients, and the suite's effective dimensionality is below ten. It closes by noting that the inter-metric correlations are mostly low, with a single near-duplicate balance pair.

IN ORDINARY LANGUAGE

The paper scores each candidate design on ten measurements. It adds them into one score and ranks the designs. An ablation test drops one measurement at a time. It then ranks all the designs again. Spearman correlation scores how well two rankings agree. A perfect match gives the top value. Dropping the free-energy term gives 0.82. That is the largest shift. Dropping spatial mixing gives 0.997. That is the smallest. The mean across the ten drops is 0.96. So the mean gap from a perfect match is 0.04. Every measurement moves the ranking at least a bit. The author then blocks a wrong reading. Moving the ranking is not the same as being separate. The ten measurements share raw parts. They draw on the same map and the same case counts. So they rise and fall together to some degree. Statisticians call fully separate measures orthogonal. These are not orthogonal. The count of truly separate directions is under ten. Even so, most pairs track each other weakly. Only one pair of balance measures is near a duplicate.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read the fact that each dropped metric moves the ranking as evidence that the ten metrics are independent measurements. The passage says the opposite, and puts the effective dimensionality below ten. Do not read the 0.82 for the free-energy term as showing that metric matters most for inference; the ablation moves the selection ranking only, and the composite ranking is a selection statistic, not a test. And do not read the mostly low correlations as freedom from SUTVA violation, the rule that one site's outcome does not depend on another site's arm. The paper claims reduced exposure and measured residual bias, not that spillover is gone.

TRY IT

Open supporting/PROPOSED_loo_ablation_MAP.csv and average the one_minus_rho column over its ten rows; you should land near 0.041, which the paragraph rounds to 0.04. Then confirm the vfe row (0.1808) and the density_symmetry row (0.0633) are the two largest in that column, and that wasser (0.0391) and euler (0.0364) sit adjacent to each other, which is the near-tie the sentence names.

E46 Six selection objectives compared at the cluster level

Discussion and Conclusions, the paragraph comparing six design-selection objectives at the cluster level together with the operating-characteristics table it presents.

WHAT THE PASSAGE IS DOING

Bridges the ten-metric design selection of the Methods to the two-criterion design of the operating-characteristics tables by running the same simulation over six pools of whole-cluster allocations drawn from the same 331 candidates, and settles whether the topological terms carry inferential value.

IN ORDINARY LANGUAGE

Six ways of picking a cluster-level design are run on the same simulation. One way draws at random from all 331 designs. Each other way keeps the 40 best under one rule. The rules are: balance of baseline traits only; leakage only; both of those; the composite score with its three topology terms cut; and the full composite. Leakage here means people moving across the arms of the trial. All six pick whole clusters at once. So the naive and core-zone estimators' coverage is much higher than in the site-level tables. The spillover-adjusted estimator is at or near nominal at both randomization units. Coverage is how often the interval holds the true effect. Only the gaps between the six rules are comparable. Three things emerge. Balance alone is no better than random choice. The full composite gives the best naive coverage at moderate and strong contamination. It gives the best core-zone coverage at strong. At moderate contamination it ties the leakage-only rule on that measure. The naive estimator ignores spillover. The core-zone one drops the most exposed sites. The full composite still beats the leakage-only rule on the naive estimator. It does so despite more cross-arm flux, that is, more movement across arms. The paper attributes this to the full composite being the coarser of the two pools: 37 of its 40 designs use the coarsest resolution, 8 clusters, against 32 for the leakage-only pool. The edge does not come through lower exposure, though, since the full composite's mean realized exposure, 0.204, is above the leakage-only pool's, 0.168. The random draw from all 331 designs mixes all three resolutions, so against it the full composite's gains combine its choice of resolution with its choice of allocation. A separate run, whose results match the six-objective table within 0.03, adds three comparators drawn at random at the full composite's own resolution. Against them its core-zone advantage disappears at moderate contamination. At strong contamination it shrinks to less than two Monte-Carlo standard errors of the difference. So most of that advantage belongs to the coarser resolution. Its naive coverage advantage survives at strong contamination, 0.406 against 0.323 to 0.340, and is only directional at moderate, 0.541 against 0.493 to 0.518. These are that run's figures. The full six-objective table in the appendix gives the full composite 0.391 at strong contamination. Cutting the topology terms costs the naive estimator 0.105 and 0.072 of coverage at moderate and strong contamination. It costs the core-zone estimator 0.057 at strong. At weak and moderate contamination that core-zone gap is inside Monte-Carlo error. That error is just the noise of the simulation. The spillover-adjusted estimator corrects the estimate for that leak. It stays near nominal under every objective, close to the level it claims. So the objective only buys it precision.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

DO NOT ACCIDENTALLY CONCLUDE

Do not conclude the topological terms drive inference directly; they act through resolution, by favoring the coarser K=8 partitions with less boundary exposure. Do not conclude they help the boundary-censored estimator at every contamination strength: at c=0.2 and c=0.35 that difference is inside Monte-Carlo error, and only at c=0.5 is it larger than the noise. Do not compare these coverages with Table 1; the randomization unit differs. Do not read them as holding when clusters vary either: the simulated outcomes carry no between-cluster variation beyond sampling chance, and whole clusters are randomized but sites are analyzed as independent, so fewer, larger clusters pay no variance penalty here. With between-cluster variation at an intracluster correlation (ICC), the share of outcome variation that lies between clusters, of 0.05 and a cluster-level analysis, the full composite's coverage at c = 0.35 matches

> that of three comparators drawn uniformly at its own resolution, its naive bias stays lower, and, like every pool concentrated at 8 clusters, it has less power than the finer pools. And do not read “best coverage” as “adequate coverage”: at strong contamination the naive estimator under the full objective still covers only 0.391.

TRY IT

From Table 5, subtract the “composite without topology” core-zone coverage from the full composite’s, and do the same for the naive estimator; then repeat both at c=0.5 using Table 16. Check your naive differences against the text, which reports 0.105 at c=0.35 as 5.8 and 0.072 at c=0.5 as 4.1 Monte-Carlo standard errors of the difference. A difference is noisier than either cell, so its standard error is larger than the per-cell coverage MCSE of 0.008 the text gives near 0.9. Say at which contamination strengths each difference clears three times the standard error of the difference, the paper’s bar for calling a difference established.

E51 Robustness as a specification multiverse, and what was not varied

Discussion and Conclusions, the paragraph titled “Robustness as a specification multiverse”, just before the Limitations subsection.

WHAT THE PASSAGE IS DOING

Frames the result as invariant across a set of defensible analytic choices, names the four additional axes varied in Appendix C, identifies the three conclusions that are not invariant, along with the directed-objective and unit-of-allocation axes on which the conclusion does not carry over, and lists three design choices left untested. It also names one more axis, variation between sites, which the last appendix varies.

IN ORDINARY LANGUAGE

A specification multiverse runs the same question down every reasonable analytic path. It then asks if the answer changes. Here the paths are three geographies, three estimators, and the leakage objective. Leakage is the modeled flow of people between the two arms, the route by which infection risk can mix from one arm into the other. The objective scores it in a symmetric and in a directed form. The appendix adds three contamination kernels, the censoring fraction, the pool size, and the random draw of the candidate space. A kernel is a rule for how spillover falls off with distance. On all three geographies the design ranks allocations by leakage. It measures that leakage with the same kernel that generates the simulated contamination. So the geography path changes the map but never that match. Only the three appendix kernels, run on the malaria map, break that match. A separate section checks an equal-weight composite against the ensemble one. Take the objective in its symmetric form. Across the geographies, the kernels and the candidate draws, the design lowers the bias of the naive and the boundary-censored estimators. For the spillover-adjusted one it lowers the RMSE instead of the bias. RMSE is the typical size of the error. In the directed form the naive bias is no better than under simple randomization. The pool size and the equal-weight check carry no simple-randomization comparator. So they bear on the design’s tuning, not on that comparison. Three conclusions are not invariant. The spillover-adjusted estimator is only nominal when the assumed kernel is the true one. Nominal means its intervals are right as often as they claim. The naive estimator’s coverage gain is the second. It does not survive redrawing the

candidate space. It also reverses at strong contamination and again under the one-over-distance kernel. The size of the core-zone coverage gain is the third. Under the one-over-distance kernel that gain sits within Monte-Carlo noise. The same holds in one of the eight candidate-space draws. The directed form of the leakage objective is named separately. It is the one axis of the main set on which the conclusion does not carry over intact. A further axis, the unit of allocation, is varied in Table 5. Suppose whole clusters rather than single sites are allocated. Then the two-criterion pool, balance plus leakage, is at or behind uniform cluster randomization. That holds on every bias, coverage and RMSE entry at c = 0.35. The full ten-metric composite stays ahead of it on naive bias and coverage and on boundary-censored bias. Its boundary-censored coverage lead, 0.919 against 0.895, is only two to three Monte-Carlo standard errors. Now set it against comparators drawn at the composite's own resolution. Then its boundary-censored, or core-zone, coverage advantage disappears. Its naive coverage advantage survives only at strong contamination. The last appendix adds one more axis, variation between sites with a site-level analysis. There the design's boundary-censored coverage advantage at site allocation persists. Its bias cut is smaller, and its RMSE gain for the spillover-adjusted estimator shrinks. Three settings were never varied: the spectral bandwidth, the cap of 50 splits per configuration, and the size of the null.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read "invariant" as covering the three untested settings; the paper names them as untested. Do not conclude that invariance of direction means invariance of size; the appendix shows the size of the coverage gain depends on the candidate draw. Do not carry the site-level advantage of the two-criterion design over to whole-cluster allocation either; at that unit its pool is at or behind uniform cluster randomization on every bias, coverage and RMSE entry at c = 0.35.

TRY IT

List the axes the multiverse varies and the three it does not. For one untested axis, say what result you would expect to move if it were varied.

E47 Nine limitations, in the paper's own words

Discussion and Conclusions, the Limitations subsection, its lead sentence and the numbered list of nine limitations.

WHAT THE PASSAGE IS DOING

Bounds what the results license: dependence on the contamination model, adaptive selection and between-pool variability, the core-zone estimand's narrower population, unevaluated randomization inference, simplified testbeds and site-level headline tables, survey sites as proxies, untested design settings, an inert weighting component, and simulated outcomes with no between-site or between-cluster variation beyond binomial sampling.

IN ORDINARY LANGUAGE

The paper lists nine things its results do not establish. Contamination here is infection risk mixing across arms. The design is tuned to the same distance decay rule, or kernel, that creates the leak. Three other kernels are also tested. The featured design won a search, so its rank is not a test. The simulation results depend on which candidate pool was drawn. The pool in Table 1 is the most favorable of the eight on boundary-censored bias, coverage and RMSE. The core-zone estimator is one formula for the effect. It answers a question about the least exposed 70 percent of sites. Under the simulated null, where the treatment has no effect, the naive and core-zone tests reject far too often. The cause is the simulation's contamination rule, the blend. It keeps mixing baseline risk across the arms. The spillover-adjusted test models that mixing. Its false-alarm rate stays at the intended level. The recommended conditional randomization test is not evaluated. On the featured partition a two-sided randomization test cannot reject at 0.05. So the paper names a two-stage cluster-level analysis as the practical primary test at that resolution. The malaria tables and the cholera and COVID testbeds all use a simpler design. It scores only balance and leakage, not the ten-metric suite. The headline malaria tables randomize single sites, not whole clusters. Survey sites stand in for communities. Water nearness is a proxy from elevation. Three design settings were never varied. One weighting part adds nothing. The simulated outcomes carry no variation between sites or clusters beyond chance. Every coverage, Type-I error and power figure outside the last appendix holds under that assumption. That appendix relaxes it. Such variation would make the model-based intervals too narrow.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not treat these as boilerplate. Three of them change how the headline numbers read: between-pool variability, the adjusted estimator's kernel dependence and the absence of variation between sites or clusters. The abstract quotes ranges for the first two and says its figures assume no variation between sites. Do not conclude the list is exhaustive; it is the set the author identified, and several items on it are untested.

TRY IT

Pick the limitation you think a referee would raise first, and find the table or appendix the paper points to for it. Say whether that evidence answers the objection or only bounds it.

E82 Three residual risks and their analysis-stage answers

Discussion and Conclusions, the analytical risk mitigation subsection, its opening paragraph and the three-row table of analytical countermeasures that the paragraph introduces.

WHAT THE PASSAGE IS DOING

This is the only passage that converts the selected design's metric standings into named analysis-stage statistical methods, one countermeasure per residual risk. It also fixes the framework's boundary, stating that the design engine chooses the allocation and does not prescribe the analysis, and that the recommendation of an estimator comes from the operating-characteristics study rather than from the metric profile, covers only the three estimators that study compares, and does not extend to the sigmoid contamination-range analysis, which keeps the boundary-zone outcomes and

against which it was not benchmarked. Its real work is that the three rows are argued asymmetrically: contiguity is declared a non-risk for this configuration and its countermeasure carried for other ones, balance at 1.4 is the largest gain yet still open on unmeasured confounders, and spatial mixing at 0.2 cannot certify robustness to long-range dispersal, because its null is heavy-tailed, and its countermeasure is the very estimator the paper already recommends.

IN ORDINARY LANGUAGE

The chosen layout beat a random layout on all ten design measures. This part of the paper makes a careful point. A better score shrinks a risk. It does not erase it. The design tool picks the map, not the analysis. The paper looks at three estimators, that is, three formulas for the treatment effect. Of those three, it recommends the boundary-censored one. It did not test that choice against a rival analysis. That rival keeps the border villages and uses them to estimate how far contamination reaches. So the recommendation does not cover that rival. Then a table lists three leftover risks with a statistical fix for each. The first risk is the treated arm breaking into separate pieces on the map. Here it does not break. It scores 1.00, one piece, where a random split averages 1.86. So the featured design does not need that fix. But a trial draws its allocation from the 40 best designs, and 37 of those 40 score worse than the random benchmark on at least one metric. So the drawn allocation's own scores decide which fixes apply. The second risk is balance, meaning the arms start out alike on traits we measured. The design's balance gain is 1.4 standard deviations of the random splits, its best on any measure. The balance score works in raw units. So elevation, measured in meters, supplies 99.1% of the squared score for this design. And the gain does not certify balance on the baseline infection rate. But traits we did not measure can still differ. The fix is a set of adjustment methods that lean less on any one model. The third risk is spread over long distances. That gain is only 0.2 standard deviations. It is small for every candidate map, because the random splits it is scored against scatter enormously. So the score cannot certify that the design holds up against long-range spread. The fix is to judge the main result on core sites. Core sites sit far from the border between the arms. Border villages would be studied as a separate group.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> #### DO NOT ACCIDENTALLY CONCLUDE
> Do not read the third row as a solution. Core-zone censoring is the boundary-censored estimator the paper already recommends. The paper reports that this estimator does not reach nominal coverage and stays biased by the residual exposure of the sites it keeps. So listing it here adds no new protection. Do not read the doubly robust row as freedom from SUTVA violation; it targets residual confounding on baseline covariates, not interference between the arms. And do not read the first row as a repair the featured design needs: its treatment arm is already a single piece at 1.00. That countermeasure is carried for allocations that fragment, and the allocation a trial draws from the elite pool may be one of them.

TRY IT

Open supporting/PROPOSED_selected_design_profile_MAP.csv and sort the Z column. Confirm the ordering this paragraph asserts: balance highest at 1.376, wasser next at 1.249, euler third at 1.019, spatial_mixing lowest at 0.183. Then read the euler row's raw_value and null_mean and check them against the b0 figures quoted in the first cell of Table "Analytical countermeasures for the residual risks of the selected configuration."

E63 Why an all-positive profile still leaves operational risk

Discussion, the operational risk mitigation subsection, the two-sentence paragraph that introduces the operational countermeasures table.

WHAT THE PASSAGE IS DOING

Frames the operational countermeasures table as addressing residual risk rather than metric deficits, stating that the risks listed are the ones that persist despite the selected configuration's all-positive standardized profile. It presents the countermeasures conditionally, as something a trialist adopting the framework could implement during a field rollout, not as a validated component of the design procedure.

IN ORDINARY LANGUAGE

The chosen design scored better than random assignment on every metric the paper uses. This paragraph makes a point that is easy to miss. A positive score does not erase a risk. Some risks are only made smaller, and what is left can still cause trouble in a real field campaign. So the table that follows lists practical fixes rather than statistical ones, covering how teams travel, enroll people, and schedule visits. These are suggestions for a future trial team, and the paper does not test whether they work.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read the all-positive profile as evidence that these operational risks were removed, and do not treat the countermeasures as validated: the paper offers them as steps a trialist could take, and no simulation in the paper evaluates them.

TRY IT

Open supporting/PROPOSED_selected_design_profile_MAP.csv and confirm the Z column is positive for every metric. Find the metric with the smallest Z, then read that metric's row in the operational countermeasures table and state, in one sentence, what risk the paper says still remains despite the positive standing.

E64 The contiguity row: one connected treatment arm, and what b0 does not certify

Discussion, the operational risk mitigation subsection, the operational countermeasure table and its Euler Characteristic (Contiguity) row.

WHAT THE PASSAGE IS DOING

Reports the selected K=8 partition's contiguity standing (Z=1.0; b0=1.00 against a null mean of 1.86) as a gain rather than a residual deficit, so the routing countermeasure is carried for configurations with b0>1 rather than applied here. It then bounds the metric: b0 is computed on adjacency of the projected Voronoi cells, not on road networks or terrain, so it does not speak to travel.

IN ORDINARY LANGUAGE

The Euler characteristic here counts b0, the number of separate connected pieces the treatment arm breaks into. The selected design scores 1.00, so its treated sites form one connected block, while random splits average 1.86 pieces. That is a mean, since any single split has a whole number of pieces. One block means no stranded island splits a field team's route. Connectedness is measured on map neighbors, using Voronoi cells, the map areas closest to each site. It is not measured on roads or terrain, so nothing here certifies travel time. The routing countermeasure is kept for designs that do break into pieces.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read a single contiguous treatment arm as evidence that field logistics are solved; the passage says explicitly that adjacency of map cells is not road access or terrain, so travel burden is untouched by this number. Do not read the routing countermeasure as something applied to this design; it is carried for fragmented configurations with b0 above 1, and this table is narrative rather than computed.

TRY IT

Open supporting/PROPOSED_selected_design_profile_MAP.csv and find the row named euler. Compare its raw value and null mean with the b0 figures quoted in this table cell, and note that the row is marked lower better, which is why fewer pieces counts as a gain.

E65 Population balance as a field-recruitment risk

Discussion, the operational risk mitigation subsection, the operational countermeasure table and its Population Balance (Mean Parity) row.

WHAT THE PASSAGE IS DOING

Converts the design's aggregate balance standing into an operational residual risk: balance closer than a typical random cluster split is asserted at the level of arm-wide means, not within every localized region, so baseline demographic imbalance between those regions can survive. The countermeasure cell proposes stratified enrollment quotas and supplementary community-engagement teams directed at underrepresented strata before baseline evaluation. Like the rest of this table, the row is narrative rather than computed.

IN ORDINARY LANGUAGE

Balance here means the two arms start out alike, on average, on four measured background traits. They are the baseline infection rate, elevation, the log of the number of people examined at each site, and a stand-in for closeness to water. The public build derives the water stand-in from elevation rank. The score works in raw units. So elevation, measured in meters, supplies 99.1% of the squared score for this design. And a good score does not certify balance on the baseline infection rate. On that score the design comes closer than a typical random split of clusters. But it does not reach equality. It also does not even out arm size. The control arm holds more than four times as many examined people as the treatment arm. The score also works on averages over the whole map. So two arms can be close overall and still differ inside one small region. This row names that leftover local gap as a risk for the field team. The suggested answer is a recruitment plan with enrollment targets by group. Extra outreach teams would work in under-enrolled areas before baseline, the measurements taken before treatment starts.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read this row as evidence that adaptive recruitment fixes local imbalance. Nothing in the paper simulates or measures enrollment quotas or outreach teams; the manuscript states that this table is narrative rather than computed, so treat the countermeasure as a suggestion for a trialist, not as a tested result. Also do not read the balance standing itself as parity: the row says closer than a typical random split, not equal.

TRY IT

Open supporting/PROPOSED_selected_design_profile_MAP.csv and read the balance row: a raw value of 40.528 against a null mean of 57.179 is the arm-wide standing this row calls closer than a typical random split. Then open supporting/PROPOSED_selected_design_arms_MAP.csv. Its mean_baseline_prevalence column (0.6013 against 0.516) is the arm mean of one of the four covariates that standing scores, and the gap shows that closer than random is not parity. Its examined_total column (2,564 against 11,233) is an arm total, which the metric does not score, since it compares arm means. The file has no covariate means inside a sub-region (sites_stratum_south is only a site count per latitude band), so the within-region imbalance this row warns about is a risk the deposit does not measure.

E66 Phased rollout as the field answer to long-range dispersal risk

Discussion, the subsection on operational risk mitigation, the Spatial Mixing (Long-Range Sensitivity) row of the operational field-countermeasures table, which pairs the residual long-range-dispersal risk with the phased-rollout countermeasure.

WHAT THE PASSAGE IS DOING

Restates the Spatial Mixing residual risk in operational rather than analytical terms, that a small standardized gain against a heavy-tailed null cannot certify robustness to long-range dispersal, so the exposure asymmetry along the arm perimeter may grow once long-range dispersal is assumed and this metric does not bound it, and pairs it with a deployment-order countermeasure: saturate the cross-arm boundary first and defer the insulated core sites to a later phase.

IN ORDINARY LANGUAGE

A trial's two arms are its treated group and its comparison group. The Spatial Mixing metric asks one question. Does the gap in cross-arm exposure grow once you assume infection travels farther? The design's gain on this metric is small. The reason is that the random-allocation yardstick it is scored against is heavy-tailed. That means the yardstick's own values scatter enormously. A score like that cannot certify that the design is robust to long-range spread. So villages on the border between arms may still face uneven exposure. The fix proposed here changes the timing of the rollout, not the map. Field teams would cover the border villages first. That forms a treated band. They would reach the far-off core villages in a later phase.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that boundary-first deployment has been shown to work. The paper states that this table is narrative rather than computed, and no simulation reported here tests a phased rollout or measures what it does to bias or coverage. Also do not read the small standardized gain as evidence that long-range dispersal is harmless. The standardized gain cannot certify robustness, and the raw metric points the other way: the selected design's exposure asymmetry at 10 km is about six times its value at 2 km (Spatial Mixing ratio 6.38), which the Results read as boundary insulation that weakens as the assumed dispersal scale widens.

TRY IT

Open supporting/PROPOSED_selected_design_profile_MAP.csv and find the spatial_mixing row. Compare its null_sd column with its null_mean, and with the same two columns on the balance row. The random-allocation spread on spatial mixing dwarfs its center, which is the heavy tail the passage blames for a standardized gain too small to certify robustness to long-range dispersal. Then read the same row's raw_value, 6.3812, the ratio the Results read as boundary insulation that weakens as the assumed dispersal scale widens.

E83 The conflict-of-interest declaration and its narrow reach

Back matter, the Conflict of Interest section, its heading and the single declarative sentence that follows it.

WHAT THE PASSAGE IS DOING

Carries the paper's required disclosure of competing interests, and carries it in the narrowest available form: the sole author disclaims financial interests and names no other class. It speaks to the author's incentives rather than to data, method, or result, as the paper's separate Funding declaration also does. That places it in how a reader appraises the reporting, not in the evidence for any claim the paper makes.

IN ORDINARY LANGUAGE

A conflict of interest is a stake that could sway how a study is run or reported. Money is the most common kind. A drug maker paying for a trial is the classic case. Here the paper's only author declares no such financial stake. That is a claim about the author, not about the data. Note the exact word used. The claim covers financial interests. It does not name other kinds. Other kinds include job pressure, a stake in a method's success, or unpaid advisory roles. Journals ask for this line so readers can judge for themselves. The line tells the reader what to weigh, not whether the work is right. Nothing in this line touches the math or the code.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read a clean declaration as support for the findings. A disclosure states the author's stake; it says nothing about whether the coverage, bias, or ranking results hold. Those still rest on the simulation evidence, and the design still does not restore nominal coverage for the naive or boundary-censored estimator. Do not stretch the line past its own wording either. It disclaims financial interests, so intellectual investment in the method, career incentives, and unpaid roles sit outside what it covers. And do not treat it as the funding statement; the paper makes that declaration separately.

TRY IT

Open 01_manuscript_publicdata.tex and grep for "competing" and then for "Funding". You will find this two-line Conflict of Interest section just before the Reproducibility and Data Availability section, and a separate two-line Funding section a few sections later, right after Ethics. Read both, then say in one sentence which question each one answers and which question neither one answers.

E67 The reproducibility statement: where every number comes from

Back matter, the "Reproducibility and Data Availability" section, its single paragraph tracing each dataset, figure, and table to a deposited file.

WHAT THE PASSAGE IS DOING

Fixes the provenance of the whole paper. It names each public data source and the retrieval date of the malaria snapshot, states that cached inputs are deposited so results regenerate without a live download once the CCR_CACHE environment variable points to them, except the directed-connectedness map, whose script downloads its inputs live and so ships pre-rendered, explains the archive's PROPOSED_ filename convention, then assigns every data-derived figure and numerical table to the script or CSV that produces it, flagging the one table with inline values and the items that are narrative or hand-drawn. It closes with the repository DOI and the three supplements.

IN ORDINARY LANGUAGE

This section is the paper's receipt for its own numbers. It names the source and date of each public data set. It says the malaria survey points were pulled on 5 September 2026. It then says where every figure and table comes from. Most are made by a deposited script or copied from a deposited results file. One table takes its coordinates from a list inside a script. One schematic is drawn by hand. Two tables and one appendix are narrative. Narrative here means written prose, not computed output. Deposited scripts and results files begin with PROPOSED_. The deposited data files, such as the cached survey points and the frozen sites table, do not. The paper says that prefix is the archive's own file name, not a sign of draft status. Everything sits in one public archive with a permanent link. Three supplements go with it. They hold checklists, a container recipe with a one-command rerun script, and the settings needed to rebuild the pipeline. The paper says the container image itself has not been test-built. The logged rerun from a clean copy ran directly on the computer, not inside the container, and it regenerates and checks the results behind the site-level operating-characteristics table.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read this as proof that the results have been independently reproduced. A data-availability statement is a promise about where the files are and what made each number, not evidence that anyone outside the author reran them. Do not treat the permanent repository link or the reporting checklists as external verification of the results either. And do not upgrade the narrative tables or the hand-drawn schematic to computed output: the passage explicitly excludes them from the generated set.

TRY IT

Open supporting/PROPOSED_third_trial_cholera.R and find the inline block of governorate names with latitude and longitude. The passage says the cholera geography table's coordinates come from there and its p0 column is the deposited attack rate divided by 1,000. Pick one governorate and check both claims yourself against the table in the paper.

E85 Ethics, funding, and AI-use declarations

Back matter, the closing declaration sections that follow the Reproducibility and Data Availability statement and precede the bibliography: Ethics, then Funding, then Acknowledgments, then Use of AI-assisted technology.

WHAT THE PASSAGE IS DOING

These are the paper's regulatory declarations, and, except for the account of AI use that the Use of AI Tools subsection at the close of the Estimand and Operating Characteristics section also gives, they do work no methods or results passage can do. The ethics paragraph explains why a study built on disease surveillance data needed no ethics-board review: every input is public, aggregate, and non-identifiable, and no one was recruited or newly surveyed. The AI paragraph repeats that subsection's account of who or what wrote and executed the analysis code, and it fixes accountability by naming each assistant, by asserting that no data were generated or altered, and by ruling the tools out as authors under COPE and ICMJE criteria.

IN ORDINARY LANGUAGE

These closing sections are the paper's compliance declarations. The first covers ethics. The study used only data that anyone can download. The data are malaria survey points, cholera case counts, and COVID-19 county counts. They are aggregate, meaning counted up by place, not by person. Nobody was recruited. No new data were collected. Nothing in the files can name a person. Human-subjects research means research on living people or their private records. That kind of work needs an ethics board to review it first. This study does not count as that kind of work. So no review was required, and none was sought. The next section covers money. No grant paid for this study. The last section covers AI. The author used AI assistants for this work. The declaration names each assistant and when it was used. They helped search the literature, drafted parts of the text and reviewed it. One of them, Claude, wrote and ran the analysis code. The author states that the tools made up no data and changed none. Each run uses a fixed random seed, so the same code returns the same numbers. The author directed the work and checked every result. The author takes full responsibility for it. The tools are not authors and do not qualify as authors.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read "no ethical review or approval was required" as a board having examined and cleared this study. No board looked at it; the author is asserting that the exemption conditions for secondary use of public, aggregate, non-identifiable data are met. Do not read the absence of human subjects as an absence of people: the parasite-rate survey points and the cholera and COVID-19 counts are real measurements from real populations, released only in aggregate form. And do not read the AI declaration as a quality claim in either direction. It does not move responsibility away from the author, and "fully scripted and reproducible from openly-available data, with fixed random seeds" describes a property of the code, not a report that anyone outside the author reran it.

TRY IT

Open 01_manuscript_publicdata.tex and read the Ethics section against the Reproducibility and Data Availability section a few paragraphs above it. Write down the data sources each one names. Ethics names the Malaria Atlas Project parasite-rate points, the Humanitarian Data Exchange cholera series, and Johns Hopkins CSSE COVID-19 county data for Louisiana; check whether that list matches the sources the reproducibility statement traces every figure and table back to.

E68 The dual-mitigation appendix: why a winner still carries risk

Appendix "Comprehensive dual-mitigation framework for sub-optimal dimensions," its opening paragraph explaining why the appendix exists and when to use it.

WHAT THE PASSAGE IS DOING

Frames the appendix by restating the consequence of a compensatory weighted composite: because ten competing objectives collapse to one score, the top-ranked configuration is a compromise, and a positive standardized gain on a metric bounds the associated risk without eliminating it. It then sets the trigger for the guide that follows, a sub-optimal standing such as a negative standardized gain on any of the ten dimensions, and instructs that the analytical and operational protocols be

applied together.

IN ORDINARY LANGUAGE

The engine scores each candidate design on ten measures, then adds them into one weighted total. That total is compensatory, meaning a strong score on one measure can cover a weak score on another. So the chosen layout is a compromise, not a design that is best on everything. A positive standardized gain, meaning the design beats random layouts on that measure, limits the matching risk but does not erase it. This appendix therefore covers all ten measures, and says what to do when a design scores poorly on one. For each, it pairs an analysis fix with a field-operations fix, and asks that both be used together.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

> #### DO NOT ACCIDENTALLY CONCLUDE
> Do not conclude that the listed countermeasures are validated, or that applying them restores the design's operating characteristics; the paper states elsewhere that this appendix is narrative rather than computed, and nothing here is simulated or measured. Do not conclude that a metric with a positive standardized gain needs no attention, since the passage says the risk is bounded, not removed.

TRY IT

Open supporting/PROPOSED_selected_design_profile_MAP.csv and read the Z column, one row per metric for the selected design. Note which metrics have a Z at or below zero, then find the matching numbered item in this appendix and read the paired analytical and operational entries. Then check whether the paper anywhere reports operating characteristics for a design after those countermeasures are applied.

E86 Three balance failures, and their paired repairs

Appendix "Comprehensive dual-mitigation framework for sub-optimal dimensions," the subsection headed Pillar 1: statistical interchangeability, covering its numbered items on population balance, Wasserstein distance, and density symmetry.

WHAT THE PASSAGE IS DOING

Opens the appendix's per-metric guide with the interchangeability dimensions, and in doing so sets the template every later item follows: a named sub-optimal condition, an analytical repair, and an operational repair. Its distinctive work is to split what a reader would loosely call balance into three separate failures, arms that differ in average level, arms that match on averages but differ in the shape of their distributions, and arms that differ in how tightly they are packed on the map. It is also where the paper names concrete estimation machinery for a poor balance standing, and the density item states its own limit, that regional strata absorb between-region confounding while within-region confounding remains.

IN ORDINARY LANGUAGE

A trial's two arms are its treated group and its comparison group. This part of the appendix covers three ways the arms can start out unlike each other. The first is a gap in averages, such as population size or the local malaria rate. The suggested analysis fix is a doubly robust method. That means the estimate stays sound if either the model for who got treated or the model for the outcome is right. The field fix is to set enrollment targets and send outreach teams to under-enrolled groups. The second way is a gap in shape. Two arms can share a mean and still differ in spread or in how lopsided the values are. The fix here compares arms across the whole range of outcomes, not just at the middle. Field teams would track the extreme sites so they do not drop out. The third way is a gap in packing. One arm sits in a dense zone; the other is spread out. The fix is to add region labels to the model, so region-wide gaps are soaked up. The paper says plainly that gaps inside a region are not. In the field, the spread-out arm would get more vehicles and staff, so both arms are reached at the same speed. The paper offers all of this as advice for a future trial team, not as tested results.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude that these repairs restore interchangeability. Doubly robust estimation and quantile methods adjust for measured baseline traits only, unmeasured ones survive, and the passage itself says that stratum fixed effects leave within-region confounding in place. Nothing in this section is simulated or measured anywhere in the paper. Also do not read these as contamination remedies: they address how alike the arms start out, not leakage of treatment between the arms, and they do not deliver freedom from SUTVA violation.

TRY IT

Open supporting/PROPOSED_selected_design_profile_MAP.csv and read the Z column for the three rows this passage governs: balance (Z = 1.376), wasser (1.249), and density_symmetry (0.943). All three are positive, while the appendix's stated trigger is a sub-optimal standing such as a negative Z. Confirm that none of the three conditions described here is actually raised by the selected design, then check whether the paper reports operating characteristics for any design after these countermeasures are applied.

E87 Pillar 2 countermeasures for fragmentation, terrain and boundary fit

Appendix "Comprehensive dual-mitigation framework for sub-optimal dimensions," the subsection headed Pillar 2 on topological and spatial cohesion, running from the Euler Characteristic (Contiguity) item through the Variational Free Energy item.

WHAT THE PASSAGE IS DOING

Extends the dual-mitigation guide to all four Pillar 2 dimensions, pairing a named analytical remedy with a named field remedy under an explicit sub-optimal condition for each. Two of these dimensions, Persistence Gap and Variational Free Energy, appear nowhere in the Discussion's countermeasure tables, so this is the only place the paper says what to do when the arms straddle mismatched terrain or when cluster borders cut across the transmission surface. Three of the four analytical remedies are modeling structures: CAR or Matern covariance for fragmented clusters, thin-plate terrain splines for mismatched topography, and GEE with a spatial working correlation

or a SAR lag model for residual spatial dependence. The fourth is not: item 6 prescribes a "fried-egg" spatial censoring design that evaluates the primary ITT endpoint only on core villages shielded from the boundary and demotes perimeter communities to a secondary cohort, so it changes which villages the primary analysis is evaluated on rather than how the outcome model is specified.

IN ORDINARY LANGUAGE

This part of the appendix deals with the shape of a design. It asks whether the treated arm forms one connected block. An arm is the treated group or the control group. It asks whether the two arms sit on similar land. It asks whether the exposure gap between the arms grows when the disease is assumed to spread farther. It asks whether cluster borders match the way disease spreads. For each of those four checks, the text says what to do when a design scores badly. Each item gets two fixes. One fix changes the analysis. The other changes the field work. For clusters broken into islands, the analysis fix is a spatial covariance model. That is a model in which nearby sites are allowed to be more alike than far ones. A single fixed within-cluster correlation can make the estimate look more precise than it is. The field fix is to route trucks along shared roads and restock at central hubs. For arms on different land, such as ridges against valleys, the fix is a smooth terrain curve added to the outcome model. The field fix is off-road vehicles and solar cold storage for the hard clusters. For rising cross-arm exposure, the fix is called fried-egg censoring. You judge the main result on core villages only. Core villages are the ones far from the border. Border villages get a separate look. The field fix is to treat the border villages first. For borders that cut across the spread pattern, the fix is a model that allows leftover spatial correlation. The field fix is to place clinics along the steepest spread gradients.

EVIDENCE STATUS **INTERPRETIVE** the author's reading of the evidence

> **DO NOT ACCIDENTALLY CONCLUDE**
>
> Do not conclude that fried-egg core-zone censoring yields a clean, spillover-free primary result. Restricting the endpoint to core sites is the same boundary censoring the operating-characteristics study simulates, and the paper reports that the design reduces that estimator's bias without restoring nominal coverage. Do not conclude either that the spatial covariance and autoregressive models listed here remove interference. They model residual dependence among outcomes, which is not the same as removing treatment effect that crosses an arm boundary, and none of these modeling remedies, the covariance, spline and autoregressive structures, is simulated anywhere in the paper.

TRY IT

Open supporting/PROPOSED_selected_design_profile_MAP.csv and find the rows named euler, persistence_gap, spatial_mixing and vfe. Those four rows are exactly the four numbered items of this subsection. Read their Z column: the values are 1.019, 0.672, 0.183 and 0.998, all positive, so none of these dimensions meets the appendix's example trigger of a negative standardized gain. That does not retire every item: the Discussion still carries Spatial Mixing as a residual risk of the selected design, because its small gain cannot certify robustness to long-range dispersal, and assigns it the same core-zone censoring as item 6. Then scan the two countermeasure tables in the Discussion and confirm that persistence_gap and vfe appear in neither, which is precisely what this appendix subsection adds.

E88 Countermeasures for vector, frontier and mobility leakage

Appendix “Comprehensive dual-mitigation framework for sub-optimal dimensions,” the Pillar 3 subsection on contamination resilience, the three numbered entries for Contamination AUC, Frontier Risk and Mobility Network Leakage.

WHAT THE PASSAGE IS DOING

Closes the dual-mitigation guide with the three dimensions that measure leakage between arms: mosquito dispersal, disease burden sitting on the border, and human commuting. Each gets a named failure condition, an analysis-side remedy, and a field-operations remedy, so a layout that scores poorly on cross-arm exposure has a stated response and not only a diagnosis. The Methods define these metrics and the Discussion reports the selected design’s standing on them; this is the only passage that prescribes what to do about them.

IN ORDINARY LANGUAGE

This part of the appendix covers the three measures of leakage between arms. An arm is a group of villages, either treated or not treated. Leakage means infection risk crossing from one arm to the other. Each item names a warning sign, then an analysis fix, then a field fix. Item 8 is about mosquitoes. If the two arms sit close together, mosquitoes can carry infection across. The analysis fix builds a table of spillover weights between villages, larger for closer pairs, and folds it into the effect estimate. The field fix is to set traps on the border and watch the insect traffic. Item 9 is about people who live on the border. If many sick people live right on the line, the trial is exposed. The analysis fix compares outcomes just inside and just outside the line. The field fix sends more case-finding staff to border villages. Item 10 is about travel. If people commute between the two arms, the treated and untreated groups blur. The analysis fix weights each person’s outcome by time spent outside their assigned zone. The field fix puts extra outposts at markets, bus depots and river crossings. For each item the paper asks that both fixes be used together.

EVIDENCE STATUS **INTERPRETIVE** the author’s reading of the evidence

DO NOT ACCIDENTALLY CONCLUDE

Do not read item 8’s distance-weighted spillover matrix or item 10’s time-weighted per-protocol analysis as repairs that remove the leakage problem. An estimator that adjusts for spillover inherits the assumed mobility kernel, so a wrong kernel gives a wrong adjustment: item 8’s remedy is the paper’s own spillover-adjusted estimator, which is unbiased with near-nominal coverage under the assumed inverse-square gravity kernel in simulations with no variation between sites beyond sampling chance (once such variation is added, the paper’s site-level analysis keeps its coverage between 0.920 and 0.947), but whose coverage falls to 0.51 to 0.89 when the contamination is generated by a kernel the analysis does not assume. The item 9 and item 10 remedies are not simulated or evaluated anywhere in this paper. Do not treat the per-protocol reweighting as equivalent to the randomized comparison; it conditions on behavior after assignment. And do not read this pillar as a route to freedom from interference between units: the design and these remedies reduce cross-arm exposure and residual bias, they do not eliminate either.

TRY IT

Open supporting/PROPOSED_selected_design_profile_MAP.csv and read the Z column for the three Pillar 3 rows: contam_auc, frontier_auc_exposure and mobility_auc_leakage. All three are positive but well under 1, far smaller than the balance or wasser rows, so by the appendix's own trigger (a negative Z) none of these three items fires for the selected design, even though it is on exactly these dimensions that the design claims its purpose.

E69 What the two testbeds hold fixed, and what they leave out

Appendix "Generalization testbeds: specifications and results," its opening paragraph setting the shared ground rules before either the cholera or COVID testbed is described.

WHAT THE PASSAGE IS DOING

States the appendix's scope: both testbeds reuse the estimand, data-generating model and three estimators of the malaria operating-characteristics simulation, varying only geography, outcome denominator, gravity masses and contamination kernel. It then narrows the claim by stating that neither testbed applies the ten-metric suite or the candidate-generation engine, that their contamination-resilient design is the simpler two-criterion balance-plus-leakage elite pool, and it fixes the replicate counts, seed, effect size and the three design labels.

IN ORDINARY LANGUAGE

This paragraph is the rulebook for the two extra testbeds. Most of the setup stays fixed. That covers the target quantity, the recipe for making fake outcomes, and the three analysis rules. Only the map, the population weights, the survey size, and the leakage formula change. Leakage here means infection risk mixing across the boundary between the two arms. Each simulated outcome comes from a made-up survey of 800 people per unit. It is not a real count. Neither testbed uses the ten-metric scoring suite. Neither uses the cluster-building engine from the Methods. Instead they keep allocations that rank well on both arm balance and leakage. Each result averages one thousand simulated trials at a fixed seed. The COVID testbed averages two thousand five hundred. The true effect is log 0.5.

EVIDENCE STATUS **SCOPE** states a boundary of applicability

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read the two testbeds as evidence that the paper's ten-metric suite or its candidate-generation engine transfers to new geographies. The paragraph says plainly that neither is used here, so whatever the cholera and COVID tables show supports only the simpler two-criterion design. Also do not treat the per-unit denominator of 800 as a real sample size; it is a stated hypothetical, so absolute precision and power in these testbeds are illustrative.

TRY IT

Open supporting/PROPOSED_third_trial_cholera.R, the script this paragraph names. Near the top you will find a single fixed seed call and a line setting the per-governorate sample to 800, flagged in the comment as hypothetical. Seeing both in the code confirms that the denominator is an assumption of the simulation and that the reported numbers are reproducible rather than measured.

E37 Yemen cholera: real map, simulated trial

Appendix “Generalization testbeds: specifications and results”, the Yemen cholera subsection, the “Data and geography” and “Contamination kernel and design” paragraphs together with the governorate centroid table.

WHAT THE PASSAGE IS DOING

Specifies exactly which parts of the cholera testbed are real data and which are simulated, and defines the gravity contamination kernel and the design pool used on this geography.

IN ORDINARY LANGUAGE

Almost everything about the setting is real and dated. Twenty Yemeni governorates take part. Governorates are provinces. Each enters at its centroid, its center point. Twenty-two units reported at the outbreak’s end. The two without usable map points are dropped. Each governorate keeps its real attack proportion, cumulative over the outbreak. That is the share of its people reported ill by 18 February 2018. The counts come from suspected-case reports by WHO Yemen and the health ministry, shared through the Humanitarian Data Exchange. Populations are worked back from cases and attack rates. So they are a relative stand-in, not a census. Contamination means infection risk mixing across a border. It follows a gravity model, which represents people traveling; shared water systems are not modeled. Flow grows when both places have more people. It shrinks with the square of the distance. The trial itself is invented. The water, sanitation, and vaccine program and its effect are only simulated.

EVIDENCE STATUS **MEASURED DATED** a real measurement from dated real data

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read the tables in this subsection as outcomes of a real cholera trial. No such trial happened; every bias, coverage, and power figure is a simulation output laid over the real geography. Also, suspected-case surveillance is not confirmed infection, so the baseline attack proportions are reported burden, not true burden.

TRY IT

From the governorate table, divide the largest baseline attack proportion (Amran, 0.090) by the smallest (Shabwah, 0.002). The spread you compute is the real baseline heterogeneity the randomization has to handle.

E70 The cholera testbed’s contamination kernel and design pool

Appendix “Generalization testbeds,” the Yemen cholera subsection, the “Contamination kernel and design” paragraph between the data-and-geography paragraph and the governorate centroid table.

WHAT THE PASSAGE IS DOING

Specifies the machinery of the cholera testbed: an inverse-square gravity kernel on back-computed governorate populations with a 5 km distance floor, and a design pool of six thousand latitude-blocked allocations from which the covariate-constrained and contamination-resilient arms are selected by a balance score and a combined balance-plus-leakage rank. It fixes the friction exponent at a single value rather than integrating over the range used by the ten-metric suite.

IN ORDINARY LANGUAGE

Cholera spreads through people and shared water, not mosquitoes. The model here represents only the human-movement route; shared water systems are not modeled. A gravity model sets flow between two areas to rise with their populations and fall with squared distance. Distances follow the earth's curve, and any pair under 5 km counts as 5 km apart. The distance power is fixed at 2, while the main metric suite sweeps a range of powers. Designs come from six thousand allocations, split north and south of the median latitude. Balance compares treatment and control averages on two baseline traits; the resilient pool keeps 40 designs, the balance-only pool the top quarter.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read the exponent of 2 as a measured travel friction for Yemen. It is one assumed setting, and because this testbed fixes it instead of sweeping it, the cholera results are not averaged over mobility uncertainty the way the ten-metric leakage AUC is. Also do not treat the populations entering the kernel as census counts: they are back-computed from reported cases and attack rates, so the flow scores are relative, not absolute.

TRY IT

Open supporting/PROPOSED_third_trial_cholera.R and check four lines against this paragraph: the two-stratum split at the median latitude, the 5 km distance floor, the gravity matrix built with a squared distance in the denominator, and the pool line that draws 6,000 allocations and then selects the constrained quartile and the 40-design elite set. Confirm for yourself that the exponent appears as one fixed value and not as a loop over a range.

E38 Cholera operating characteristics

Appendix "Generalization testbeds: specifications and results", the Yemen cholera subsection, its two operating-characteristics tables and the "Findings" paragraph that reads them.

WHAT THE PASSAGE IS DOING

Reports the cholera testbed's operating characteristics at moderate contamination and across the contamination sweep. The contamination-resilient design reduces bias and lifts coverage on the two estimators contamination biases, naive and boundary-censored; at the spillover-adjusted estimator, which is near nominal under every design, its gain is precision instead, RMSE 1.041 to 0.480. Covariate-constrained randomization does not improve on simple randomization at those two estimators, though its balance does cut spillover-adjusted RMSE from 1.041 to 0.876.

IN ORDINARY LANGUAGE

These tables show operating characteristics. That is how each design behaves over many replayed trials. Here that means one thousand simulated trials at one fixed random seed. The seed is fixed, so anyone can rerun them and get the same numbers. Bias is the average gap between the estimate and the true effect. Coverage is how often the confidence interval catches the truth. It should sit near the rate it claims. Take moderate contamination, c equals 0.35. There the resilient design cuts boundary-censored bias from 0.278 to 0.216. The boundary-censored estimator first drops the 30 percent of governorates most exposed to the other arm. Its coverage rises from 0.506 to 0.648. Covariates are traits measured before the trial. Balancing them alone does not help the two estimators that contamination biases. It does make the spillover-adjusted estimate more precise, though. On the naive estimator, the balance-only design's coverage is in fact lower, 0.143 against 0.211. With no contamination, the boundary-censored estimator has near-nominal coverage under every design. That coverage is 0.947 to 0.958. A separate deposited run at that level puts its bias, and the naive estimator's, within 0.005 of zero. The resilient design's boundary-censored coverage lead grows as contamination gets stronger. It reaches 0.142 at c equals 0.35. Then it narrows at c equals 0.5, where inference is poor for every design.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> #### DO NOT ACCIDENTALLY CONCLUDE
> Do not conclude that the design restores valid inference under contamination. Coverage of 0.648 is still far below the advertised rate, so bias is reduced, not eliminated. Also do not take the naive test at face value: at c equals 0.35 it shows mild size inflation, a Type-I error of about 0.08 to 0.11, which the boundary-censored and adjusted estimators bring close to nominal. The paper traces that inflation to the contamination blend and not to the small number of clusters. Even when the true effect is zero, the blend keeps pulling each governorate's risk toward the baseline risk of its neighbors in the other arm, and at c equals 0, where there is no blend, the same twenty governorates give a naive Type-I error of 0.044 to 0.052. The paper also does not interpret the naive-size difference between designs, 0.113 against 0.079.

TRY IT

In the sweep table, note that at c equals 0 the resilient design's coverage (0.958) and simple randomization's (0.950) are both near nominal and within Monte-Carlo noise of each other. Then, using the Monte-Carlo standard error of each difference at 1,000 replicates, find the lowest contamination level at which the resilient design's lead over simple randomization reaches three standard errors of the difference, the paper's bar for calling a difference established.

E39 The cholera Type-I caveat, traced to the contamination blend

Appendix, the Yemen cholera testbed subsection, the closing 'Findings' paragraph, where the naive test's mild size inflation is discussed and traced to the contamination blend.

WHAT THE PASSAGE IS DOING

Flags a caveat on the naive test: its model-based Type-I error is mildly inflated at moderate contamination, traces that inflation to the contamination blend by showing that the same twenty governorates give near-nominal size at c = 0, and uses it to motivate randomization inference as the

primary test.

IN ORDINARY LANGUAGE

Type-I error is the false-positive rate when the true effect is zero. The tables aim for 0.05. That target is called the nominal level. This testbed has 20 clusters. In the run at moderate contamination, c = 0.35, the naive model-based test lands near 0.08 to 0.11. Model-based means the test trusts the standard errors from a fitted regression. The paper traces this inflation to the contamination blend, and not to the small number of clusters. The blend is the simulation's rule that pulls each place's infection risk toward that of its neighbors in the other arm. Under the simulated null the true effect is zero, but the blend is still active, so it still mixes baseline risk across the arms. With no contamination, c = 0, the same twenty governorates give a naive Type-I error of 0.044 to 0.052, close to nominal. So the inflation is the blend's. The boundary-censored and spillover-adjusted estimators bring the size back close to nominal. This is why the paper recommends randomization inference for its main test. The paper does not run that test here. In its plain form, that means you shuffle the treatment labels many times and rank the real result among the shuffles. The paper warns that this plain form is not automatically valid under the simulated null, and points to conditional versions built for interference.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> #### DO NOT ACCIDENTALLY CONCLUDE
> Do not count this inflation against the contamination-resilient design. It appears under every design, at Type-I of roughly 0.08 to 0.11, and the paper traces it to the contamination blend, since the same governorates give a naive Type-I error of 0.044 to 0.052 at c = 0. Do not push that into a claim that the design leaves size untouched: the paper declines to interpret the naive-size difference across designs, 0.113 against 0.079. And do not read randomization inference as the only remedy. Here the boundary-censored and adjusted estimators already bring size close to nominal; randomization inference is what the paper recommends for the primary treatment-effect test.

TRY IT

In Table 9, compare the naive Type-I entries (0.113, 0.095, 0.079) with the core-zone entries (0.066, 0.060, 0.059). Notice that under the core-zone estimator every design sits between 0.059 and 0.066, while the naive entries still differ by design; the paper does not interpret that difference across designs.

E40 COVID Louisiana: directed geometry, confirmed cases

Appendix "Generalization testbeds: specifications and results", the Louisiana COVID-19 subsection, the "Data, geography and kernel" paragraph.

WHAT THE PASSAGE IS DOING

Specifies the directional testbed: real parish geography, populations, and confirmed-case attack proportions; a simulated intervention and survey; a directed gravity kernel with its symmetric counterpart; and four competing designs whose leakage diagnostics are tabulated.

IN ORDINARY LANGUAGE

The map is real. It covers 64 Louisiana parishes. For each one the author uses its true center, its 2019 estimated population from the U.S. Census Bureau, and its total confirmed COVID-19 cases through 15 April 2020. The attack proportion is cases divided by population. It runs from 0.0001 to 0.0148. Confirmed is the key word. Testing in early 2020 was scarce, and it was aimed at cities. So these counts mix real infection with access to testing. The trial itself is simulated. The author made up a parish-level program. They also made up a survey of 800 people per parish. Contamination flows through a directed gravity kernel. A kernel is just the formula that gives flow from one place to another. Directed means that flow from i to j need not match flow from j to i. Production is the sender's infected mass. Attraction is the receiver's mass of people still at risk. On this surface, Orleans parish is the largest net exporter. St. Tammany is the largest net importer. The paper calls that a description of reported cases. It is not a reconstruction of true spread. All 64 parishes had at least one confirmed case. So no parish is dropped. No continuity correction is used. Four designs then compete. Cross-arm flow is leakage that crosses between the treated group and the control group. One design aims at the total of that flow. One aims at the net imbalance in its direction. An allocation is one way of splitting the parishes into the two arms. Each resilient pool holds the 40 allocations that rank best on balance and on its own flow measure. The contamination itself comes from the directed kernel. So the symmetric design targets the total of that flow. The directed design targets its net direction.

EVIDENCE STATUS **MEASURED DATED** a real measurement from dated real data

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not conclude the analysis showed how COVID actually spread through Louisiana. The source and sink labels are computed from confirmed-case counts under uneven testing, so they describe the reported-incidence surface only. Also do not quote absolute power from this testbed: the survey size is illustrative, many parishes carry few events, and the paper restricts its conclusions to relative design behavior.

TRY IT

Production for a parish is its attack proportion times its population. Ask yourself: if early-2020 testing had been uniform across the state, which parishes' production terms would rise, and could the net-source ranking change?

E89 Reading the Louisiana operating-characteristics grid

Appendix "Generalization testbeds: specifications and results", the Louisiana COVID-19 subsection, the moderate-contamination operating-characteristics table and its caption, sitting between the data-geography-and-kernel paragraph and the naive-ITT coverage sweep table.

WHAT THE PASSAGE IS DOING

Presents the whole four-design by three-estimator grid for the Louisiana geography at one contamination setting, so the symmetric and directed contamination objectives can be compared row against row on identical columns. It is the only place in the paper where those two objectives meet across all three estimators and all five operating characteristics at once; the sweep table that follows shows only naive coverage. Its caption also fixes the reading rule, giving a coarser

Monte-Carlo noise level on bias for the spillover-adjusted rows than for the naive and core-zone rows, with coverage noise near 0.01 throughout, and deferring model-based size and the case for randomization-based primary inference to the Findings.

IN ORDINARY LANGUAGE

This table is a scorecard. Each row pairs one way of analyzing the trial with one way of assigning the arms. Each column scores that pair over many repeat runs, called replicates. There are two thousand five hundred runs behind every number. The map is real Louisiana geography. The trial and its effect are simulated. The true effect is set at log 0.5. Leakage is held at a moderate level, c equals 0.35. Leakage means infection risk mixing across the line between the two arms. Now the columns. Bias is the average gap between the estimate and the truth. RMSE is the typical size of the error. Coverage is how often the interval holds the truth. An interval is a range of plausible values. Coverage should sit near the rate it claims. Type-I is how often the test flags an effect that is not there. Power is how often it finds one that is. Now the rows. The three blocks are three ways to analyze. The naive way ignores leakage. The core-zone way first drops the units near the edges. The adjusted way models the leakage instead. Each block holds the same four ways to assign arms. A plain random draw comes first. Then a draw balanced on baseline traits. Then two draws that also aim at leakage. The bold rows target total leakage. The last rows target only its net direction, and they end up with more total leakage than a plain random draw. The caption gives the noise in each number. It is about 0.01 on coverage. It is about 0.008 on bias in the top two blocks. It is 0.02 to 0.03 on bias in the adjusted block. So a small bias gap in the adjusted rows may be noise. The same bias gap higher up may be real. On coverage the noise is about 0.01 in every block. The top two blocks never reach the claimed rate. The best naive value is 0.663. The best core-zone value is 0.845. The bold design nudges both up, the naive value from 0.627 and the core-zone value from 0.816 under a plain random draw. Each nudge is only about 2.7 times the noise in the difference, so the paper places it at the two-MCSE threshold, not established. Neither block is fixed. The adjusted block sits near the claimed rate under every design. Its coverage runs 0.947 to 0.956. So coverage there is not a design win. There the design shows up as lower error, 1.487 down to 1.136.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
>
> Do not read the adjusted block's near-nominal coverage, 0.947 to 0.956, as this design's achievement or as a working fix for contamination. That block sits near the claimed rate under simple randomization too, so coverage there is not a design effect; it is bought at an error size no smaller than 1.136 and power of only 0.085 to 0.101, and it leans on the mobility kernel that generated the contamination being roughly right and on the simulation having no variation between parishes beyond sampling chance. Do not read the naive and core-zone blocks as repaired either: their best coverage anywhere in the table is 0.663 and 0.845, short of the claimed rate under every design. And do not rank designs down the Power column, because realized Type-I differs across rows, from 0.050 to 0.104 in the naive block alone, so higher power can mean a looser test rather than a better design.

TRY IT

In Table 11, subtract the bold symmetric row's naive bias from the naive simple-randomization row: 0.296 minus 0.272 is 0.024, about three times the caption's bias noise of 0.008 for that block. Now do the same inside the adjusted block: 0.011 minus negative 0.031 is 0.042, against a bias noise of 0.02 to 0.03 there. Then say why the paper reads the design's adjusted-block gain as an RMSE reduction rather than a bias reduction.

E41 The c = 0 calibration check

Appendix, the Louisiana COVID-19 testbed subsection, the naive-ITT coverage sweep table and the first 'Findings' paragraph that follows it.

WHAT THE PASSAGE IS DOING

Uses the zero-contamination column as a calibration check: at c = 0 naive model-based coverage is near nominal under every design, about 0.94 to 0.96, and a separate deposited run at c = 0 puts every design and estimator at coverage 0.946 to 0.958, with naive and boundary-censored bias within 0.024 of zero. It names the modeling condition that check rests on, that the baseline covariate enters on the log-odds scale so the fixed-effects binomial GLM is correctly specified, and it recommends randomization inference, not a naive permutation test, as the primary treatment-effect test.

IN ORDINARY LANGUAGE

Setting c to 0 means the simulation injects no contamination at all. It works as a calibration check. With nothing leaking, a sound design and model should hit the advertised coverage. They do. Naive coverage sits near 0.94 to 0.96 under every design. A separate run at c equals 0 puts every design and estimator at coverage 0.946 to 0.958. It puts their Type-I error at 0.041 to 0.055. Type-I error is the false-alarm rate when there is no effect. In that run, the naive estimator and the boundary-censored one are within 0.024 of zero bias. The boundary-censored estimator drops the most exposed sites. But that requires one modeling choice. The baseline attack proportion must enter the regression on the log-odds scale, meaning the log of p over 1 minus p. Otherwise the fixed-effects binomial model is misspecified. Under contamination the naive test's size inflates only mildly here, up to about 0.10. The paper still recommends randomization inference, not a naive permutation test, as the primary test. It does not evaluate it here.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> #### DO NOT ACCIDENTALLY CONCLUDE
>
> Do not conclude the naive model-based test is safe as the primary analysis because the c equals 0 column looks nominal. That column shows the machinery is calibrated under the simulation's binomial model, where at c equals 0 the analysis model is the data-generating model; it does not show that the test holds its size under contamination, or when outcomes vary between parishes beyond binomial sampling. Under contamination the naive test's Type-I error reaches about 0.10, which is why the paper recommends randomization inference for the primary test; separately, naive coverage falls for every design as contamination grows, because the naive estimate is pulled toward zero.

TRY IT

Read the simple-randomization row of Table 12 from 0.955 at c equals 0 down to 0.396 at c equals 0.50. Confirm the collapse tracks contamination strength and is absent at baseline, so it is a contamination effect, not a specification defect.

E42 Bias transfers; the directed objective backfires

Appendix, the Louisiana COVID-19 testbed subsection, the two closing 'Findings' paragraphs on the symmetric design's bias gains and on the directed objective.

WHAT THE PASSAGE IS DOING

States where the generalization holds and where it weakens, shows that minimizing net directional flux leaves the highest total cross-arm flux of any design and more bias than the symmetric design, demotes the directed refinement to a descriptive diagnostic, and closes with the kernel-alignment caveat and the three pieces of evidence the paper names against it: the CCR≈SR negative control, the symmetrization contrast, and the kernel-misspecification sensitivity of the appendix.

IN ORDINARY LANGUAGE

The clean transfer is bias. Bias is how far an estimate sits from the truth on average. The symmetric resilient design lowers it. Naive bias goes from 0.296 to 0.272. Boundary-censored bias goes from 0.268 to 0.232. Under simple randomization and the symmetric design, the spillover-adjusted estimator's bias is already within simulation noise of zero. So there the design shows up as a smaller spread, not a smaller bias. Coverage is how often the interval holds the truth. The spillover-adjusted estimator is already near target under every design, 0.947 to 0.956. On this map, though, that coverage comes with wide intervals and little power. Its RMSE, the typical size of the error, is 1.136 to 1.487 against a true effect of size 0.693. Its power, how often the test finds that effect, is only 0.085 to 0.101. The naive estimator's coverage rises from 0.627 to 0.663 at moderate contamination. That is about 2.7 units of simulation noise. So the paper places the gain at the two-MCSE threshold, not established. At strong contamination the naive gain is larger, 0.396 to 0.512. But inference is poor under every design there. Now the directed objective. Net flux is inflow minus outflow between the arms. The directed design cuts net flux more than twentyfold. Yet its total cross-arm flux is the highest of any design. The paper ties attenuation bias, which pulls the estimate toward zero, to that total flux. The symmetric objective controls that flux, and the directed one does not. In line with that, the directed design is worse on bias than the symmetric one at the naive and boundary-censored estimators. The boundary-censored gap is under three units of simulation noise, so it is not established. The tie is not one for one. Against simple randomization, the directed design carries more total flux. Yet its naive bias is about the same, 0.293 against 0.296. And its boundary-censored bias is lower, 0.250 against 0.268.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read the directed design's low net imbalance as an advantage. Its power at the spillover-adjusted estimator is 0.085. That is no higher than simple randomization's and below the symmetric design's 0.101. Do not read its slightly lower boundary-censored RMSE, 0.366 against 0.374, as a bias gain; it is at most a small variance refinement. Do not treat

> the source-and-sink reading as a validated transmission pathway; it is a feature of how incidence is reported. And do not carry the size of the resilient design's advantage to settings where the gravity kernel is wrong. The paper calls that size conditional on the contamination model, with the balance-only null, the symmetric-versus-directed contrast, and the appendix's kernel-misspecification sensitivity as its safeguards.

TRY IT

Open supporting/PROPOSED_covid_directional_diagnostics.csv and compare the columns mean_net_flux_imbalance and mean_total_cross_arm_flux for CRR_dir and SR. Confirm that the directed design has by far the smallest net imbalance and the largest total cross-arm flux, then check in Table 11 that it also carries the worse bias of the two contamination-aware designs at the naive and boundary-censored estimators.

E48 Kernel misspecification: what holds and what breaks

Appendix, Sensitivity analyses, the opening paragraph together with the 'Kernel misspecification' paragraph.

WHAT THE PASSAGE IS DOING

Generates the contamination from three kernels the design and the analysis do not assume, and reports that the design's bias reduction holds under all of them, while the core-zone coverage gain is established only under the one-over-distance-cubed kernel, sits at the two-MCSE threshold under exponential decay and is directional only under the one-over-distance kernel, where the naive estimator's coverage gain reverses; the spillover-adjusted estimator's nominal coverage survives none of the three.

IN ORDINARY LANGUAGE

A kernel is the rule for how much people mix at a given distance. The design and the analysis both assume that mixing drops as one over distance squared. This appendix makes the contamination three other ways. They are one over distance, one over distance cubed, and exponential decay over ten kilometers. Coverage means how often the interval holds the true value. Under each of these three kernels, the resilient design lowers the bias of every estimator. It also raises core-zone coverage. But the gain is established, at three or more units of simulation noise, only under the one-over-distance-cubed kernel. Under the decay kernel it clears two units but not three. Under the one-over-distance kernel it is only directional. Those labels are judged against each run's own simulation noise, not against the spread between candidate pools. The spillover-adjusted estimator is a different story. Under the longer-range kernel it corrects too little. Its bias is about 0.11 to 0.15, and its coverage about 0.73 to 0.75. Under the shorter-range kernel it corrects too much. Its coverage is about 0.88 to 0.89. Under the decay kernel its intervals are the wrong width. There coverage falls to 0.51 under the resilient design, against 0.729 under simple randomization. Meanwhile its typical error size, the RMSE, is nearly unchanged, 0.267 against 0.273. So on this kernel the design lowers the adjusted estimator's coverage. This is the shape term of the paper's anchoring decomposition made visible.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

DO NOT ACCIDENTALLY CONCLUDE

Do not conclude the resilient design is immune to kernel error; its advantage is measured under four kernels, not all kernels. Do not rank the spillover-adjusted estimator above the design on the strength of the main tables; those tables assume the kernel is right, and this appendix is where that assumption is charged.

TRY IT

For the simple-randomization rows of Table 13, compare the spillover-adjusted bias under the one-over-distance kernel with the naive bias. Then say what a positive bias of 0.145 means for an estimator that is supposed to remove contamination.

E50 Censoring fraction and elite pool size

Appendix, Sensitivity analyses, the 'Censoring fraction and elite pool size' paragraph.

WHAT THE PASSAGE IS DOING

Varies two tuning choices, how many highest-exposure sites the core-zone estimator drops and how many allocations the elite pool keeps, and shows the design's advantage holds at every setting while the spillover-adjusted estimator is unaffected.

IN ORDINARY LANGUAGE

Two settings were fixed by choice, and the appendix turns each one. The resilient design is the one picked to hold down cross-arm flow. That flow is contamination passing between treated and untreated sites. The first setting is how many of the most exposed sites the core-zone estimator drops. Dropping more of them moves that estimator closer to the truth. Coverage is how often the interval holds the true value. Under the resilient design, coverage is 0.29, 0.35 and 0.48 at 20, 30 and 40 percent dropped. The price is the data thrown away. For that estimator the design still beats simple randomization, which assigns sites at random, at every fraction. The second setting is the elite pool, the set of best allocations kept as candidates. Keeping 10 instead of 80 cuts the pool's average cross-arm flow threefold. Core-zone coverage then rises from 0.30 to 0.40. The spillover-adjusted estimator, which models that flow and subtracts it, stays near 0.95 throughout. The paper's choice of 40 sits in the middle and was fixed before these runs.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

DO NOT ACCIDENTALLY CONCLUDE

Do not conclude that dropping 40 percent of sites is free; the estimand narrows to the remaining 60 percent. Do not conclude a pool of 10 is better; a smaller pool trades leakage for less design variability to average over, and these runs use their own candidate-pool draw, so read them against the between-pool spread.

TRY IT

Explain why coverage rises as more exposed sites are dropped, and name the price paid in the estimand.

E49 Between-pool variability: why the headline is a range

The sensitivity-analyses appendix, the paragraph headed “Between-pool variability” together with the eight-seed between-pool variability table it reports.

WHAT THE PASSAGE IS DOING

Redraws the six thousand-allocation candidate space under eight seeds and reports the spread of every simple-randomization and resilient-design cell, showing that the resilient design’s core-zone coverage varies from 0.28 to 0.48 and that Table 1’s seed is the most favorable, while bias, Type-I error and adjusted-estimator precision are stable.

IN ORDINARY LANGUAGE

The candidate pool the design chooses from is itself random. The appendix draws it eight times. Simple randomization barely moves. Its core-zone coverage runs from 0.26 to 0.29. The resilient design moves a lot. Its core-zone coverage runs from 0.28 to 0.48, with a mean of 0.38. Its naive-estimator coverage, 0.11 to 0.23, is not reliably better than simple randomization’s 0.17 to 0.20. Some things are stable. Bias is always lower under the resilient design. Type-I error is always lower too. And the spillover-adjusted estimator’s precision is always better. The seed used for Tables 1 to 3 gives the most favorable of the eight pools. That holds on boundary-censored bias, coverage and RMSE. It is not the most favorable pool on every measure. So the abstract and conclusions quote the range for boundary-censored coverage. The conclusions also give the between-pool means of boundary-censored bias. Their other numbers come from the favorable pool. They should be read against these ranges.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read the 0.481 in Table 1 as representative; re-simulated, its pool gives 0.479, the top of the eight-pool range. Do not conclude the design’s coverage gain for the naive estimator is established; across pools it is not. And do not conclude the design does nothing: its bias reduction holds in every pool.

TRY IT

From Table 15, compute the resilient design’s mean core-zone coverage minus simple randomization’s mean, then compare that gap with the resilient design’s own range, 0.283 to 0.479. Say which number you would report if you had to choose one.

E90 The six-objective table at every contamination strength

Appendix, the sensitivity-analyses section, the full six-objective operating-characteristics table together with its caption, which follow the short paragraph on selection objectives across contamination.

WHAT THE PASSAGE IS DOING

This is the complete version of the six-objective comparison that the Discussion presents at one contamination strength only. It adds the null block, the weak block and the strong block, and a Type-I error column for each of the three estimators, so a reader can see whether the ranking of objectives holds as leakage grows and whether test size decays alongside coverage. The caption

also fixes the reading rules for the grid: the spillover-adjusted block reports RMSE where the other two report bias, the core-zone cells rest on fewer replicates than the rest because censoring removes an entire arm under some allocations, and power is omitted because it sits at 1.000 in nearly every cell.

IN ORDINARY LANGUAGE

This table is four blocks stacked in one grid. Each block is one contamination strength. Contamination means infection risk mixing across the border between the two arms of the trial. The strength is set by a dial called c. It runs from 0, no leak, to 0.5, the strongest leak in this table. Each block holds the same six ways to pick a design. The first is plain random choice among all 331 candidates. The other five keep the top designs under a stated rule. Three estimators are scored in every row. The naive one ignores leakage. The core-zone one first drops the sites most exposed to it. The spillover-adjusted one corrects for the leak with a mobility model. For the first two, the table gives bias and coverage. Bias is the average gap between the estimate and the truth. Coverage is how often the stated interval holds the truth. For the third, RMSE stands in for bias. RMSE is the typical size of the error. That estimator has almost no bias here. So the designs differ in precision instead. Every estimator also gets Type-I error. That is how often a test claims an effect when there is none. Power is left out of the grid. It was 1.000 in almost every cell. Read one block at a time. Compare the six rows inside it. At c = 0 nothing leaks, and every method is calibrated, with coverage from 0.935 to 0.955. That holds because the simulation has no variation between sites or clusters beyond sampling chance. When the paper adds such variation, sized by an intracluster correlation of 0.02 to 0.05, the same kind of analysis covers only 0.225 to 0.561 even with no leak.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

DO NOT ACCIDENTALLY CONCLUDE

Do not read the c = 0 block as evidence that the design delivers valid inference. Those rows only show that every method behaves when nothing leaks, and the naive and core-zone columns fall away as the blocks move to 0.2, 0.35 and 0.5. The best naive coverage in the strongest block is 0.391, and that same row rejects a true null 0.267 of the time, which is a broken analysis rather than a rescued one. Do not read the row order inside a block as a test of one objective against another. These rows are selection statistics from a simulation, and this table prints no Monte-Carlo error, so neighboring rows can be separated by noise alone. Do not credit the full ten-metric composite's lead over the plain random row to its choice of allocation alone: that row mixes all three resolutions (8, 12 and 16 clusters), and in a separate run against comparators drawn at random from the composite's own resolution its core-zone advantage disappears (0.928 against 0.914 to 0.931 at c = 0.35, 0.905 against 0.882 to 0.894 at c = 0.5), while its naive advantage survives at c = 0.5 (0.406 against 0.323 to 0.340) and is directional at c = 0.35 (0.541 against 0.493 to 0.518). Do not read the steady spillover-adjusted coverage column as safety in general, because every cell here assumes the same mobility kernel that generated the leak. And do not read the omitted power column as strength of evidence. Power is 1.000 because the simulated effect is large and the simulation has no between-cluster variation, not because contaminated analyses are trustworthy; with that variation added at an intracluster correlation of 0.05, the full composite's naive power at c = 0.35 is 0.630 under a cluster-level analysis.

TRY IT

Open supporting/PROPOSED_objectives_oc_MAP.csv, the deposited source for this table, and look at the nrep column. Confirm that every row with estimator “corezone” sits between 1244 and 1500 while every naive and adjusted row is exactly 1500, which is the uneven replicate count the caption warns about. Then scan the power column across the 72 rows and confirm that only three values are below 1, each of them 0.999, exactly as the caption states.

E92 Adding variation between sites: which analyses stay honest

The last appendix, Between-Site and Between-Cluster Heterogeneity: its opening paragraph and the paragraph headed Calibration.

WHAT THE PASSAGE IS DOING

Relaxes the assumption every main table makes, that sites differ only through their measured baseline and random sampling. It adds a random site effect, and for whole-cluster allocation a cluster effect as well, then asks which analyses still give honest intervals when there is no contamination at all.

IN ORDINARY LANGUAGE

In the main tables, every site’s infection rate is fixed by its baseline, its arm and the contamination rule. Real sites also differ for reasons nobody measured. This appendix adds that. Each site gets a random push up or down, and so does each cluster when whole clusters are allocated. The size of the push is set by the intracluster correlation, or ICC. ICC measures how alike two people in the same place are, beyond what the measured baseline explains. The paper tries ICC values of 0, 0.02, 0.05 and 0.10 for single sites, and up to 0.05 for whole clusters. Then it analyzes each simulated trial in several ways. With no contamination at all, the paper’s main analysis gives intervals that are far too narrow once ICC is above zero. That main analysis is a binomial model that treats every examined person as independent. At site allocation its coverage falls to between 0.200 and 0.622. An analysis at the level of the site, which treats each site as one data point, stays on target, covering 0.938 to 0.961. The two-stage cluster-level analysis first averages within each cluster. It comes close, covering 0.928 to 0.959, with a false-positive rate a little above its target. The mixed-model alternative, called a GLMM, is too liberal, so the paper does not use its intervals.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> #### DO NOT ACCIDENTALLY CONCLUDE
> Do not read the ICC values as estimates for any real trial; they are a grid chosen to show sensitivity, and the paper does not say which value applies. Do not read ‘stays on target’ as proof that the site-level analysis is right under contamination; this paragraph checks the case with no contamination. And do not read the binomial model’s failure here as a coding error: at ICC 0 it is the true model, and it fails only because the added variation is real in this simulation.

TRY IT

Open supporting/PROPOSED_icc_site_MAP.csv in the deposit. Keep the rows with c equal to 0 and compare the cover_binom and cover_lm columns at icc 0 and at icc 0.05. The binomial column should drop well below 0.95 while the site-level column stays near it.

E93 Site allocation with variation between sites

The last appendix, the paragraph headed Site allocation under contamination.

WHAT THE PASSAGE IS DOING

Re-measures the site-level stress test with site-to-site variation switched on, separating what the design does to bias from what the analysis does to coverage, and showing how the design's precision gain for the spillover-adjusted estimator shrinks.

IN ORDINARY LANGUAGE

Now contamination is switched on, at the moderate setting, c equal to 0.35. Two things happen. First, how much the design cuts bias depends on how the analysis counts sites. The main analysis counts each examined person, so big sites weigh more. The site-level analysis counts each site once. The design's contamination score also gives big sites more weight. So the design's bias cut looks larger in the main analysis at ICC 0, from 0.285 to 0.163, than in the site-level one at ICC 0.05, from 0.278 to 0.239. Second, the design still helps coverage once the analysis is honest about site-to-site variation. For the boundary-censored estimator, which drops the most exposed sites, the design raises coverage by 0.150, 0.120 and 0.084 at ICC 0.02, 0.05 and 0.10. Neither the naive nor the boundary-censored estimator gets near its target at this contamination strength. The spillover-adjusted estimator keeps coverage between 0.920 and 0.947. In the mixed-model estimate, the design's cut in its error shrinks from 40% to between 16% and 28%. In the site-level analysis's own estimate, the cut is no smaller at ICC 0.02. That is because site-to-site variation is error that no design can remove.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read the smaller bias cut as the design failing: both analyses still show the design lowering bias, and the difference comes from how sites are weighted. Do not read a coverage gain as coverage restored; the boundary-censored estimator stays far below its target here. And do not carry the error cut of the main tables into a real trial without the site-to-site variation this paragraph adds.

TRY IT

In supporting/PROPOSED_icc_site_MAP.csv keep the rows with c equal to 0.35, a nonzero delta and estimator corezone. For each ICC, subtract the SR row's cover_lm from the CRR row's. You should get about 0.150, 0.120 and 0.084 at ICC 0.02, 0.05 and 0.10.

E94 Whole clusters: the composite against fair comparators

The last appendix, the paragraph headed Whole-cluster allocation.

WHAT THE PASSAGE IS DOING

Separates what the ten-metric composite buys through its preference for coarse maps from what it buys through its choice of allocation, by adding comparators drawn at random at the same coarseness, and charges the coarse maps the variance that cluster-to-cluster variation adds.

IN ORDINARY LANGUAGE

This paragraph asks what the ten-metric composite really buys when whole clusters are allocated. The earlier comparison mixed coarse and fine maps. And the composite prefers coarse maps, with fewer, bigger clusters. So the paper adds three fair comparators. These are allocations drawn at random only from maps as coarse as the composite's picks. Against them, the composite's better coverage for the boundary-censored estimator disappears, 0.928 against 0.914 to 0.931. That advantage came from coarseness, not from the composite's choice of allocation. Its advantage for the naive estimator holds at strong contamination, 0.406 against 0.323 to 0.340. But at moderate contamination it is only directional. Now add variation between clusters, and analyze at the cluster level. Then naive coverage rises for every pool. The composite now matches the fair comparators. It keeps the lowest naive bias, 0.117 against 0.128 to 0.132. But coarse maps cost power, the chance of detecting a real effect. Power is 0.630 for the composite against 0.711 for random allocation over maps of every coarseness.

EVIDENCE STATUS **MODELED LOCKED** a number from a stated, fixed-seed simulation

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not read 'matches' as 'the composite is useless': it still has the lowest naive bias, and how firm that lead is depends on the comparator and on the contamination strength. Do not read the power numbers as a forecast for a real trial; they come from one variance setting, ICC 0.05. And do not read coarse maps as free: fewer, larger clusters cut contamination but leave fewer units to compare, which is where the lost power comes from.

TRY IT

Open supporting/PROPOSED_icc_objectives_MAP.csv. Keep the naive rows with c equal to 0.35, a nonzero delta and icc equal to 0.05, and read cover_cl and reject_cl for FULL, SR-K8, SR-FULLmix, SR-Spec8 and SR-cluster. Coverage should be close across the first four and lower for SR-cluster, while power, the reject_cl column, should be highest for SR-cluster.

E95 Reading the two heterogeneity tables

The last appendix, the two tables after its three paragraphs: site-level allocation with between-site variation, then whole-cluster allocation with between-cluster variation.

WHAT THE PASSAGE IS DOING

Lays out the numbers behind the appendix's three paragraphs, with a column structure that keeps each analysis and each variance setting separate, so a reader can check every quoted figure against its row.

IN ORDINARY LANGUAGE

These two tables hold the numbers behind the last three paragraphs. The first covers allocating single sites at moderate contamination. Each row is one estimator under one design. The columns give bias under the two analyses, and coverage under the main analysis with no extra variation. They also give coverage under the site-level analysis at three ICC values, and the mixed model's coverage. Then come the site-level false-positive rate and power at ICC 0.05. The second table covers whole clusters. Each row is one way of choosing allocations. The rows are the six objectives from the main comparison, then the three fair comparators drawn at the same coarseness. A column called K mix counts how many of a pool's allocations use 8, 12 or 16 clusters. For the comparator drawn with the composite's own mix, it shows that mix instead. The two blocks give moderate and strong contamination, c equal to 0.35 and 0.5. A column n counts the simulated trials in which the boundary-censored analysis still had at least two clusters in each arm.

EVIDENCE STATUS **DEFINITION** defines a term

> **DO NOT ACCIDENTALLY CONCLUDE**
> Do not compare numbers across the two tables: they come from different allocation units and different variance settings. Do not read the binomial columns at ICC 0 as a fresh result; they reproduce the main tables, which checks the code. And do not read a larger n as a better design: it only counts how often censoring left enough clusters to analyze.

TRY IT

Run tools/make_icc_tables.py in the deposit with four arguments. The first three are outputs/PROPOSED_icc_site_MAP.csv, outputs/PROPOSED_icc_objectives_MAP.csv and outputs/PROPOSED_icc_objectives_pools_MAP.csv. The fourth is a folder for the results. It rebuilds both tables, captions included, and the result should match the manuscript exactly.